WaveFEM

SOLVER VALIDATION

Validation Report

WaveFEM compared against Ansys HFSS on S-parameters, far-field patterns, adaptive mesh convergence and runtime.

22 examples · Generated 03 November 2026

INTRODUCTION

About WaveFEM

WaveFEM is a three-dimensional electromagnetic simulator based on the finite element method (FEM). It is built for microwave and RF design: filters, waveguide components and antennas. Everything runs from one desktop application, from importing geometry to plotting fields, S-parameters and radiation patterns.

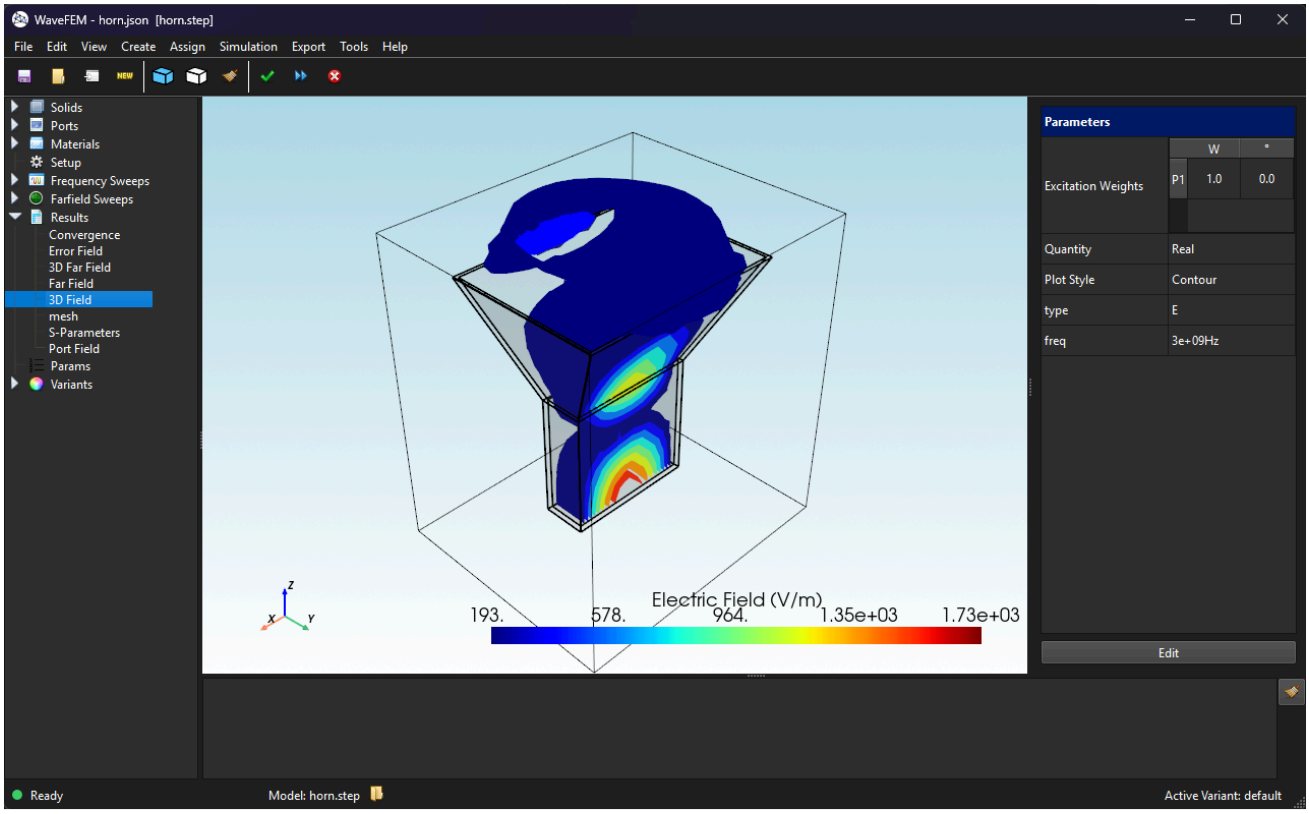


Figure 1 — Electric field inside a horn antenna

INTRODUCTION

About this document

Ansys HFSS is widely regarded as the reference tool for high-frequency electromagnetic simulation, with a mature and feature-rich environment. Its licensing costs, however, put it out of reach for many individual engineers, small teams and academic researchers. WaveFEM was developed to close this gap: a finite element solver with the accuracy and capability needed for the same class of problems, at a much lower cost.

This report presents 22 simulation examples solved in both WaveFEM and HFSS, ranging from canonical benchmark structures to practical RF and microwave designs. Each example compares S-parameters, far-field patterns, adaptive mesh convergence and runtime, so that solution accuracy, convergence behaviour and the range of supported problem types can be judged directly. Together, the examples serve as both a validation study and a practical reference for engineers evaluating WaveFEM as their primary simulation tool.

OVERVIEW

Summary of results

Each example is solved in WaveFEM and in Ansys HFSS with the same geometry, adaptive refinement target (maximum delta S) and frequency sweep. Runtimes are wall-clock times from the solver logs; element counts are for the final adaptive pass.

Example	Category	WaveFEM time	HFSS time	WaveFEM elements	HFSS elements	Page
Coupled Line Filter	Filter	00:01:03	00:01:33	29,677	17,994	4
Hairpin Filter	Filter	00:00:23	00:01:02	11,082	4,559	6
Interdigital Bandpass Filter	Filter	00:00:37	00:01:14	20,407	16,596	8
L Resonator Notch Filter	Filter	00:02:32	00:02:28	58,011	39,070	10
Stepped Impedance Filter	Filter	00:00:04	00:00:38	3,368	2,975	12
Waveguide Filter	Filter	00:01:09	00:03:38	34,387	48,845	14
IC Filter	Filter	00:00:36	00:00:53	17,283	9,453	16
Corrugated Horn	Antenna	00:01:11	00:01:24	52,861	32,687	18
Dielectric Resonator Antenna	Antenna	00:00:04	00:00:30	4,810	3,191	21
Dipole	Antenna	00:00:24	00:00:32	19,906	9,761	24
Dual Band Slot Antenna	Antenna	00:00:10	00:00:57	6,908	8,598	27
Planar Vivaldi	Antenna	00:02:47	00:01:17	54,475	15,177	31
Probe Fed Patch	Antenna	00:00:29	00:01:14	26,213	19,159	34
Slotted Waveguide Antenna	Antenna	00:00:05	00:00:31	5,409	4,496	37
Two Element Patch Array	Antenna	00:01:54	00:02:16	73,331	42,337	40
Inset Fed Patch	Antenna	00:00:31	00:01:26	23,642	24,003	43
Pyramidal Horn	Antenna	00:00:10	00:00:36	7,991	5,248	46
Exponential Horn	Antenna	00:00:11	00:01:16	14,197	23,138	49
Spiral Antenna	Antenna	00:04:21	00:05:24	40,711	36,454	52
Spiral Inductor	Planar	00:00:16	00:00:49	8,423	8,894	56
Coax to Waveguide Adapter	Transmission Line	00:00:05	00:00:38	3,486	3,732	60
Waveguide	Transmission Line	00:00:10	00:00:45	4,909	5,915	62

FILTER

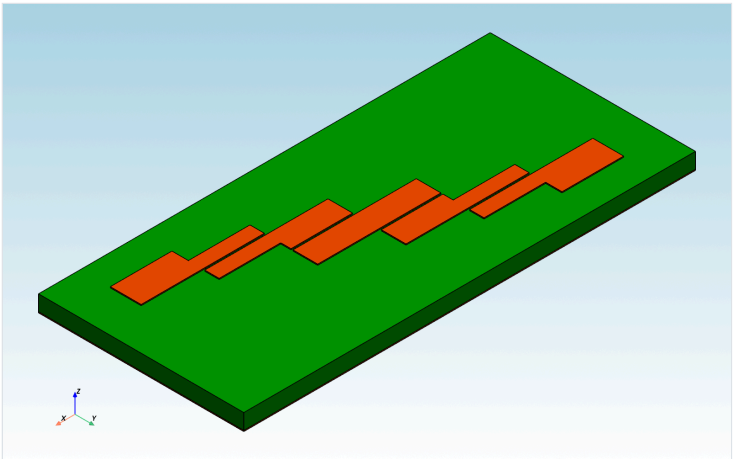
Coupled Line Filter

A 5.5-7.5 GHz bandpass filter. Using a 1.5 mm substrate with e_r=4.3

Simulation Setup

Refinement frequency	7 GHz
Delta S	0.02
Frequency range	2 - 10 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:01:03	00:01:33
Adaptive meshing	00:00:29	00:00:39
Frequency sweep	00:00:28	00:00:52
Adaptive passes	12	10
Final elements	29,677	17,994
Matrix size	169,338	110,368

Convergence

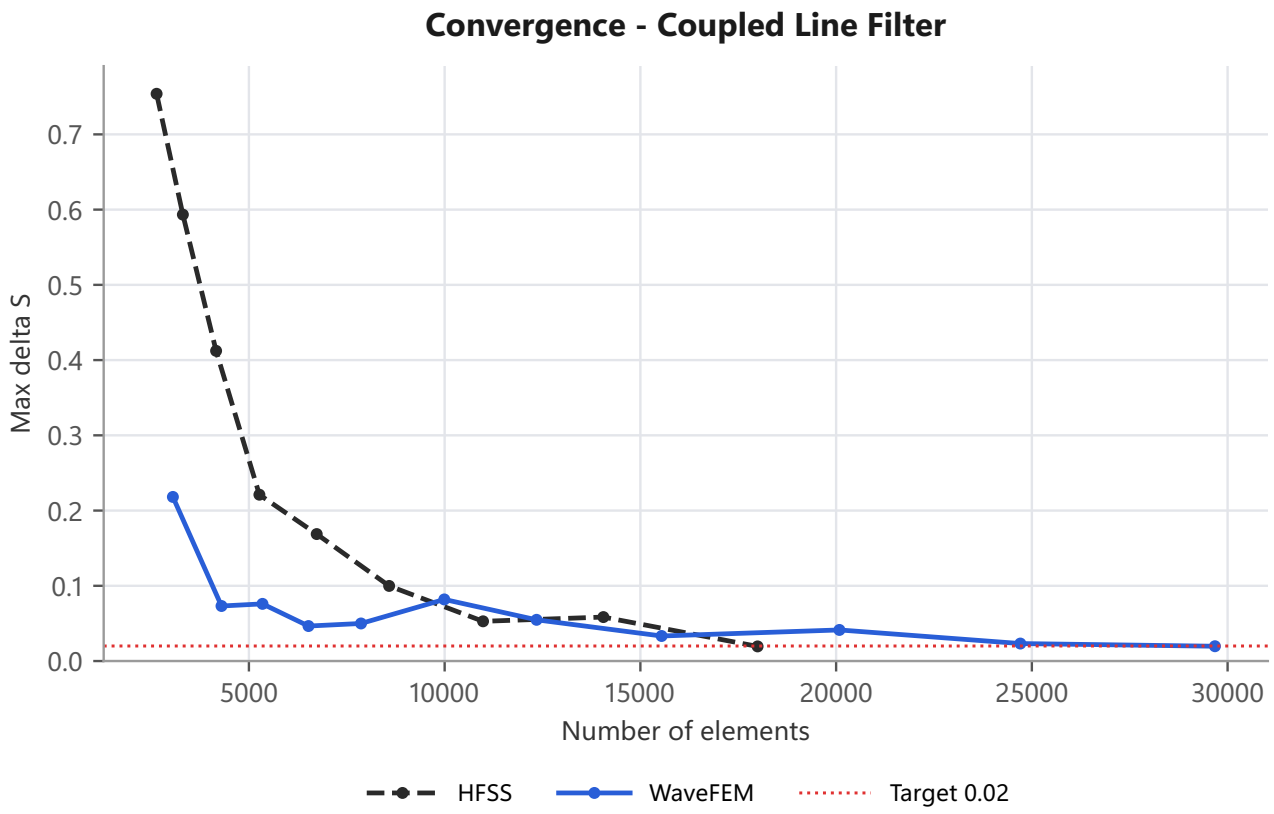


Figure 2 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

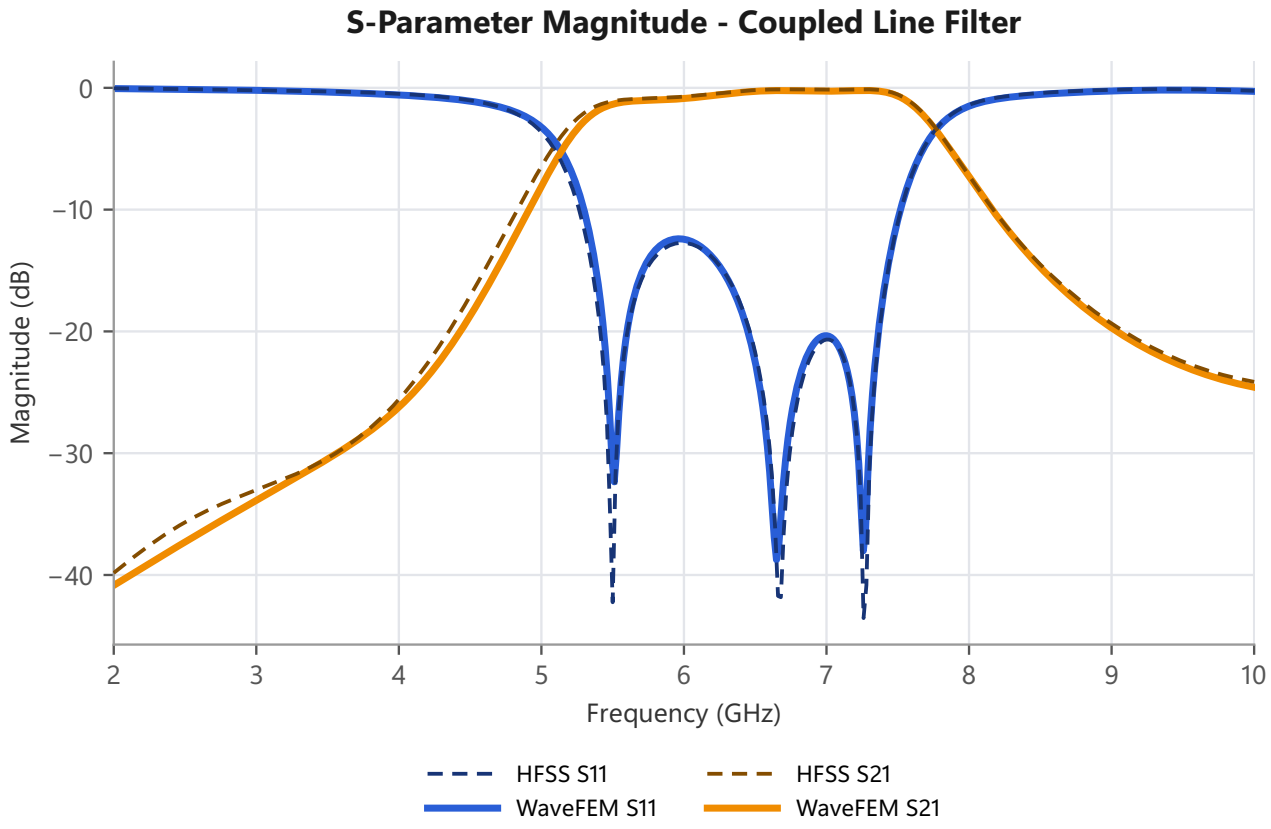


Figure 3 — S-parameter magnitude.

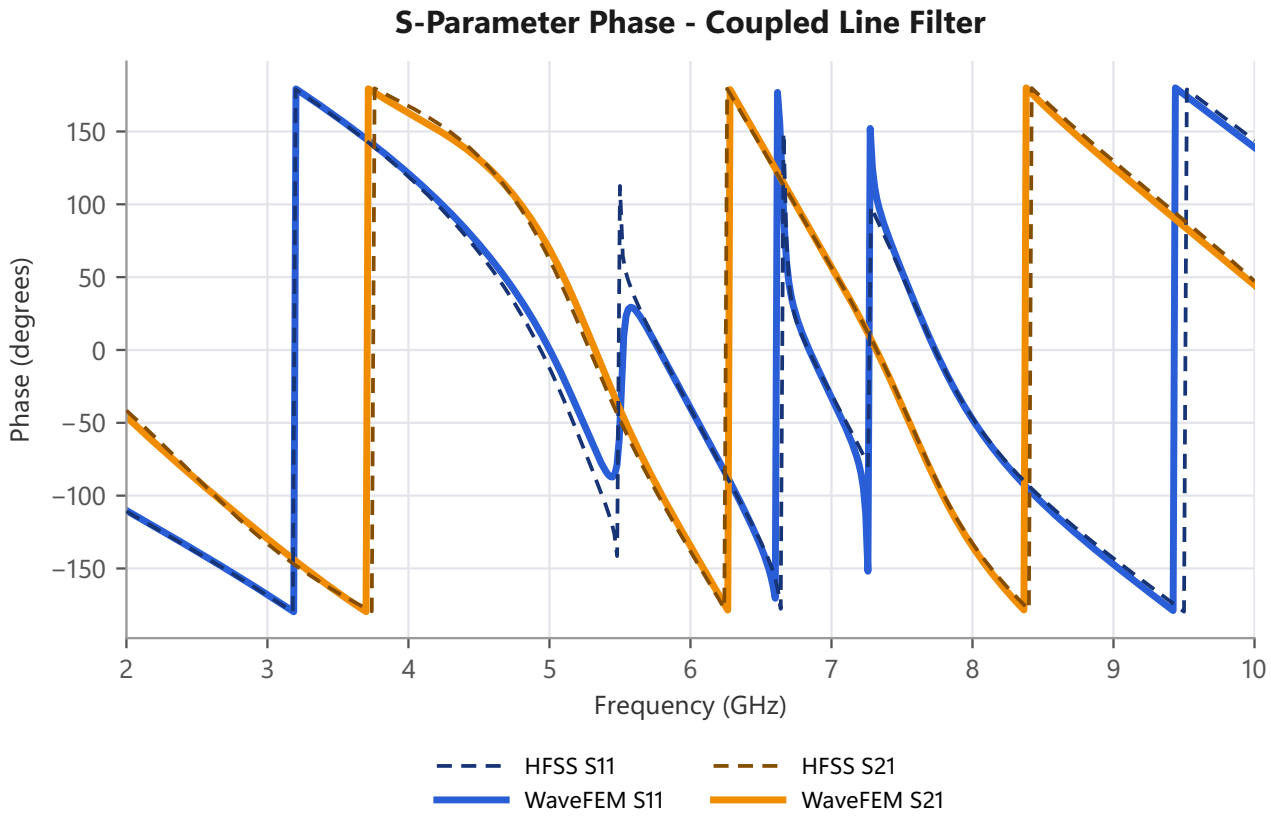


Figure 4 — S-parameter phase.

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FILTER

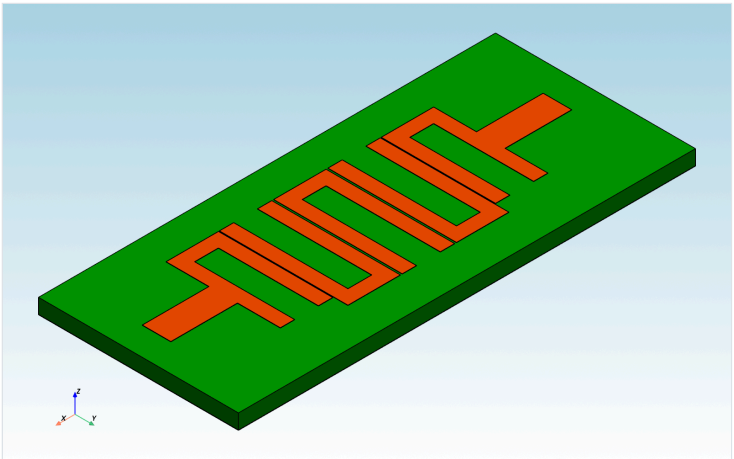
Hairpin Filter

A 2-4 GHz bandpass hairpin filter. Using a 1.55mm substrate with e_r=4.4 & tand=0.01

Simulation Setup

Refinement frequency	3.5 GHz
Delta S	0.02
Frequency range	1 - 5 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:00:23	00:01:02
Adaptive meshing	00:00:11	00:00:28
Frequency sweep	00:00:10	00:00:32
Adaptive passes	9	10
Final elements	11,082	4,559
Matrix size	61,046	26,618

Convergence

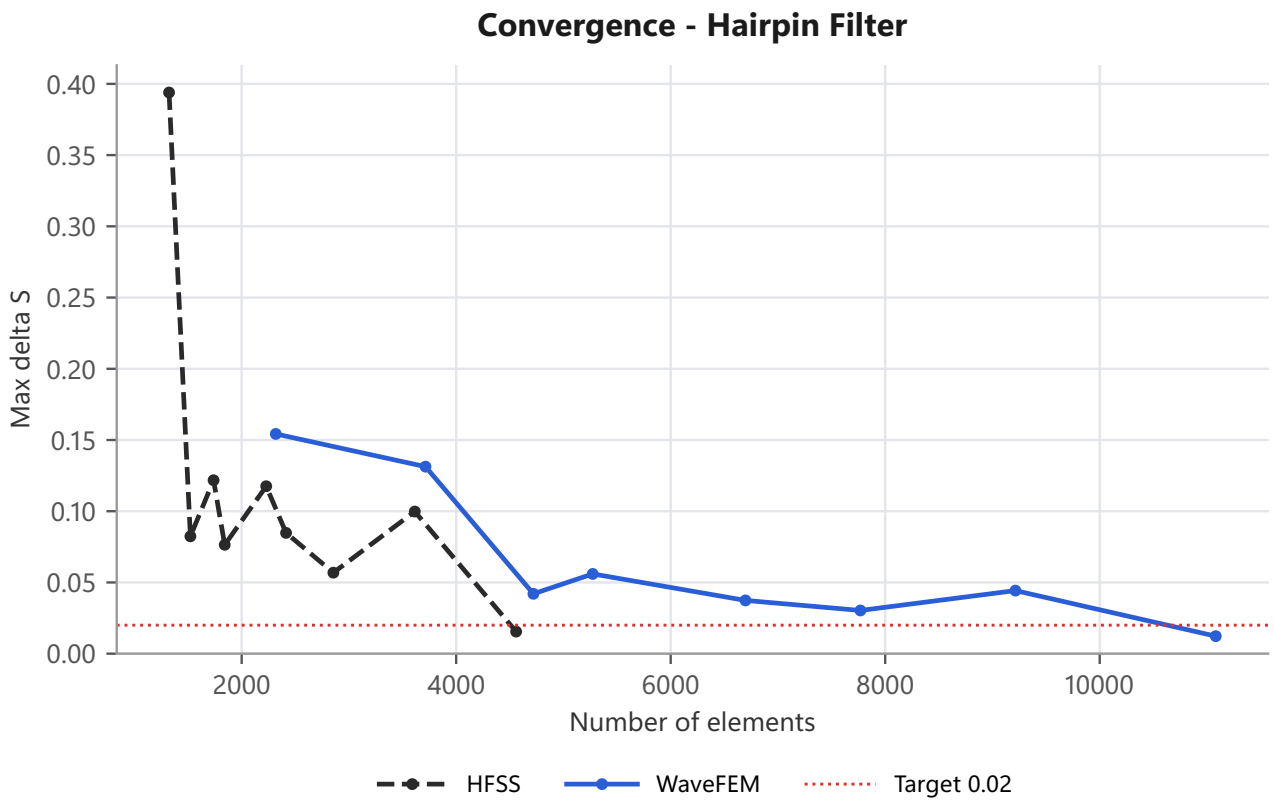


Figure 5 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

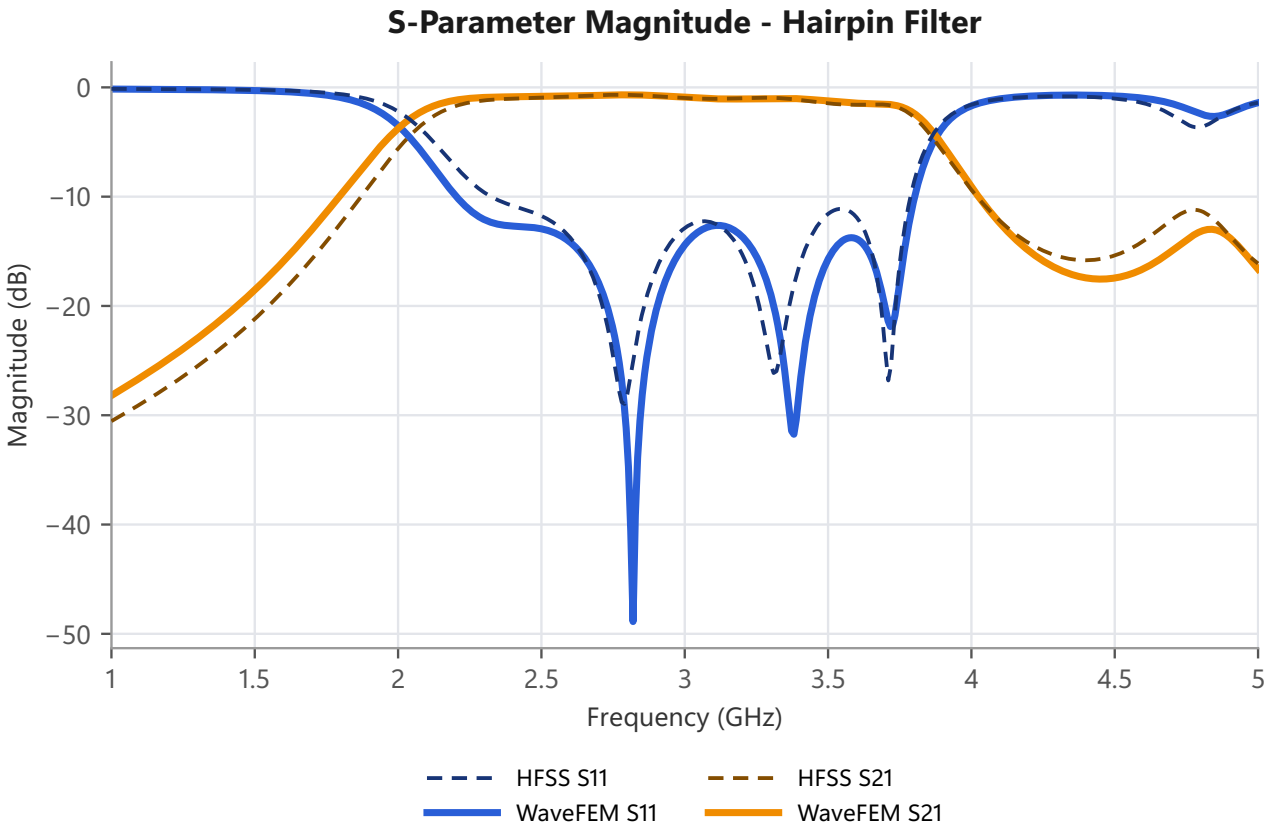


Figure 6 — S-parameter magnitude.

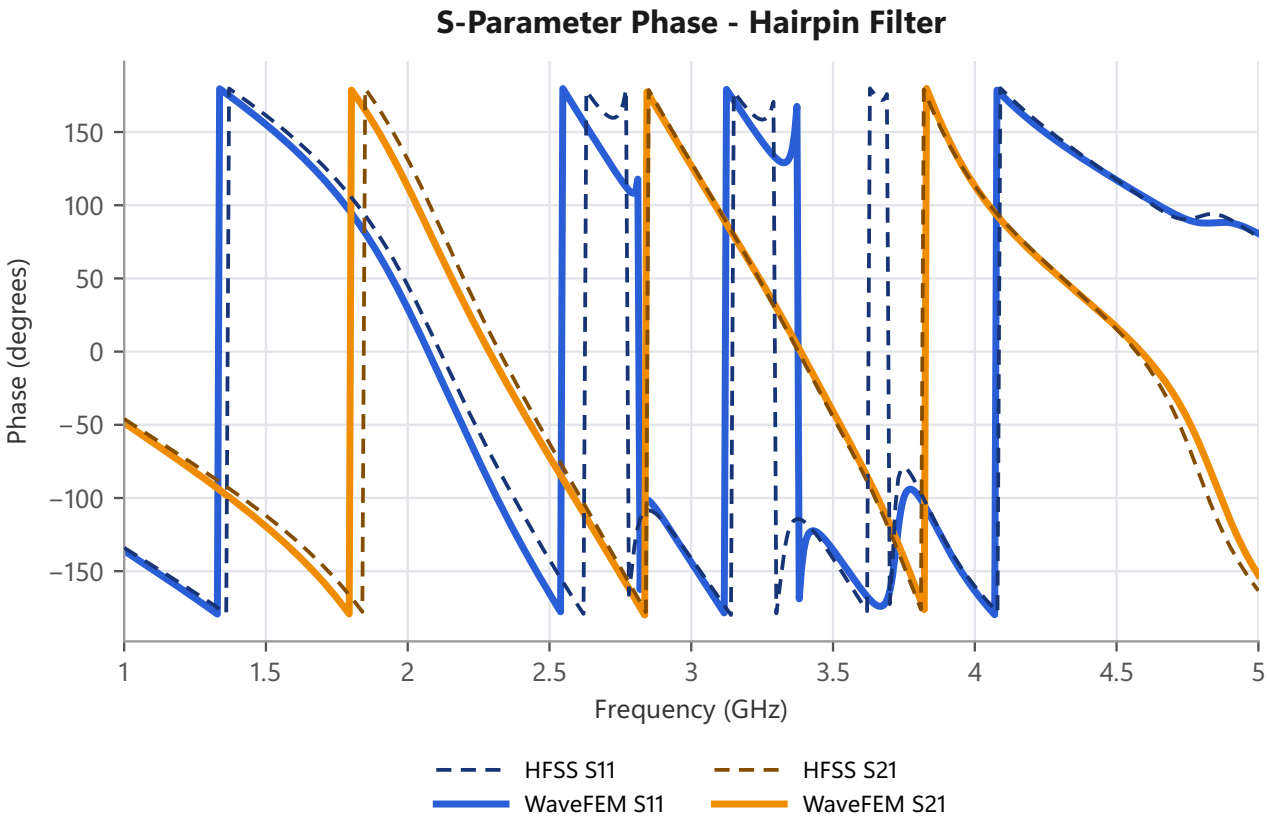


Figure 7 — S-parameter phase.

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FILTER

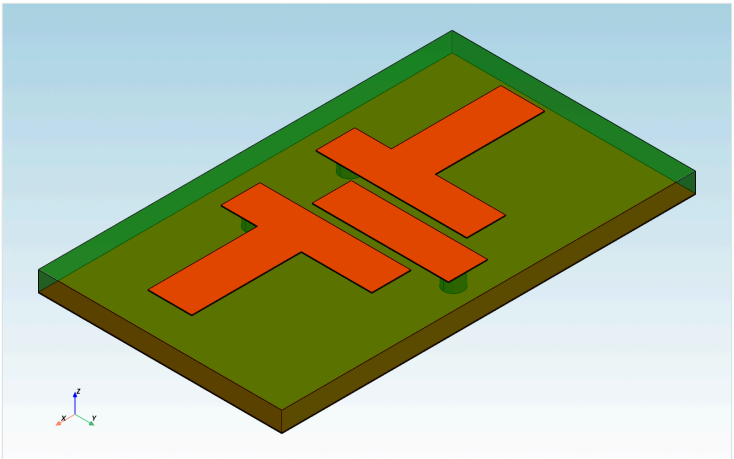
Interdigital Bandpass Filter

A 7-9 GHz bandpass filter. Using a 0.8mm substrate with e_r=3.55 tand=0.0027

Simulation Setup

Refinement frequency	8 GHz
Delta S	0.02
Frequency range	4 - 12 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:00:37	00:01:14
Adaptive meshing	00:00:18	00:00:27
Frequency sweep	00:00:15	00:00:41
Adaptive passes	10	8
Final elements	20,407	16,596
Matrix size	116,192	100,730

Convergence

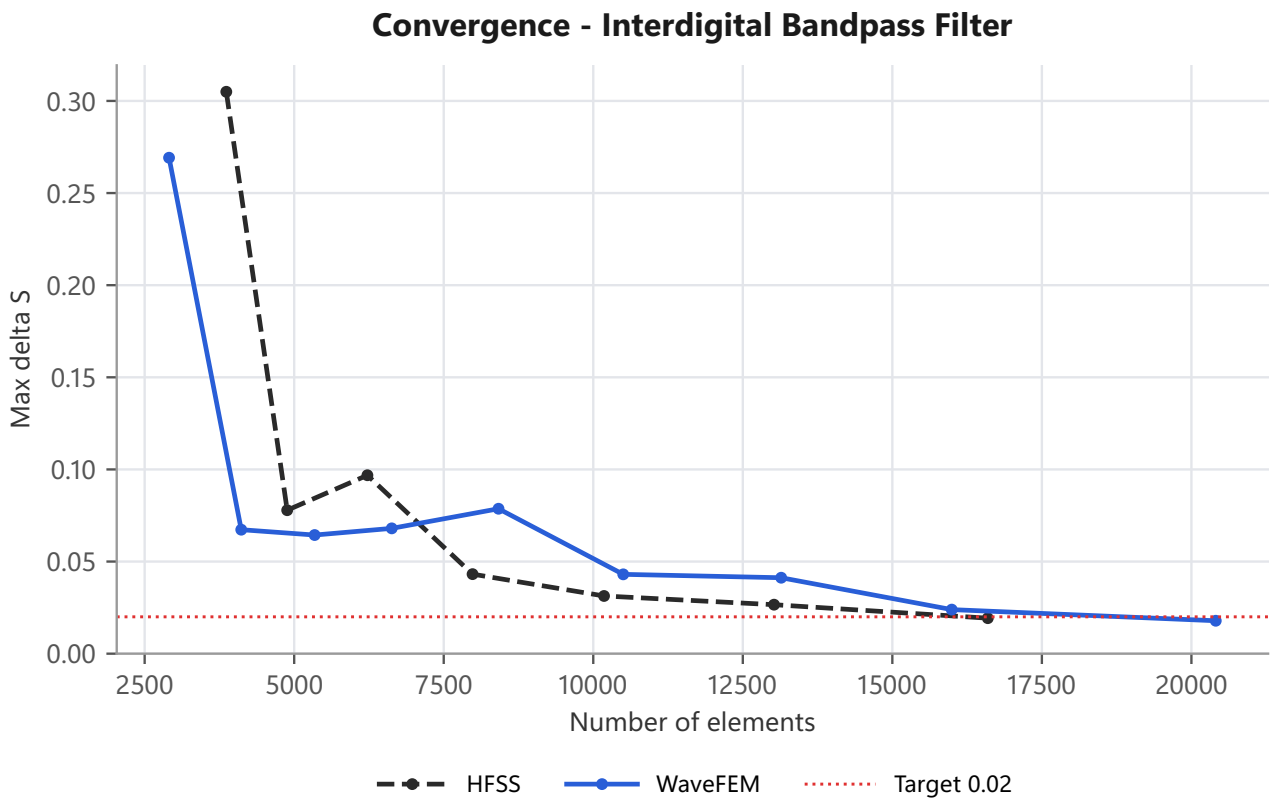


Figure 8 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

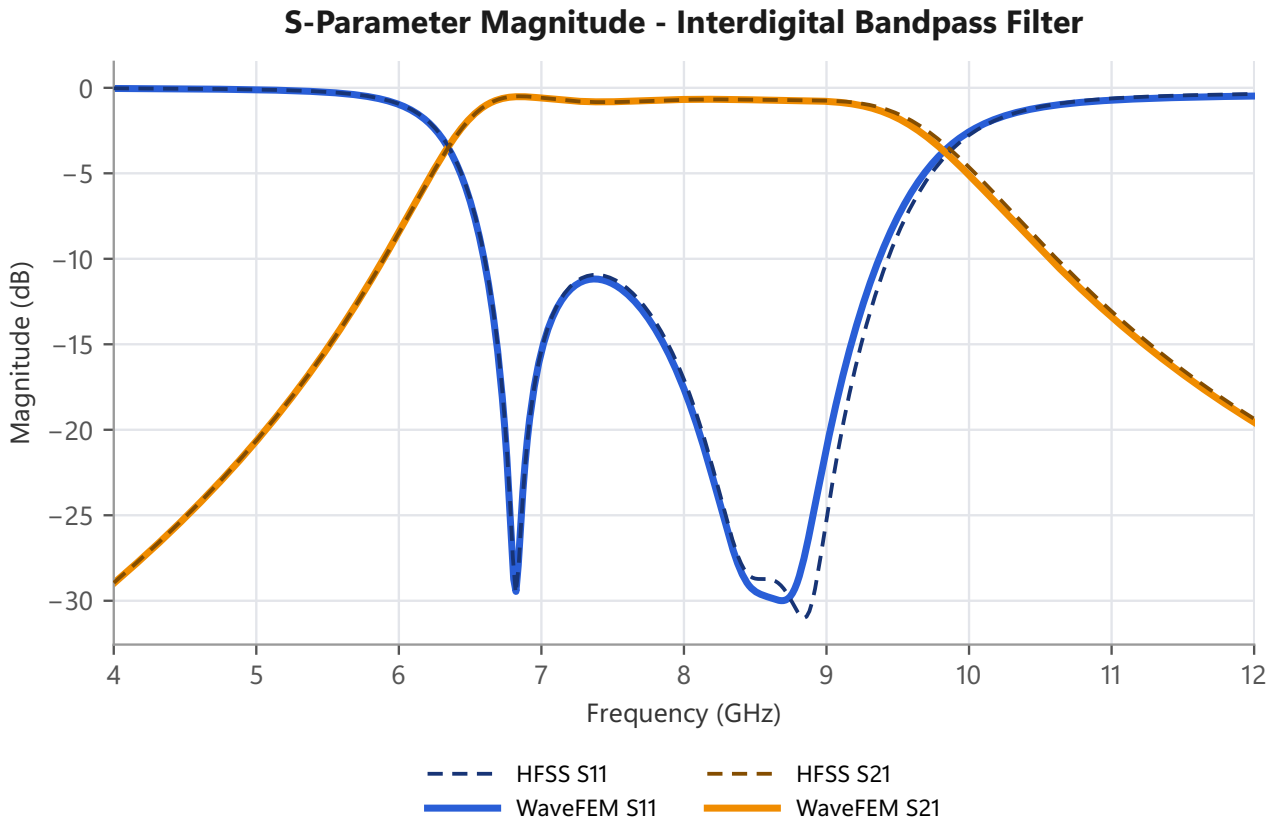


Figure 9 — S-parameter magnitude.

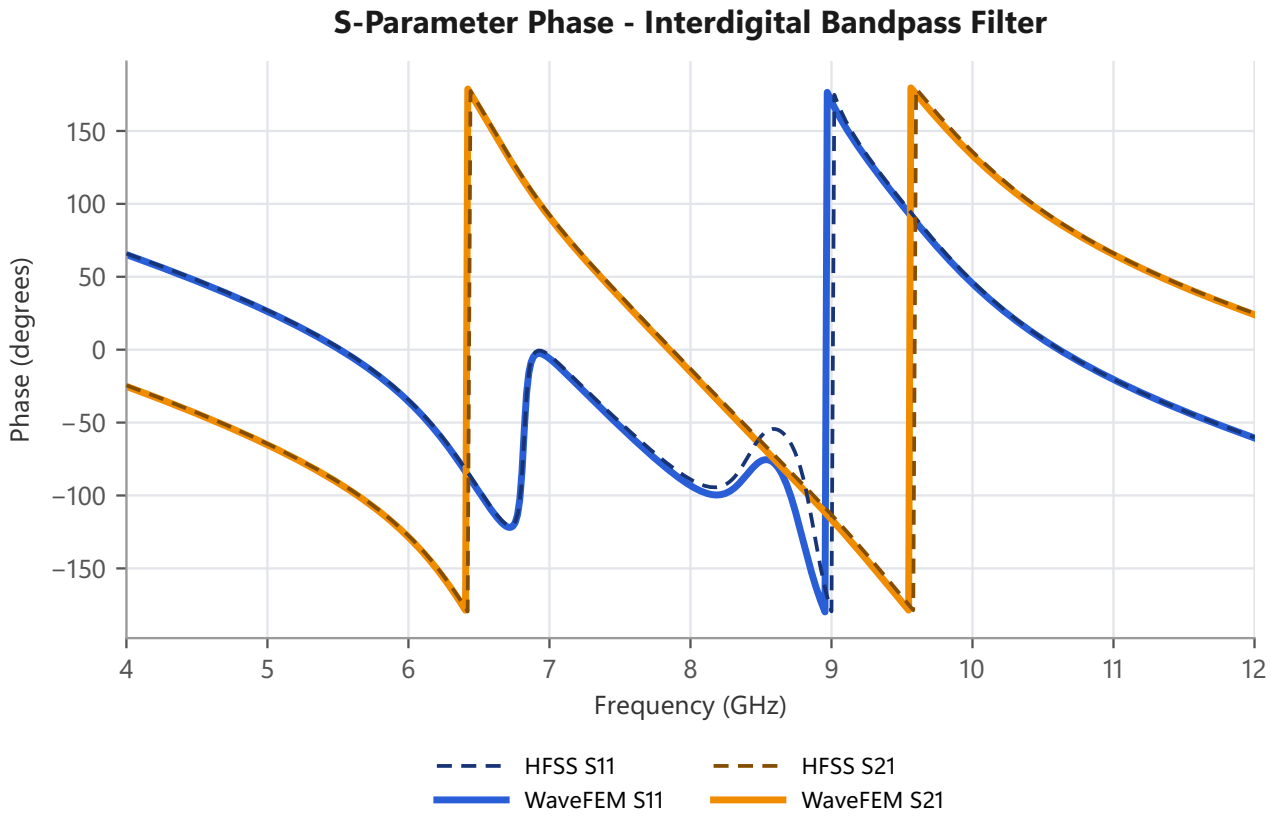


Figure 10 — S-parameter phase.

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FILTER

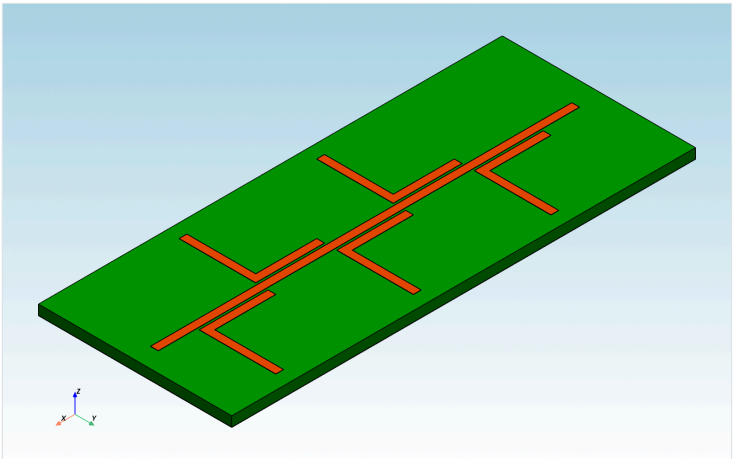
L Resonator Notch Filter

A 3.2 GHz notch filter. Using a 1.27mm substrate with e_r=10.8

Simulation Setup

Refinement frequency	3.4 GHz
Delta S	0.02
Frequency range	2 - 4 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:02:32	00:02:28
Adaptive meshing	00:01:00	00:01:02
Frequency sweep	00:01:18	00:01:24
Adaptive passes	14	15
Final elements	58,011	39,070
Matrix size	329,498	239,094

Convergence

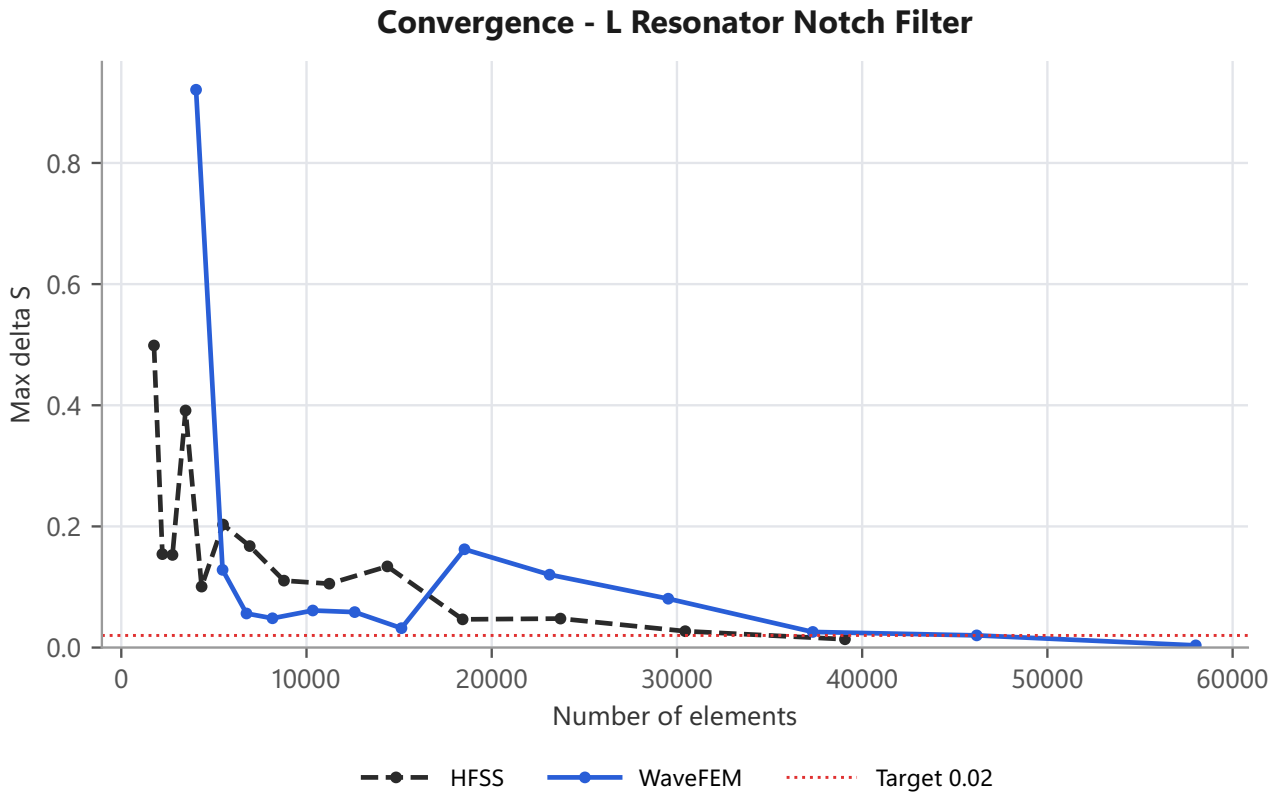


Figure 11 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

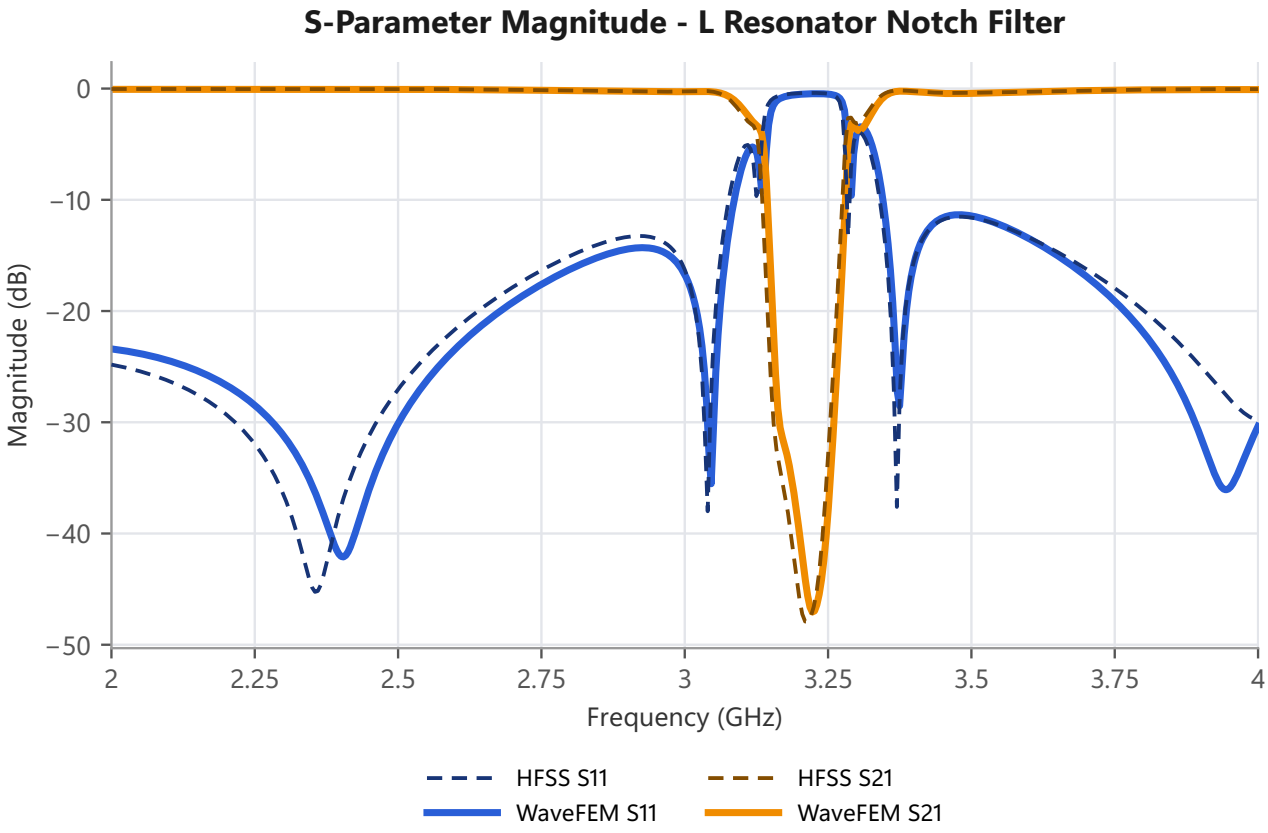


Figure 12 — S-parameter magnitude.

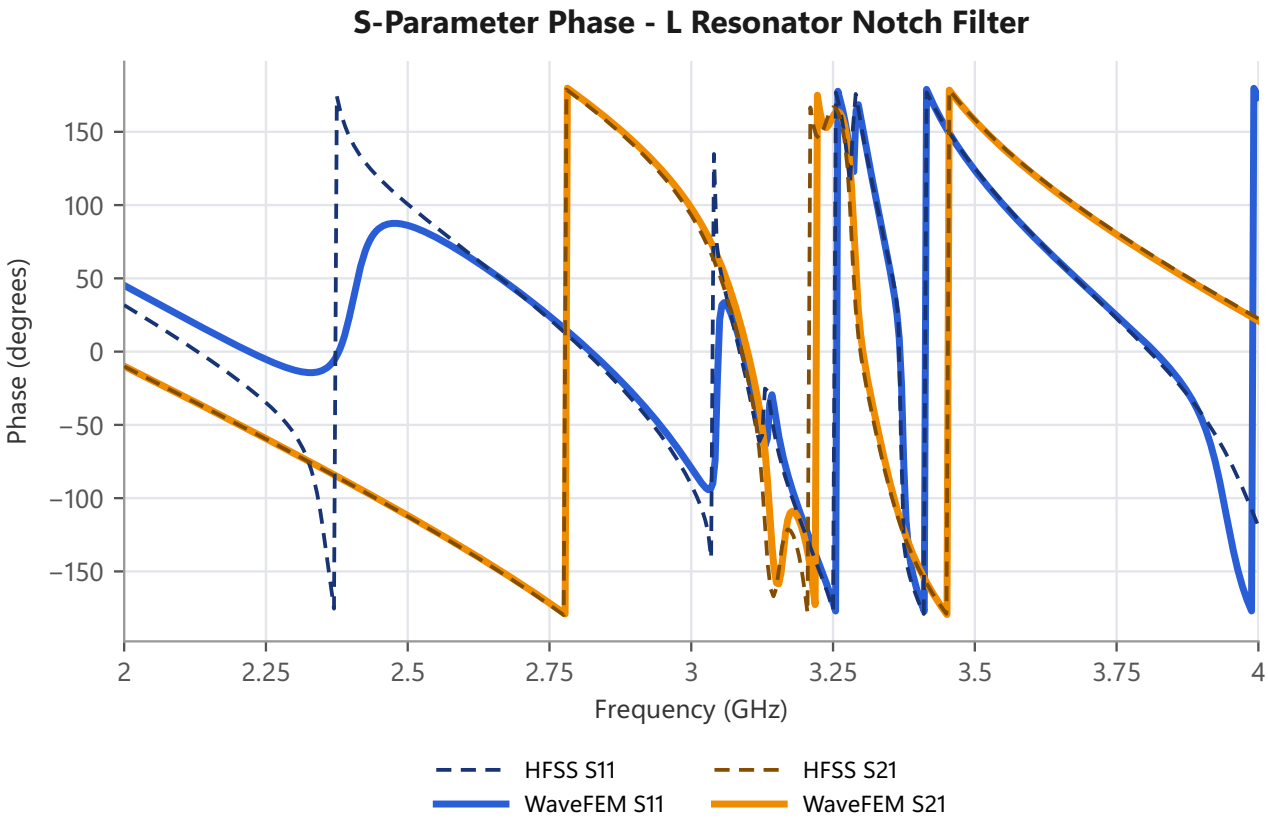


Figure 13 — S-parameter phase.

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FILTER

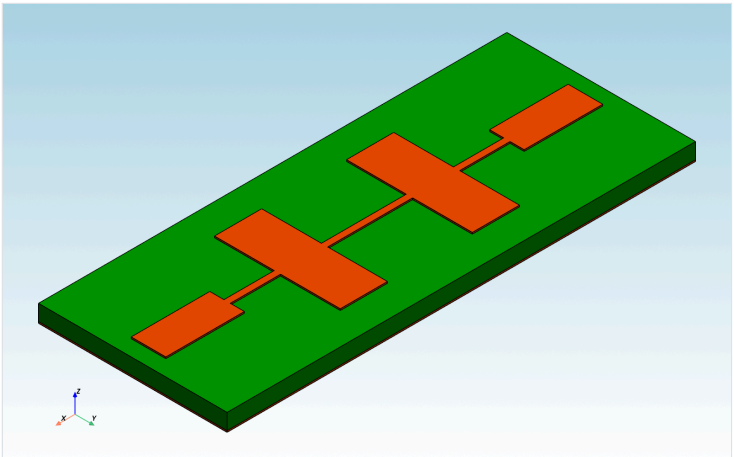
Stepped Impedance Filter

A 9 GHz lowpass filter. Using a 0.5mm substrate with ε_r=3.55 tand=0.0027

Simulation Setup

Refinement frequency	7 GHz
Delta S	0.02
Frequency range	0.1 - 20 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:00:04	00:00:38
Adaptive meshing	00:00:02	00:00:17
Frequency sweep	00:00:02	00:00:19
Adaptive passes	3	7
Final elements	3,368	2,975
Matrix size	19,102	17,656

Convergence

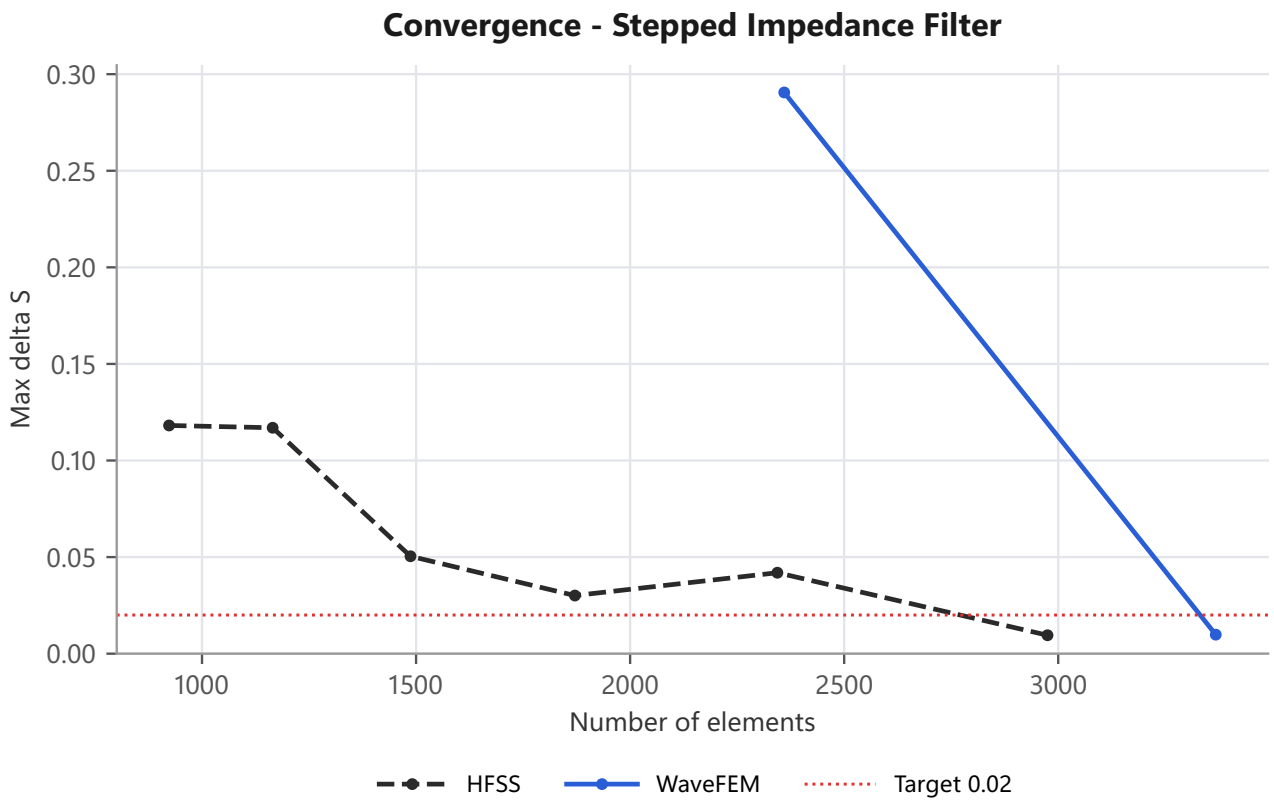


Figure 14 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

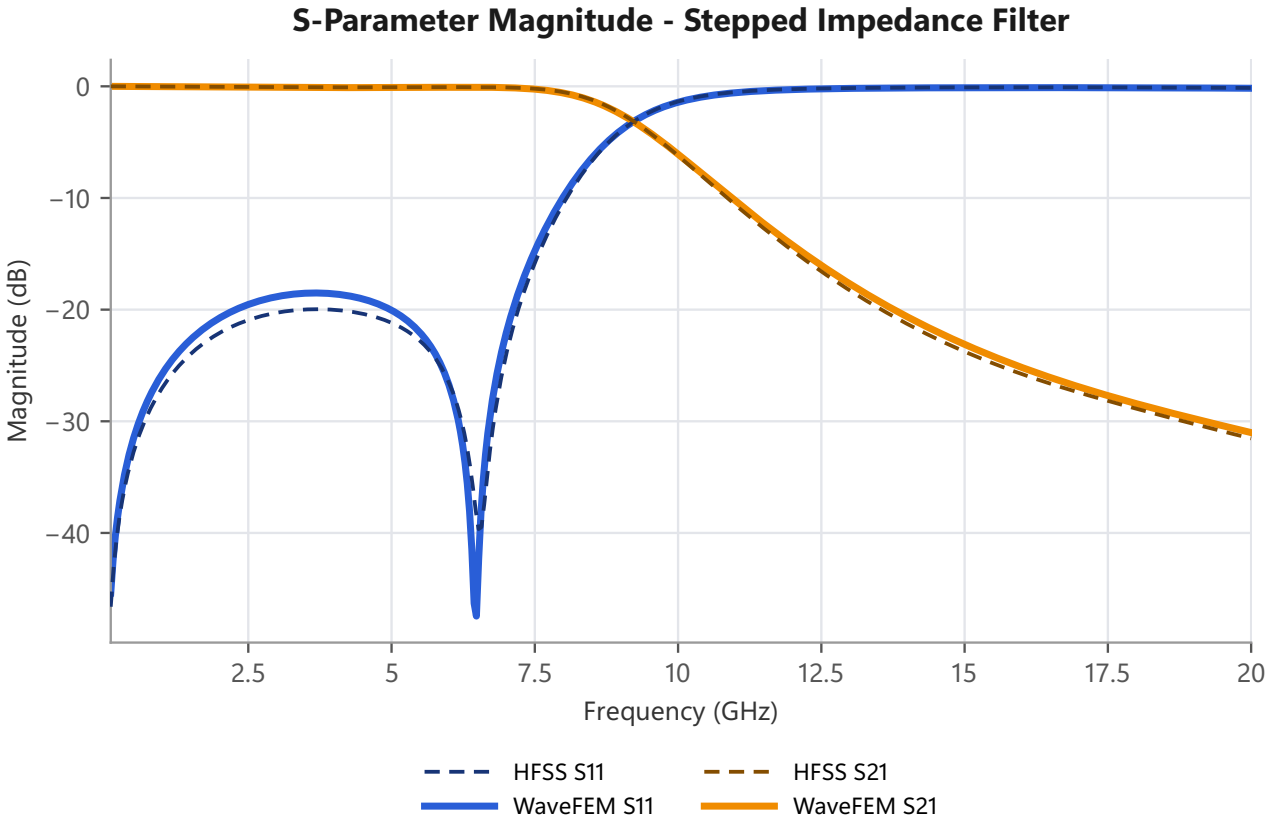


Figure 15 — S-parameter magnitude.

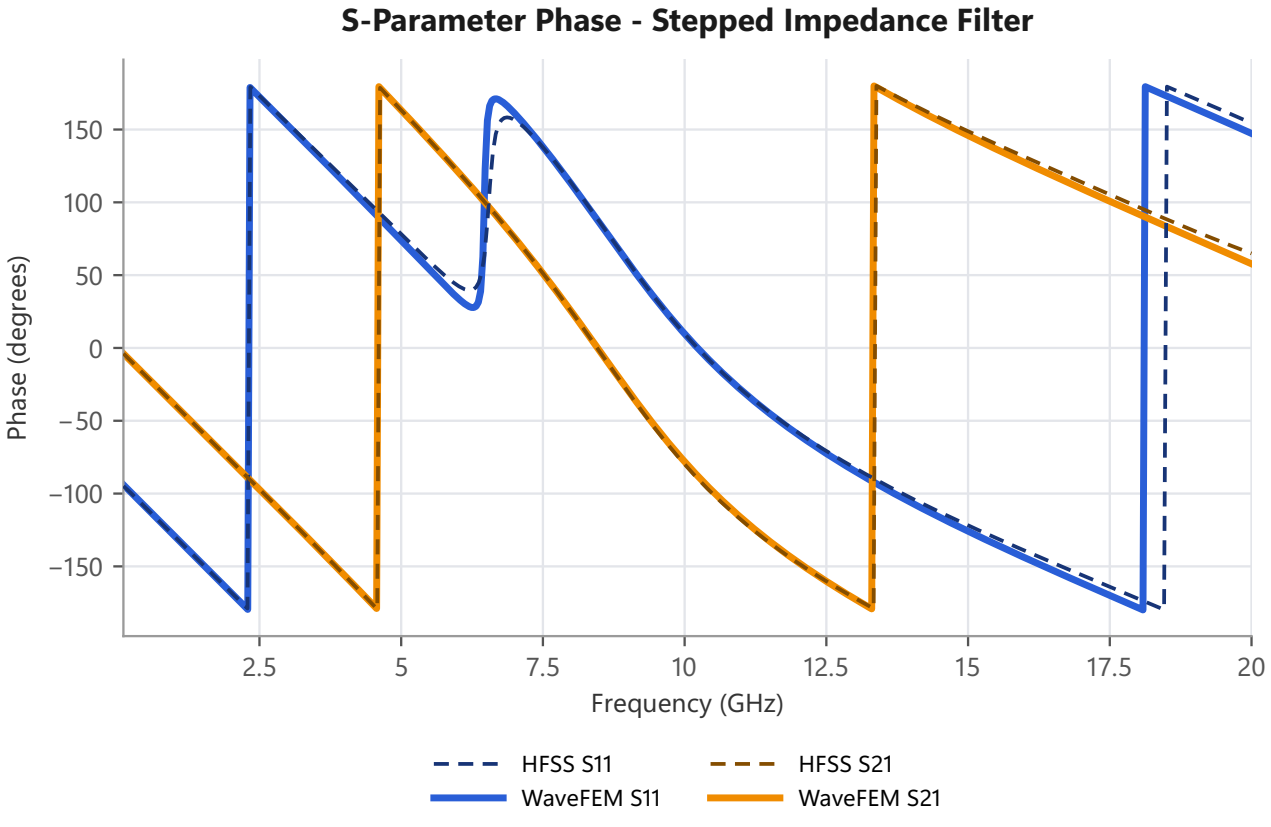


Figure 16 — S-parameter phase.

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FILTER

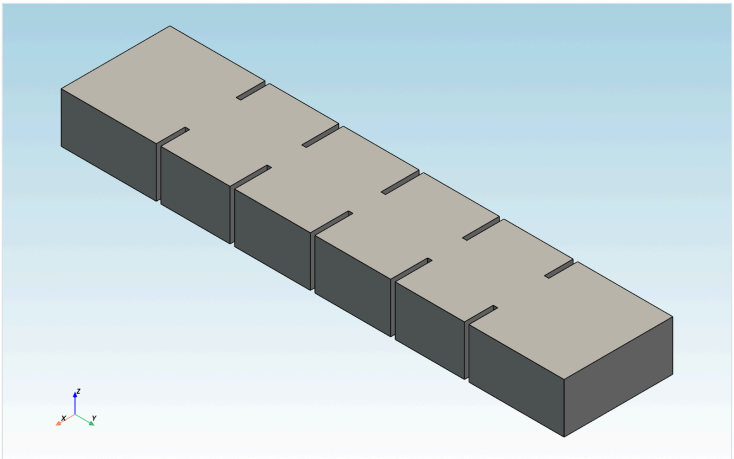
Waveguide Filter

A five-cavity inductive iris bandpass filter in a 23 x 10 mm rectangular waveguide

Simulation Setup

Refinement frequency	11 GHz
Delta S	0.02
Frequency range	9 - 13 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:01:09	00:03:38
Adaptive meshing	00:00:29	00:01:26
Frequency sweep	00:00:31	00:02:08
Adaptive passes	15	19
Final elements	34,387	48,845
Matrix size	194,696	298,116

Convergence

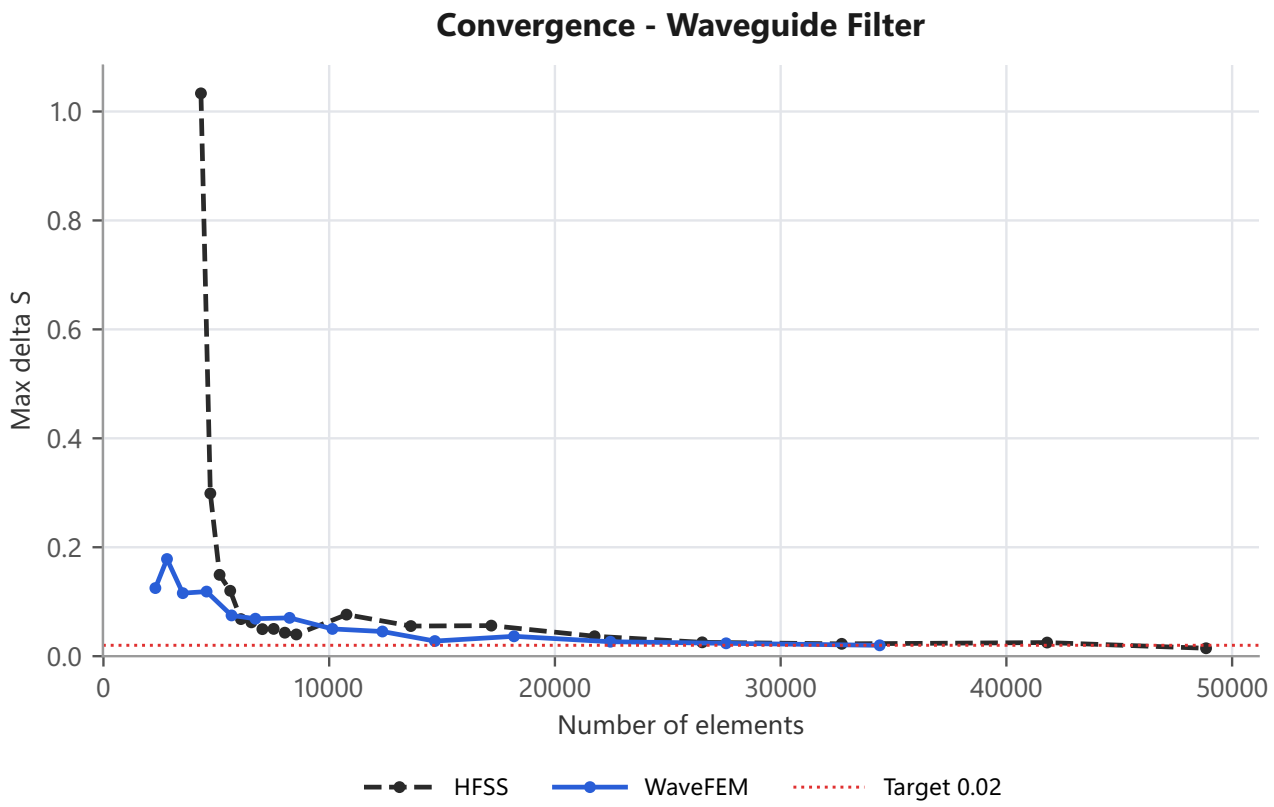


Figure 17 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

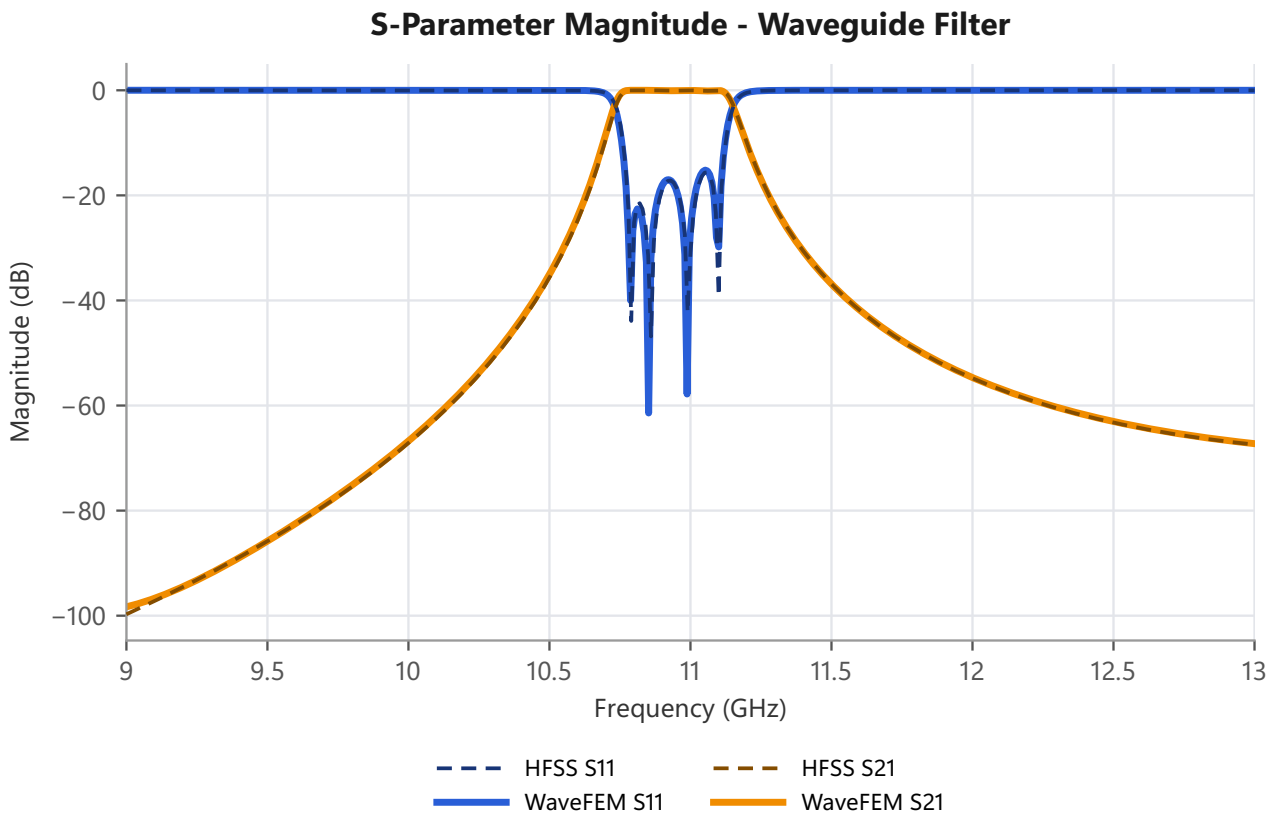


Figure 18 — S-parameter magnitude.

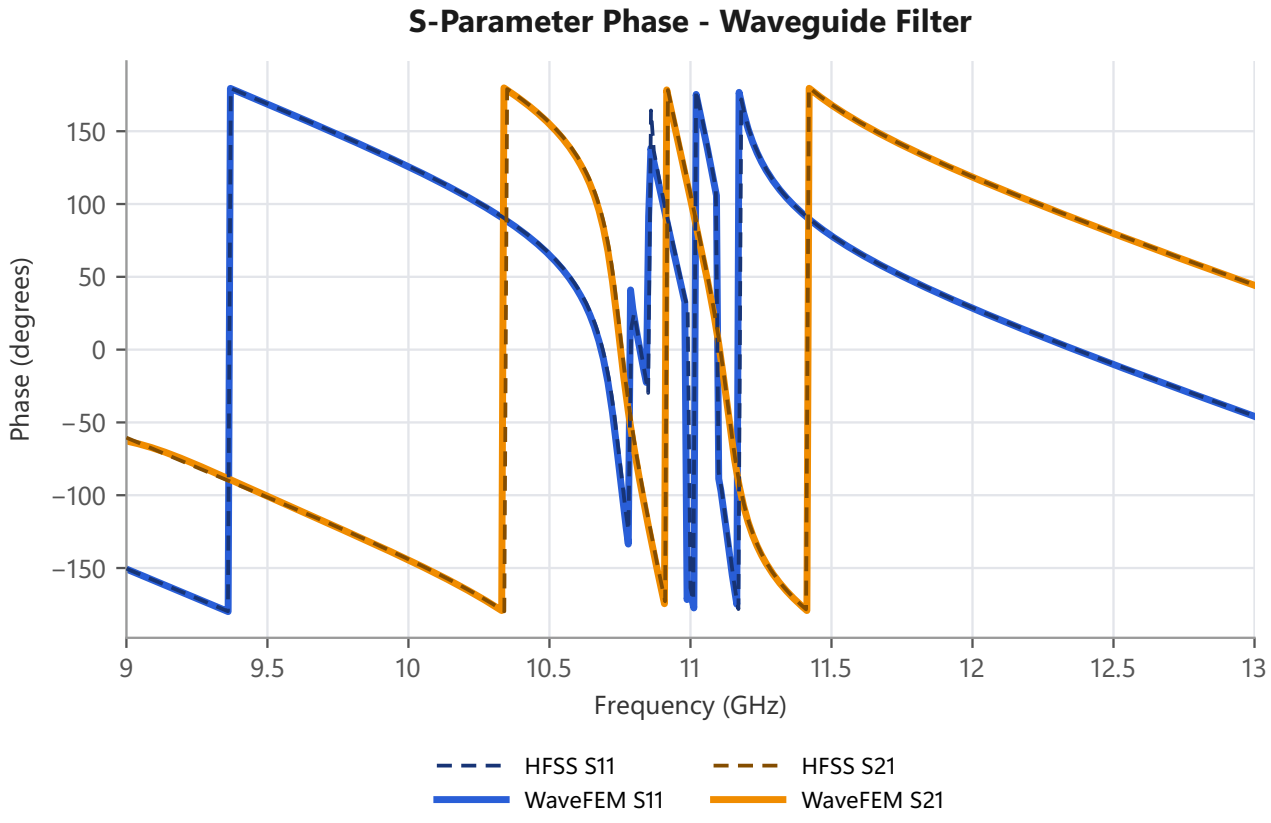


Figure 19 — S-parameter phase.

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FILTER

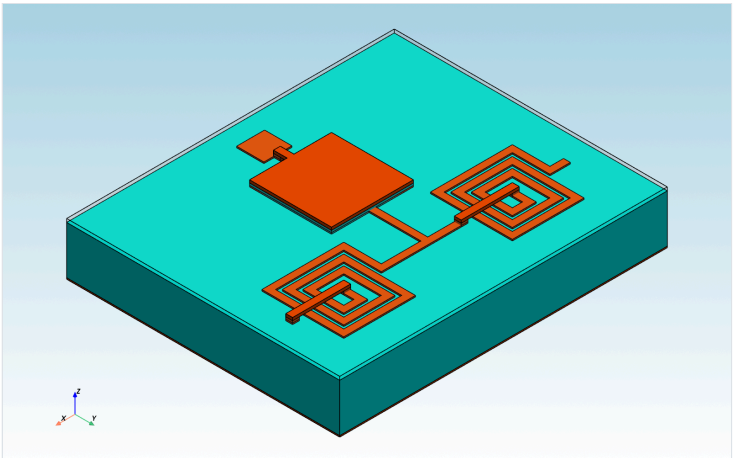
IC Filter

An MMIC-style lumped low-pass T filter: two square spiral inductors and a shunt metal-insulator-metal capacitor grounded through a via, on a 70 µm substrate.

Simulation Setup

Refinement frequency	50 GHz
Delta S	0.02
Frequency range	1 - 50 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:00:36	00:00:53
Adaptive meshing	00:00:20	00:00:23
Frequency sweep	00:00:14	00:00:27
Adaptive passes	7	8
Final elements	17,283	9,453
Matrix size	120,320	63,002

Convergence

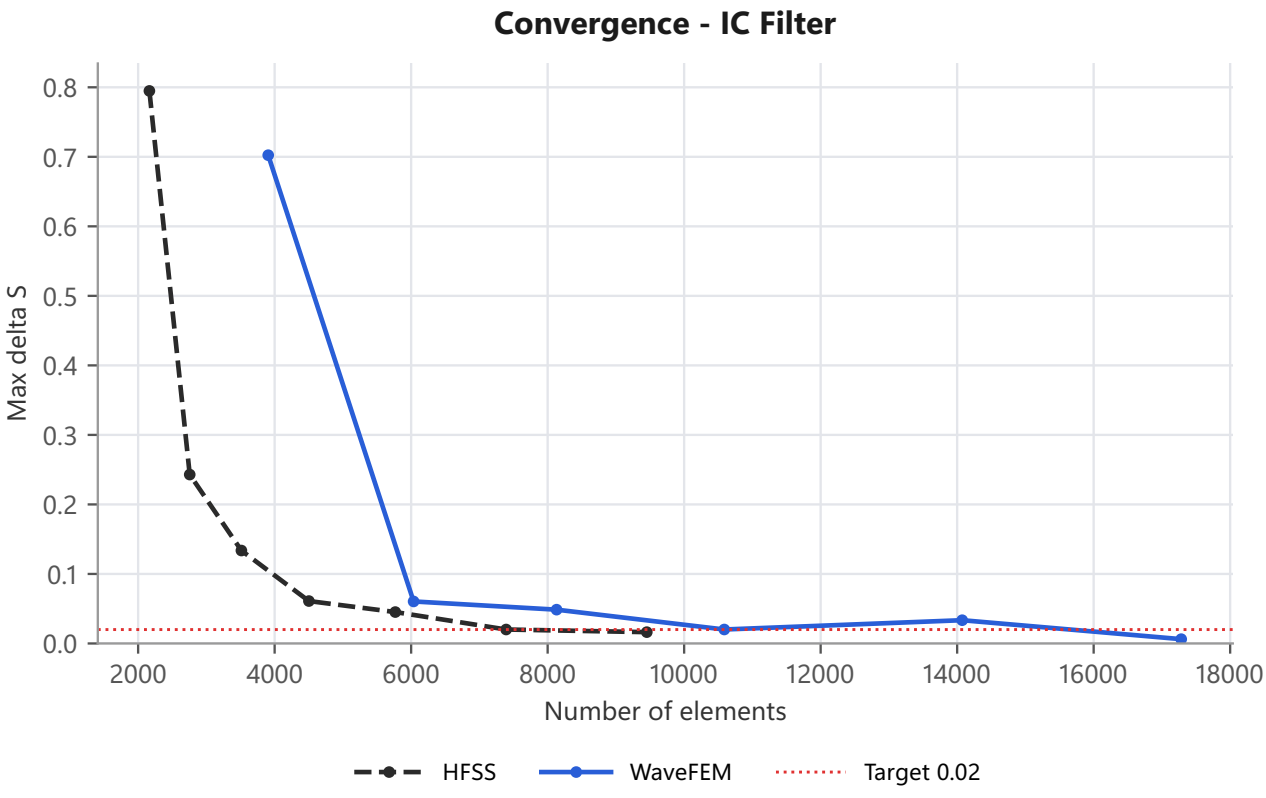


Figure 20 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

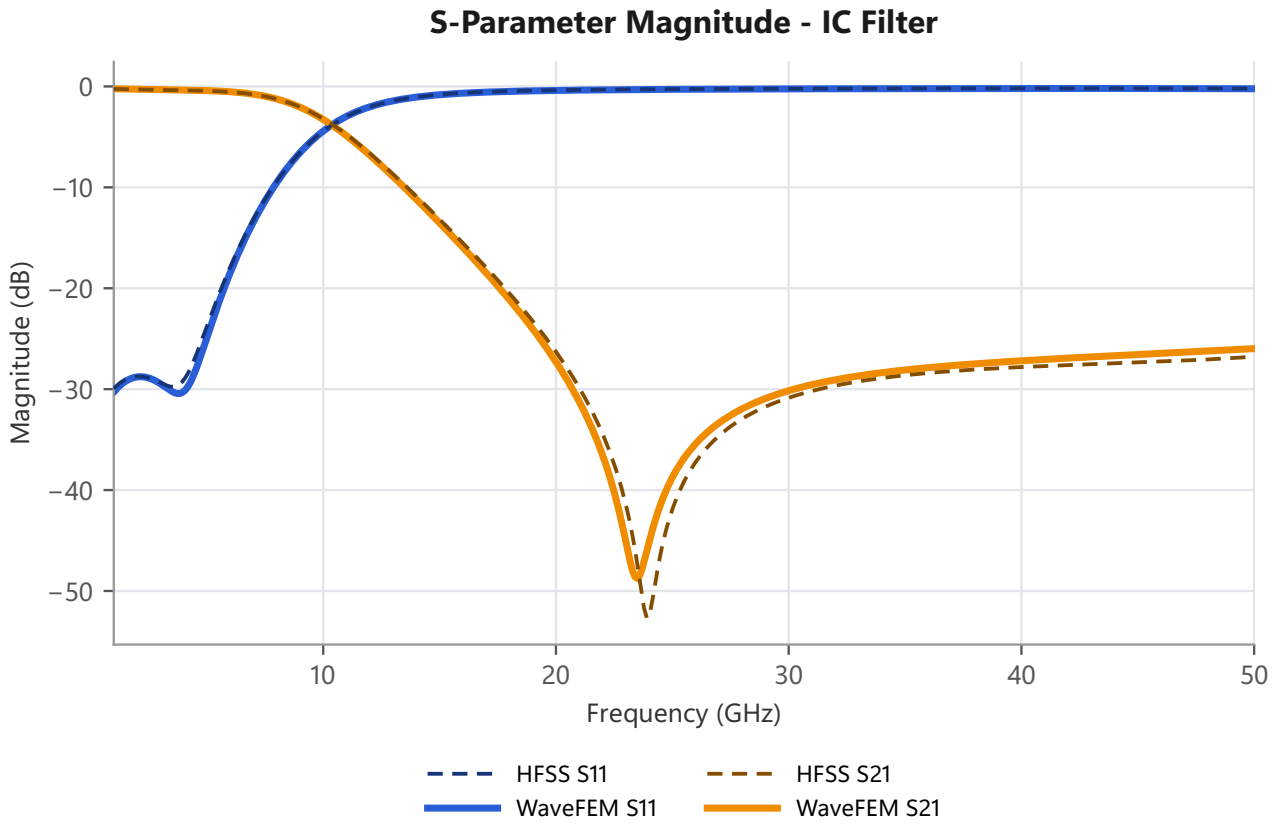


Figure 21 — S-parameter magnitude.

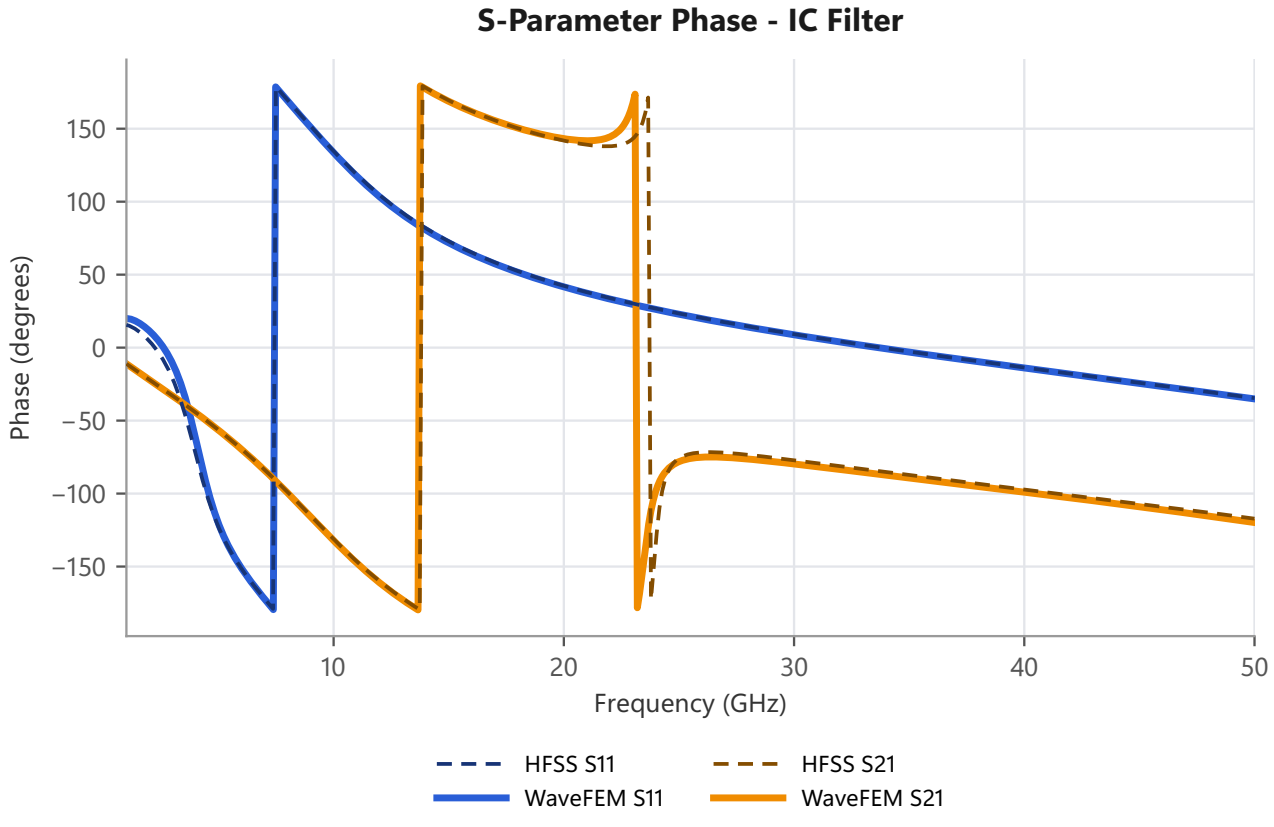


Figure 22 — S-parameter phase.

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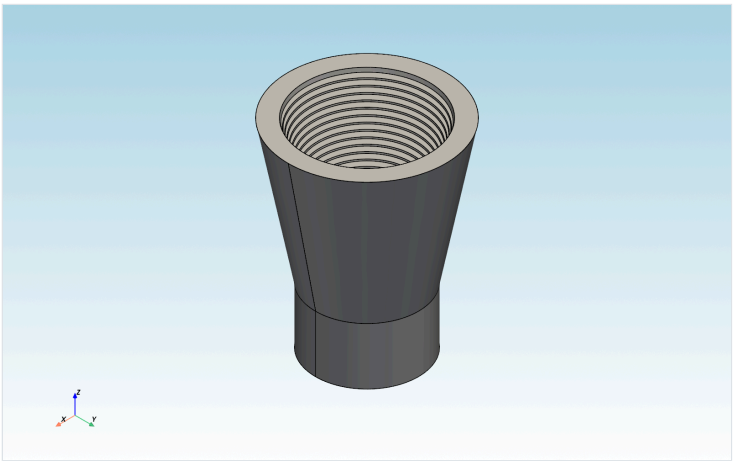
Corrugated Horn

A X-band corrugated horn antenna designed to have low side lobes

Simulation Setup

Refinement frequency	11 GHz
Delta S	0.02
Frequency range	9 - 11 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:01:11	00:01:24
Adaptive meshing	00:00:30	00:00:17
Frequency sweep	00:00:36	00:00:45
Adaptive passes	3	2
Final elements	52,861	32,687
Matrix size	307,740	196,483

Convergence

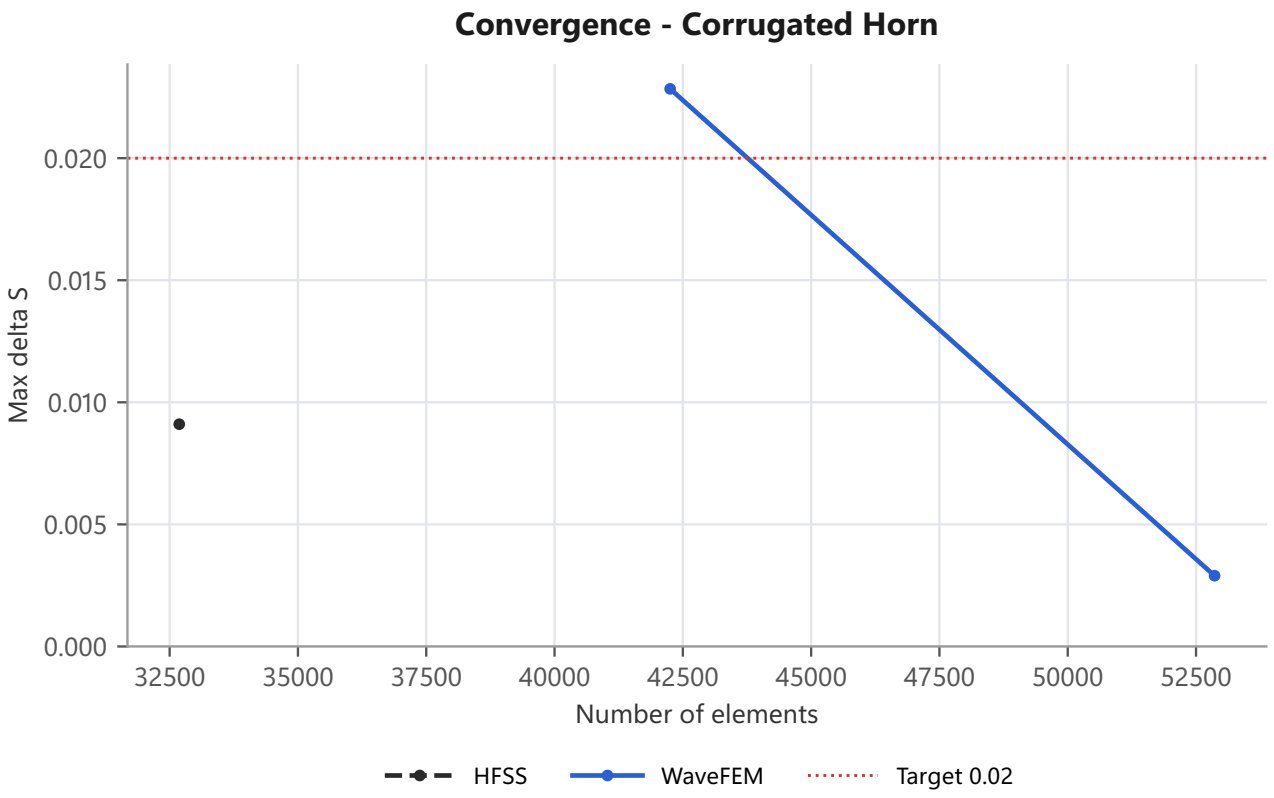


Figure 23 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

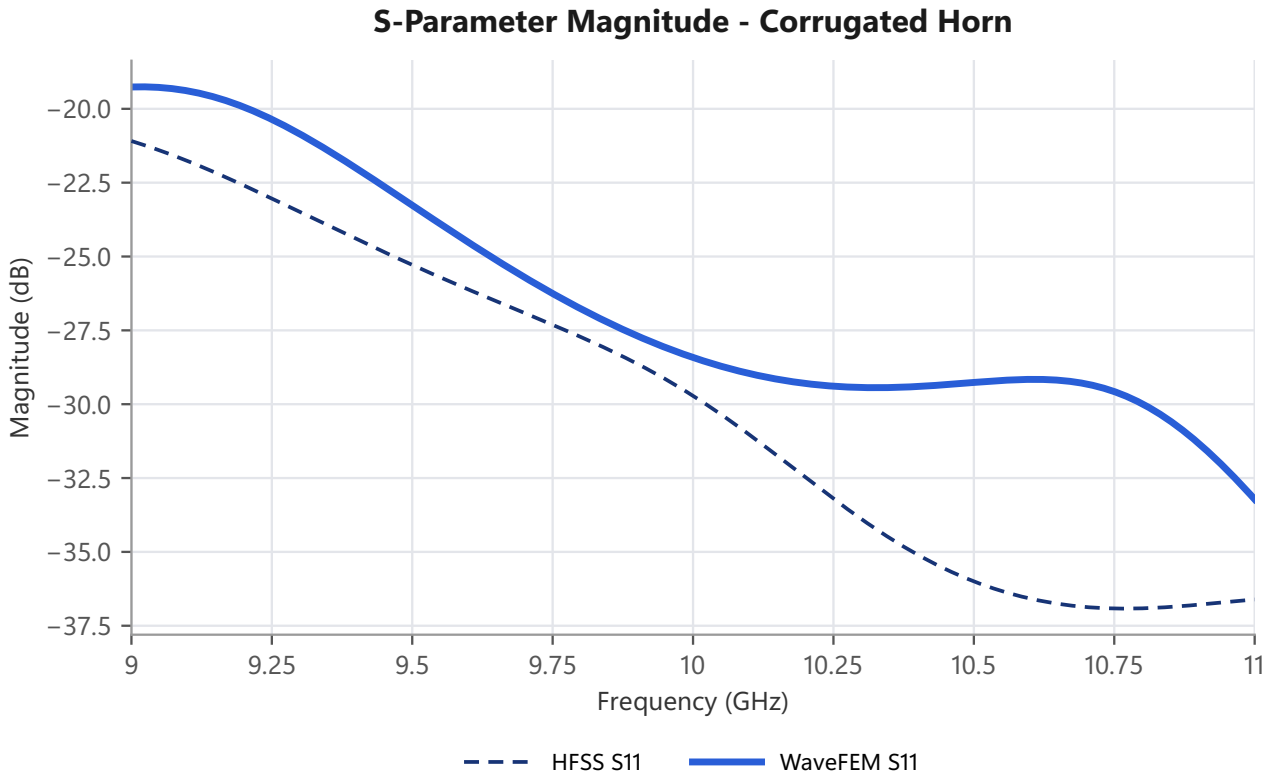


Figure 24 — S-parameter magnitude.

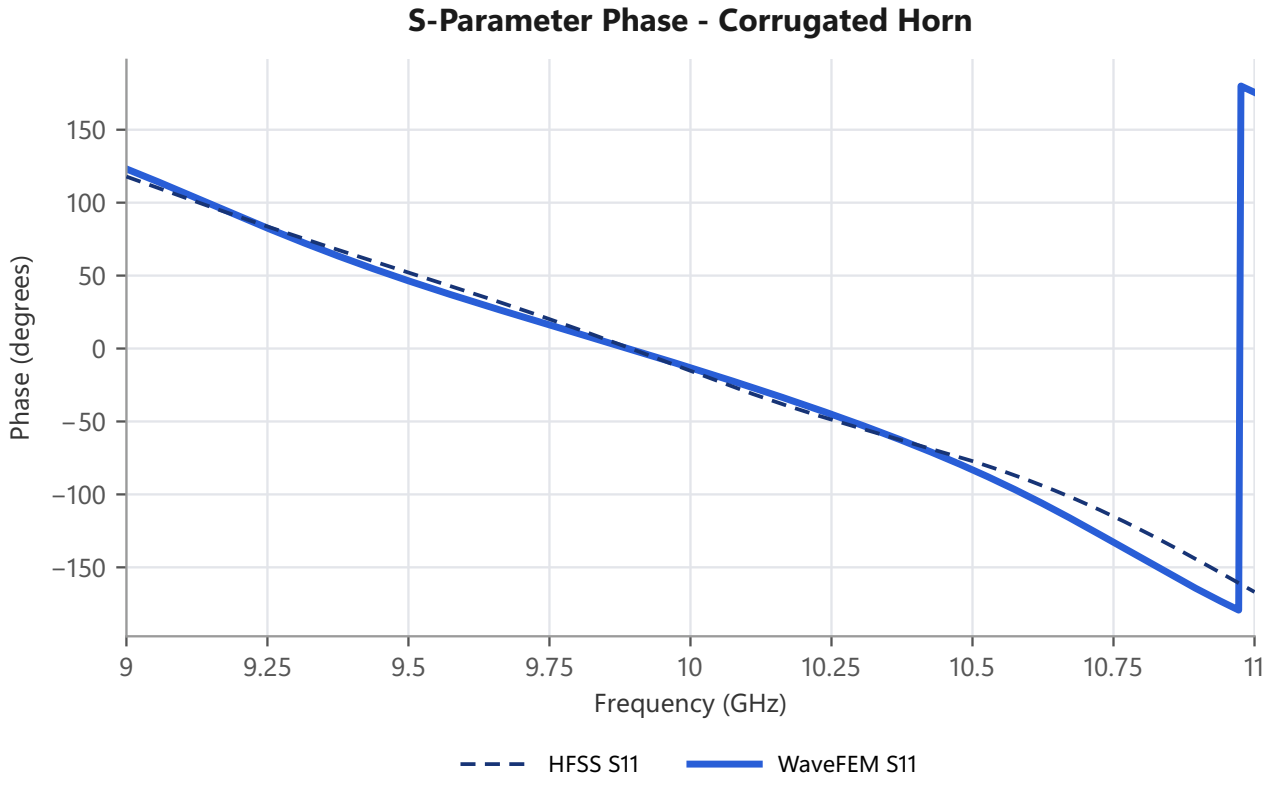


Figure 25 — S-parameter phase.

Far Field

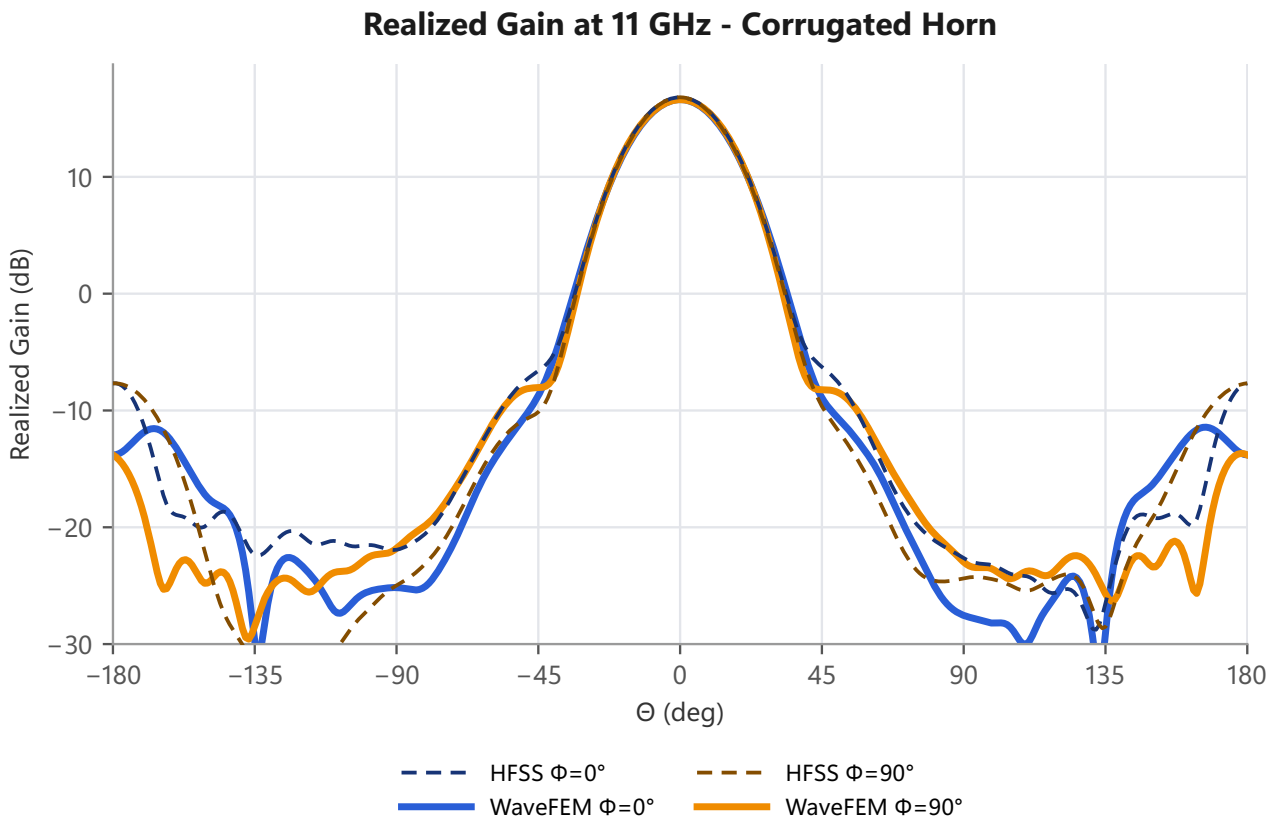


Figure 26 — Total realized gain cuts at 11 GHz.

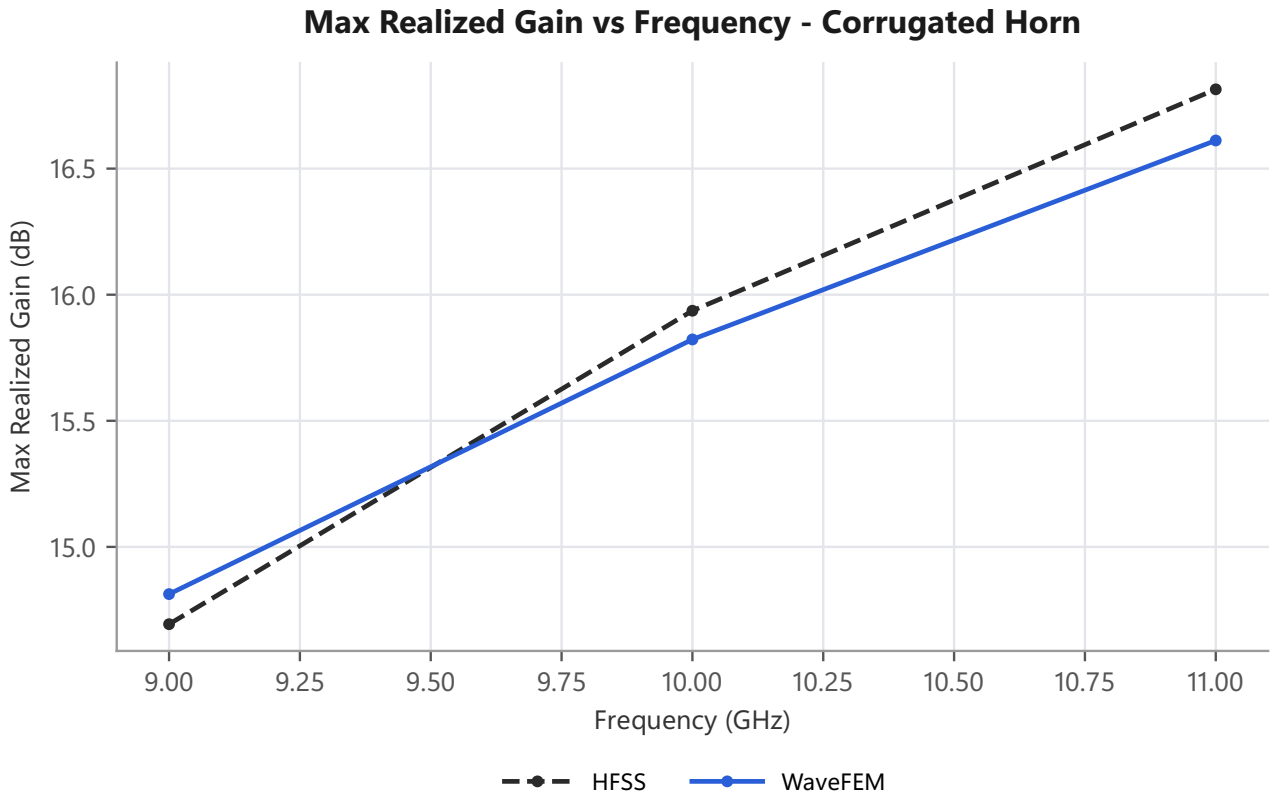


Figure 27 — Maximum realized gain against frequency.

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ANTENNA

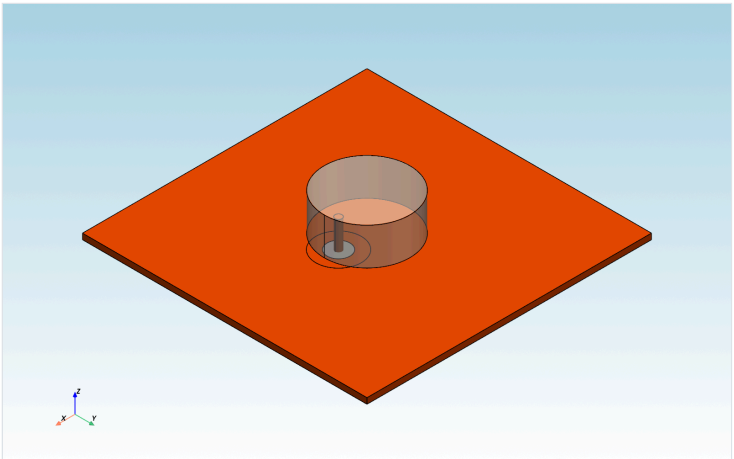
Dielectric Resonator Antenna

A 5.6 GHz probe fed dielectric resonator antenna, using ceramic with e_r=10 & tand=0.002

Simulation Setup

Refinement frequency	5.6 GHz
Delta S	0.02
Frequency range	4 - 7 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:00:04	00:00:30
Adaptive meshing	00:00:02	00:00:07
Frequency sweep	00:00:02	00:00:19
Adaptive passes	3	3
Final elements	4,810	3,191
Matrix size	28,820	18,923

Convergence

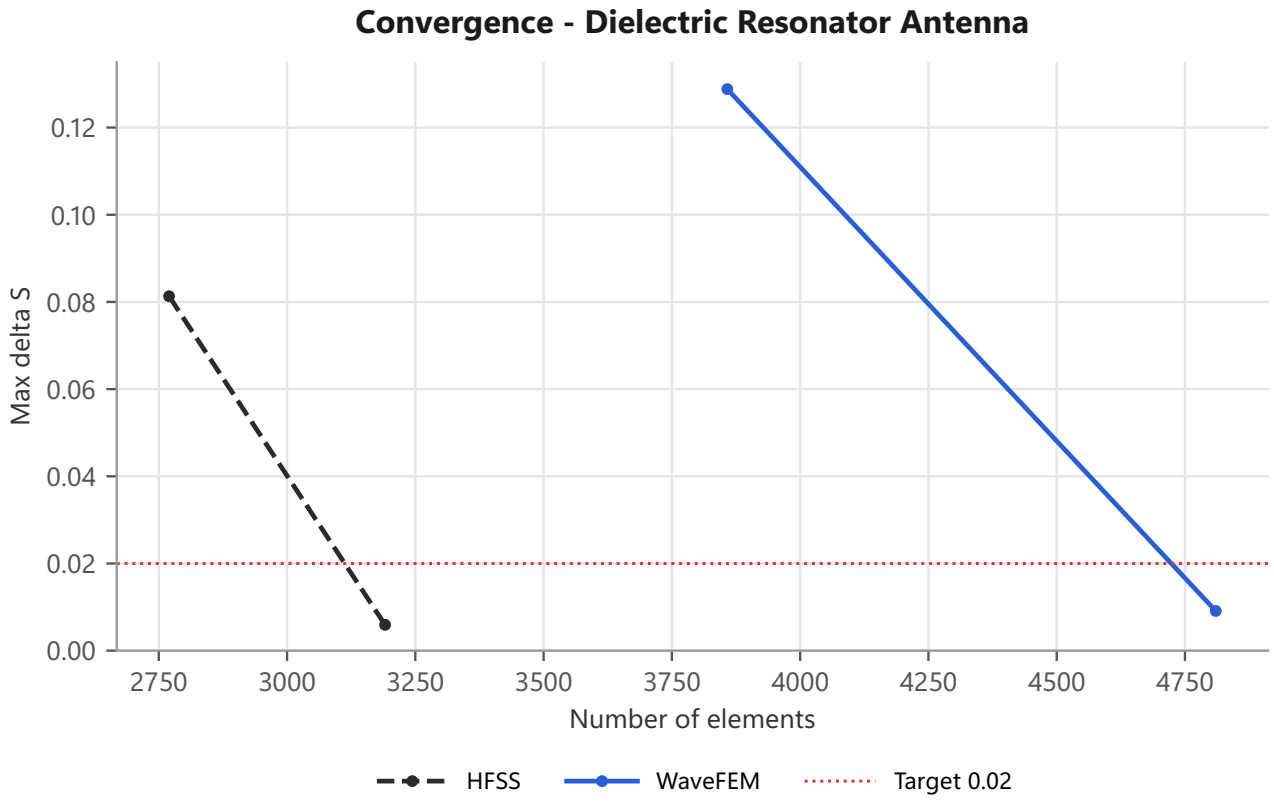


Figure 28 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

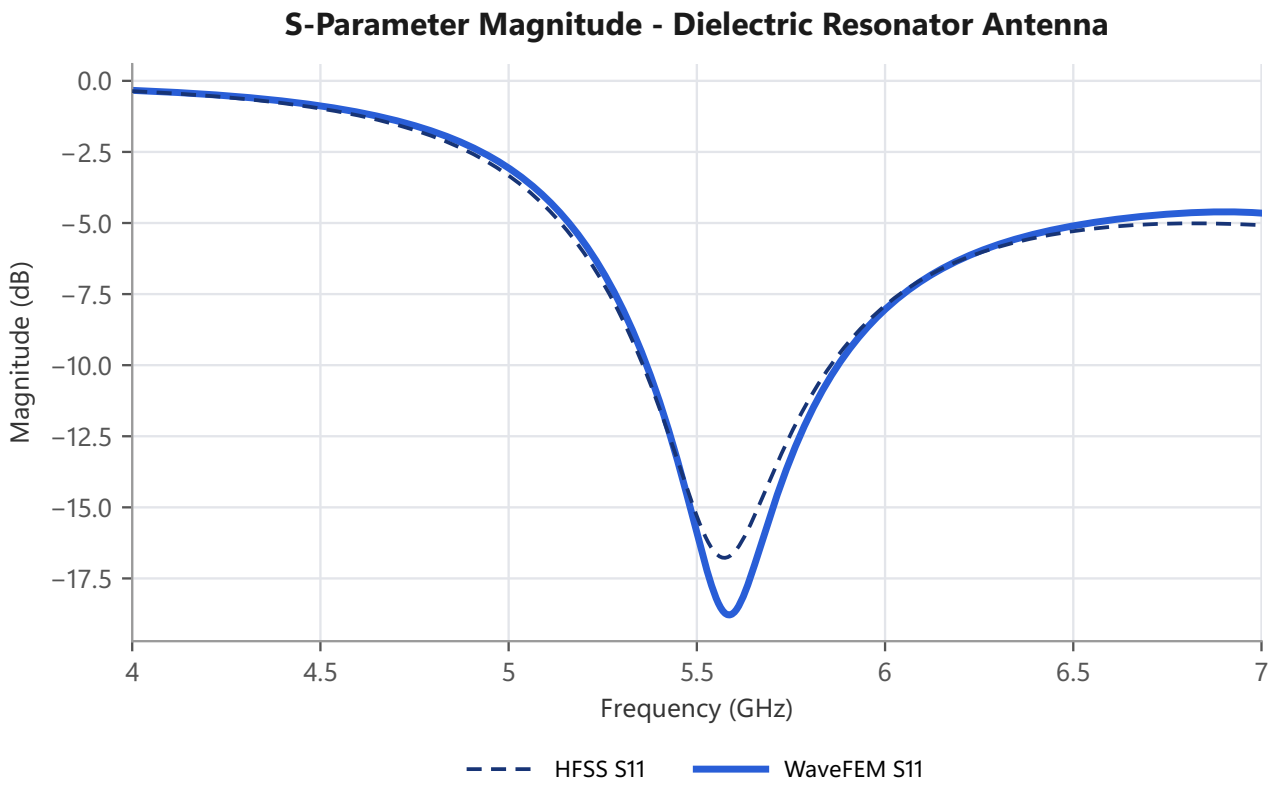


Figure 29 — S-parameter magnitude.

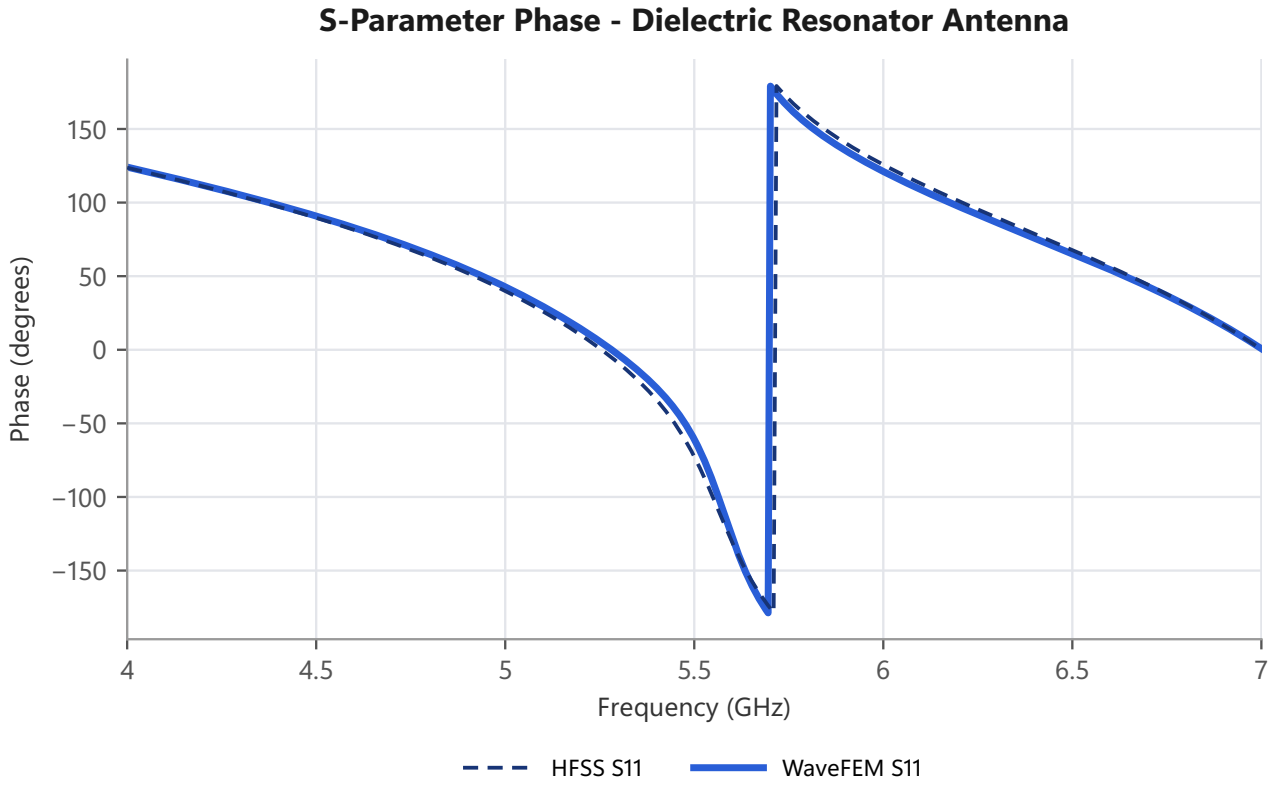


Figure 30 — S-parameter phase.

Far Field

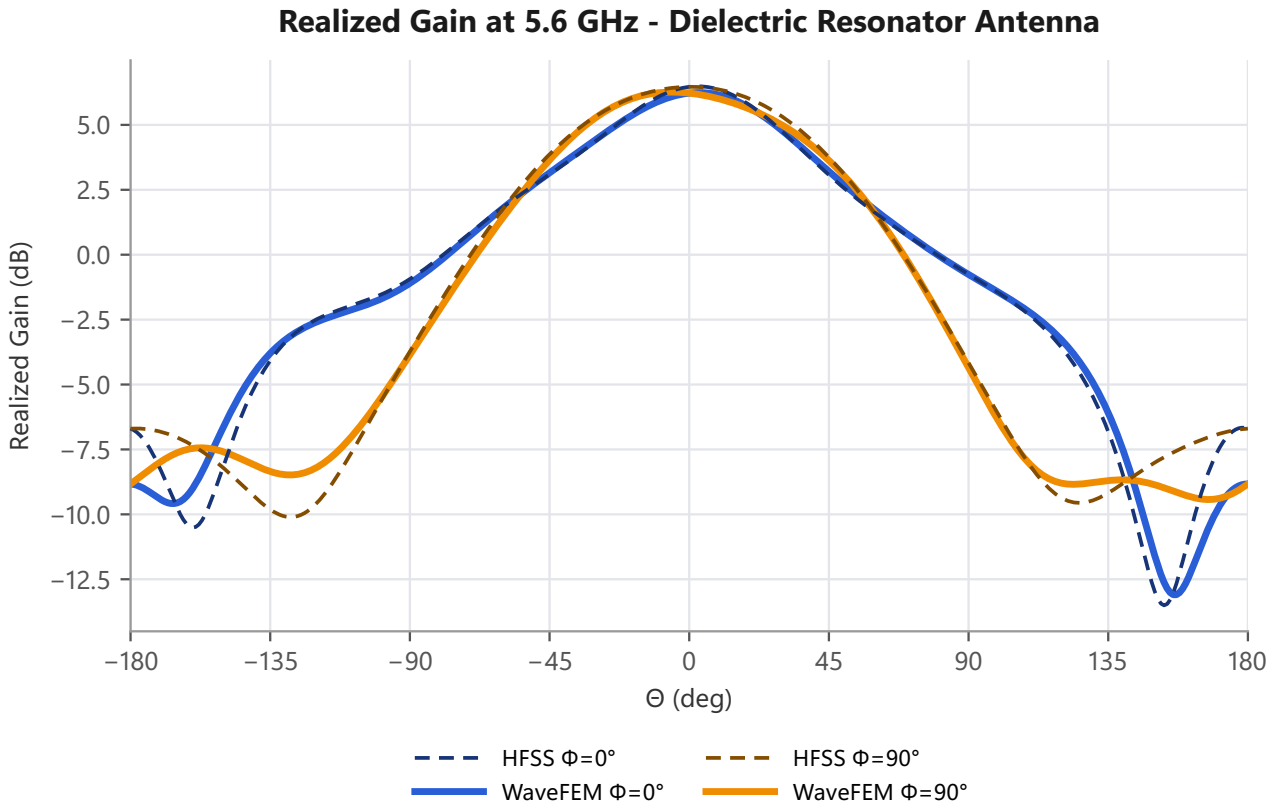


Figure 31 — Total realized gain cuts at 5.6 GHz.

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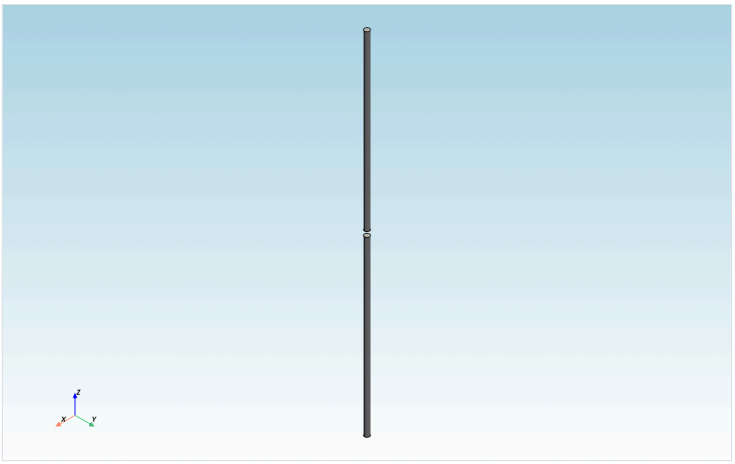
Dipole

A half-wave wire dipole along z: two PEC arms of radius 1 mm and length 69.2 mm (140.4 mm overall) across a 2 mm gap, fed by a 73 ohm lumped port and refined at 1 GHz.

Simulation Setup

Refinement frequency	1 GHz
Delta S	0.02
Frequency range	0.8 - 1.2 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:00:24	00:00:32
Adaptive meshing	00:00:13	00:00:21
Frequency sweep	00:00:09	00:00:08
Adaptive passes	4	8
Final elements	19,906	9,761
Matrix size	120,250	60,767

Convergence

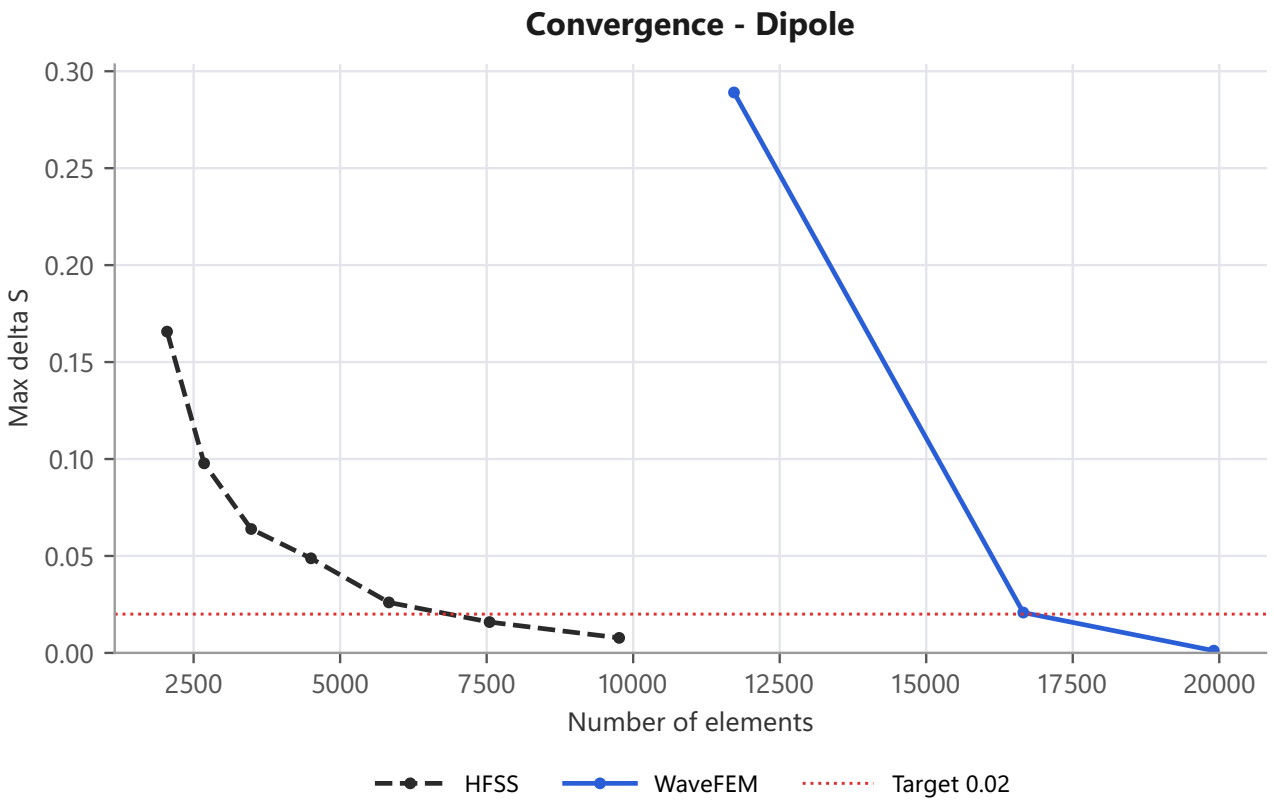


Figure 32 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

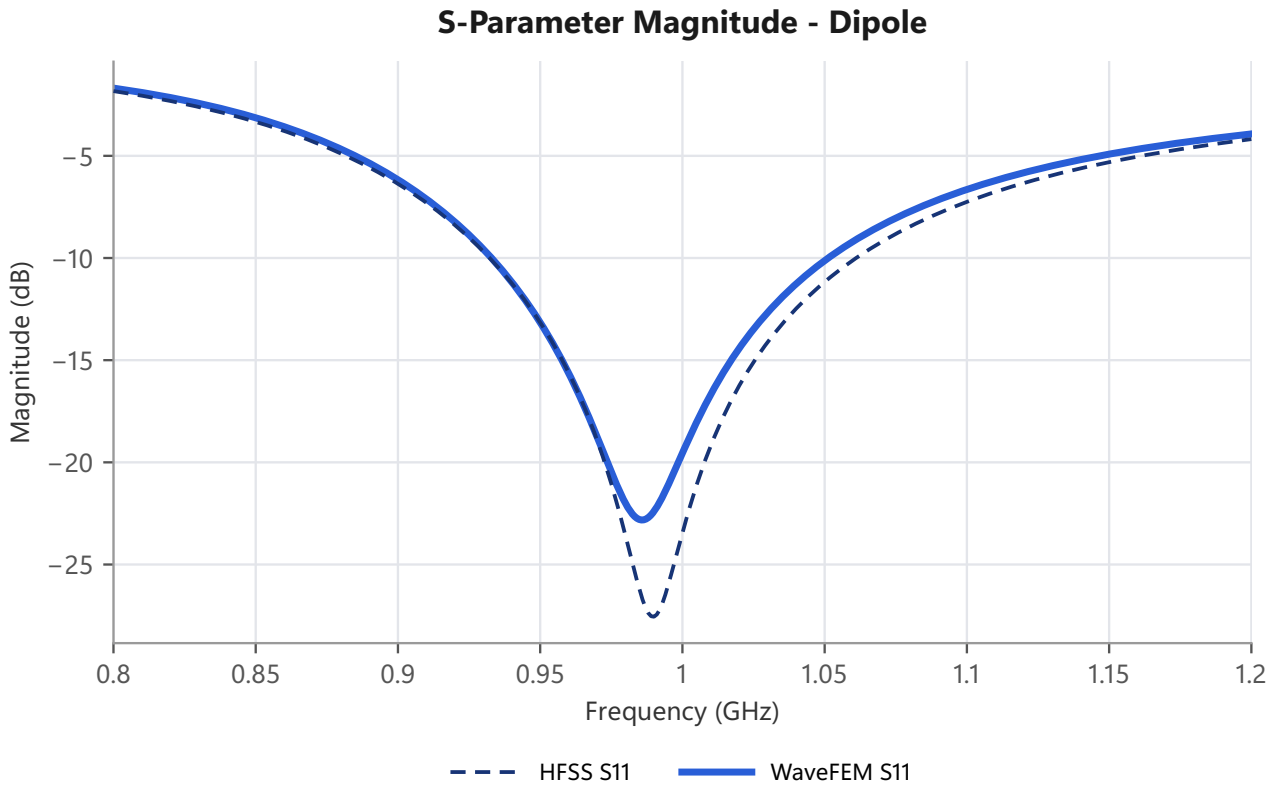


Figure 33 — S-parameter magnitude.

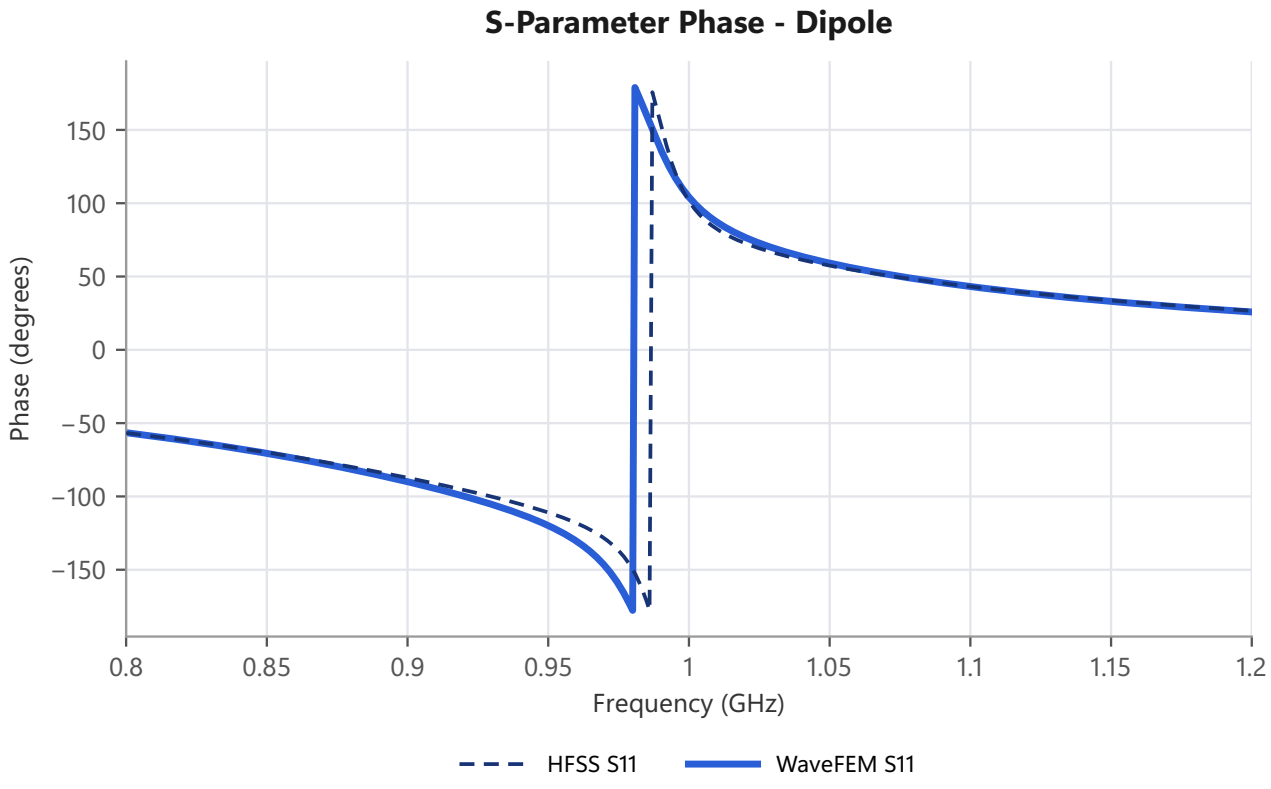


Figure 34 — S-parameter phase.

Far Field

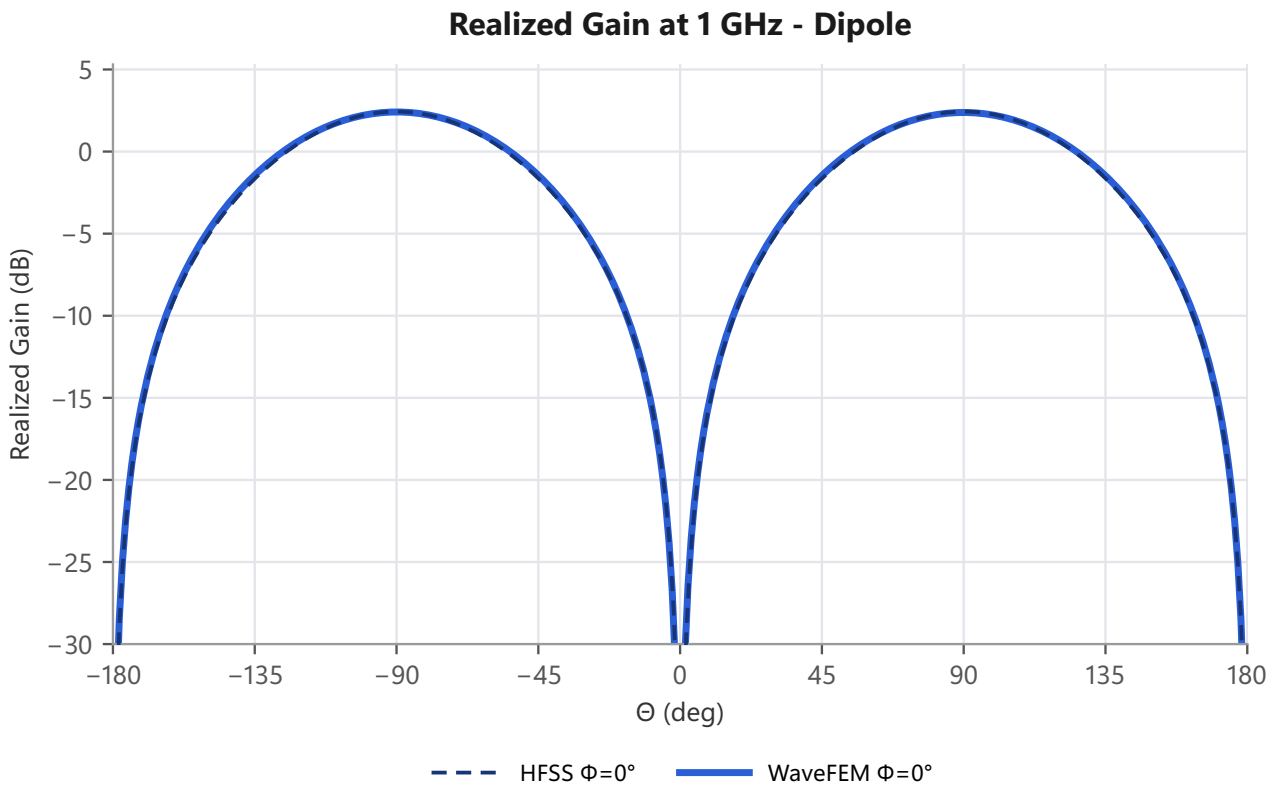


Figure 35 — Total realized gain cuts at 1 GHz.

ANTENNA

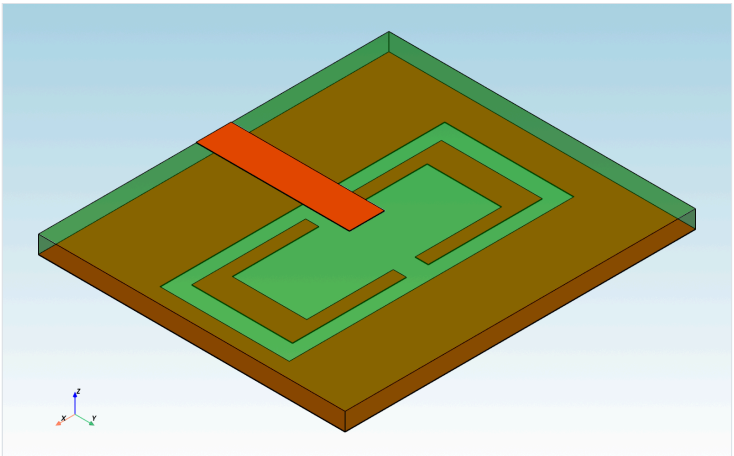
Dual Band Slot Antenna

A microstrip-fed dual-band WLAN slot antenna: a rectangular slot in the ground plane with two L-shaped strips inside it, on a 32 x 28 mm, 1.6 mm FR4 board

Simulation Setup

Refinement frequency	5.5 GHz
Delta S	0.02
Frequency range	2 - 6.5 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:00:10	00:00:57
Adaptive meshing	00:00:03	00:00:22
Frequency sweep	00:00:06	00:00:33
Adaptive passes	4	8
Final elements	6,908	8,598
Matrix size	40,954	53,169

Convergence

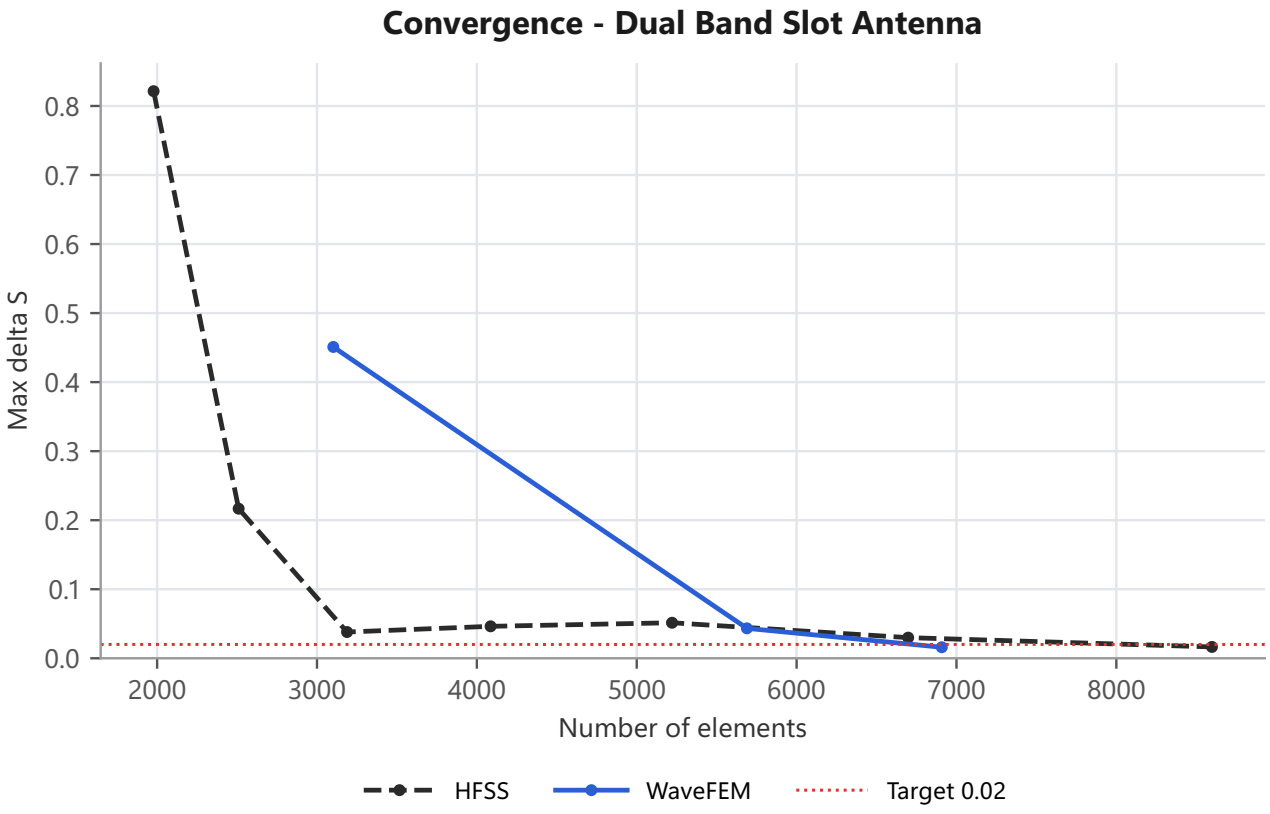


Figure 36 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

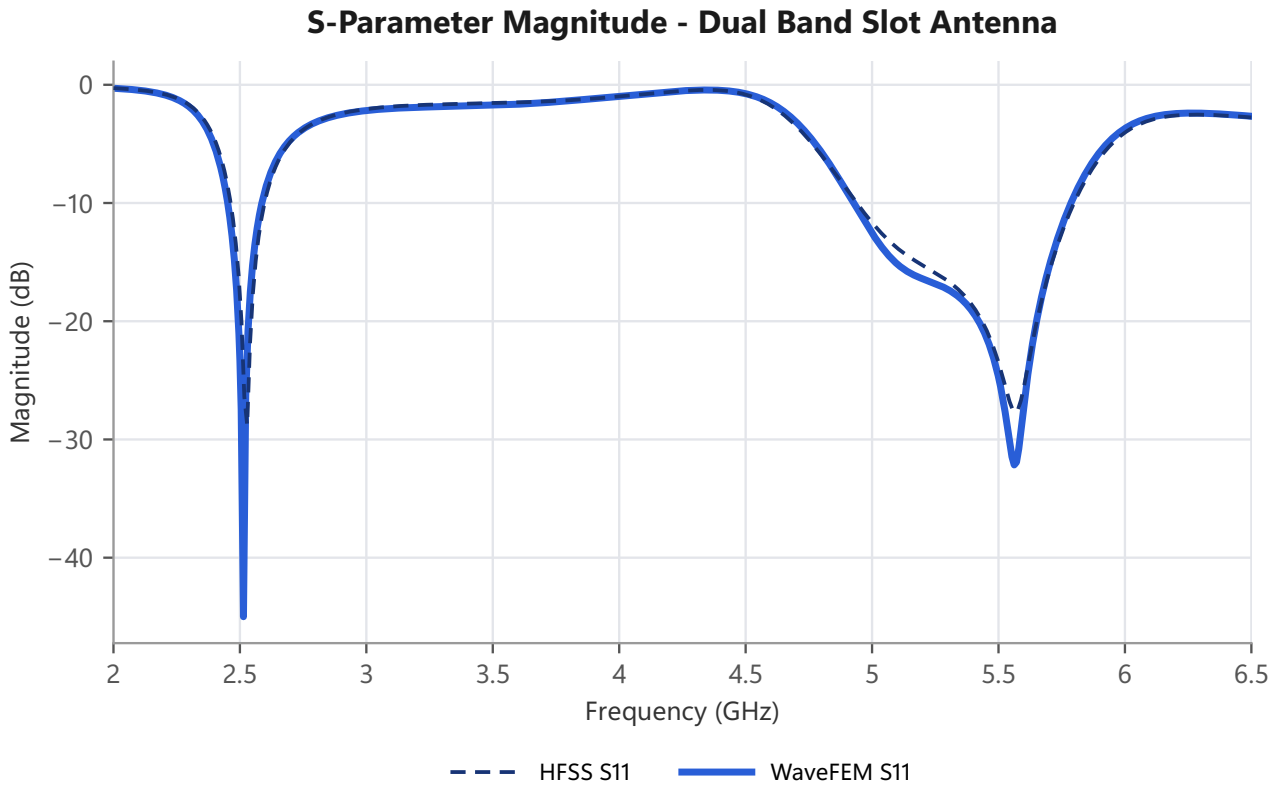


Figure 37 — S-parameter magnitude.

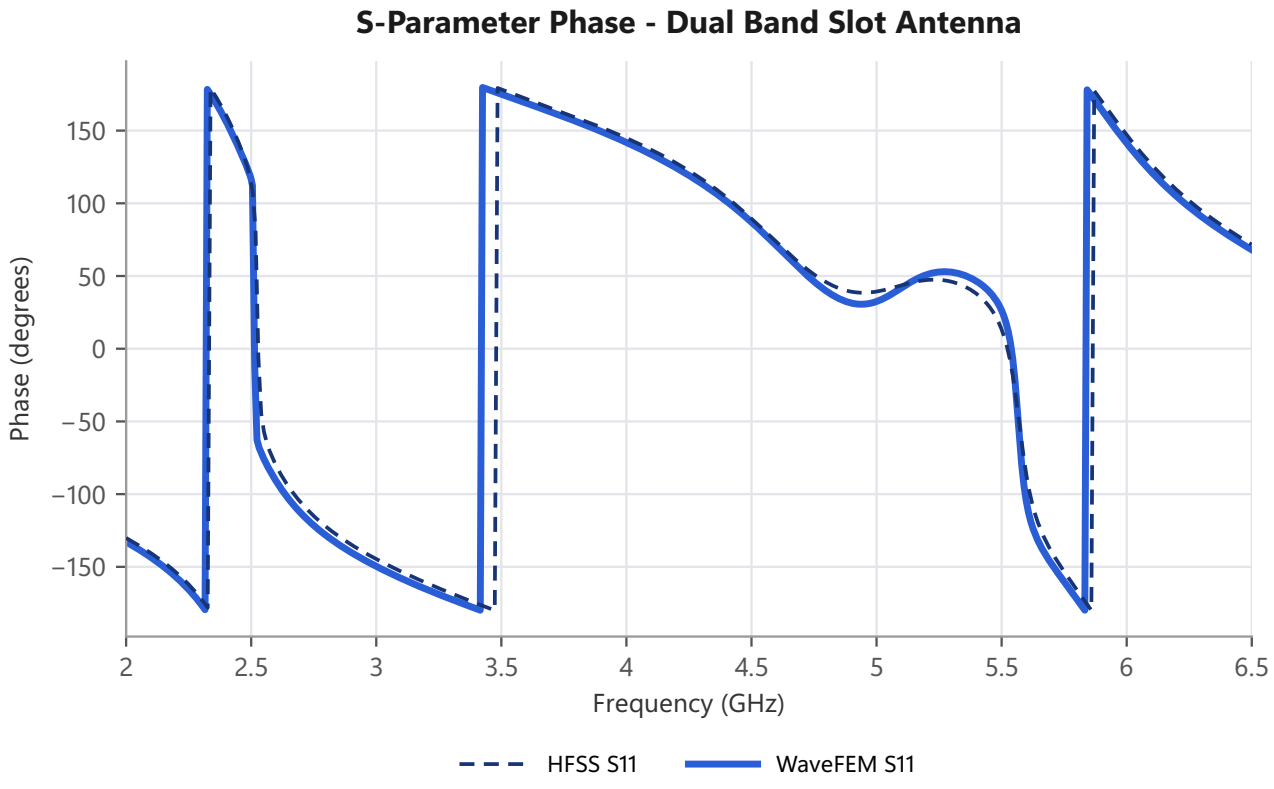


Figure 38 — S-parameter phase.

Far Field

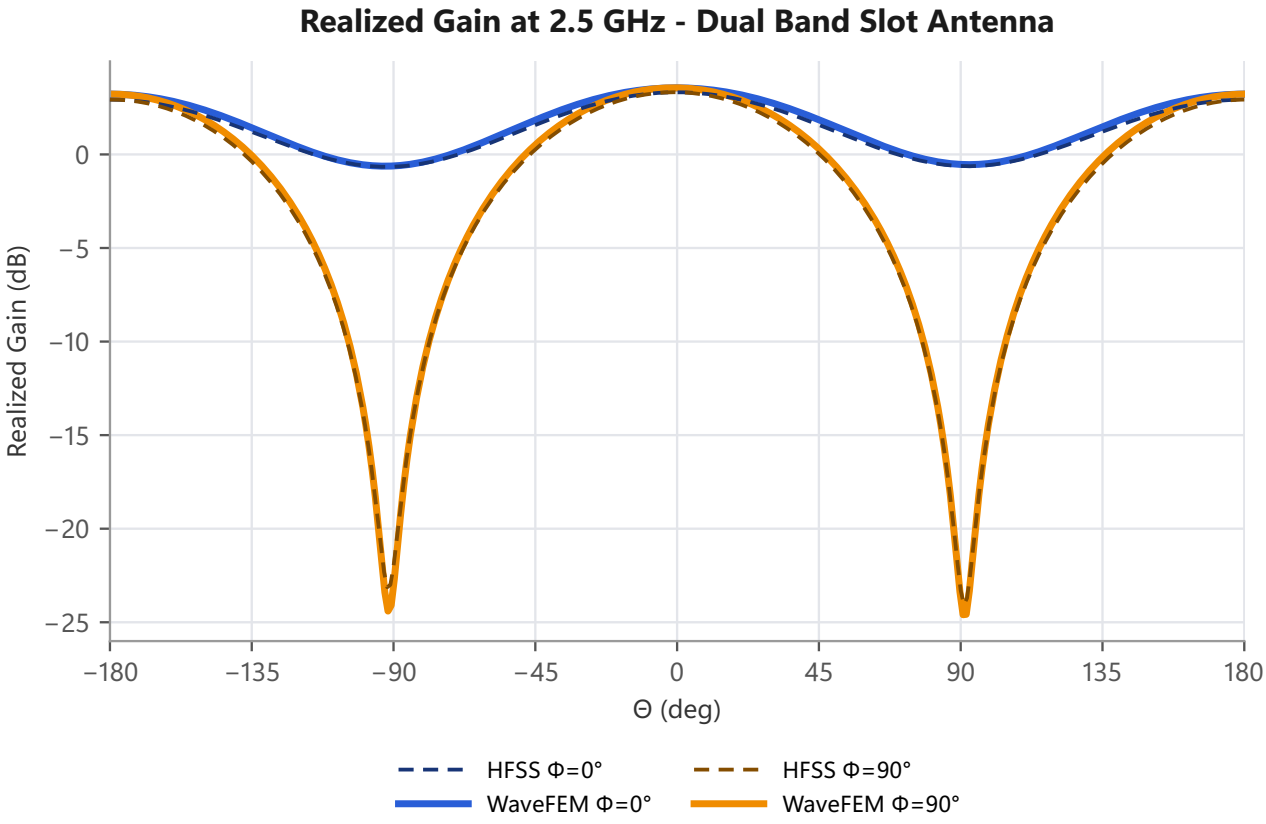


Figure 39 — Total realized gain cuts at 2.5 GHz.

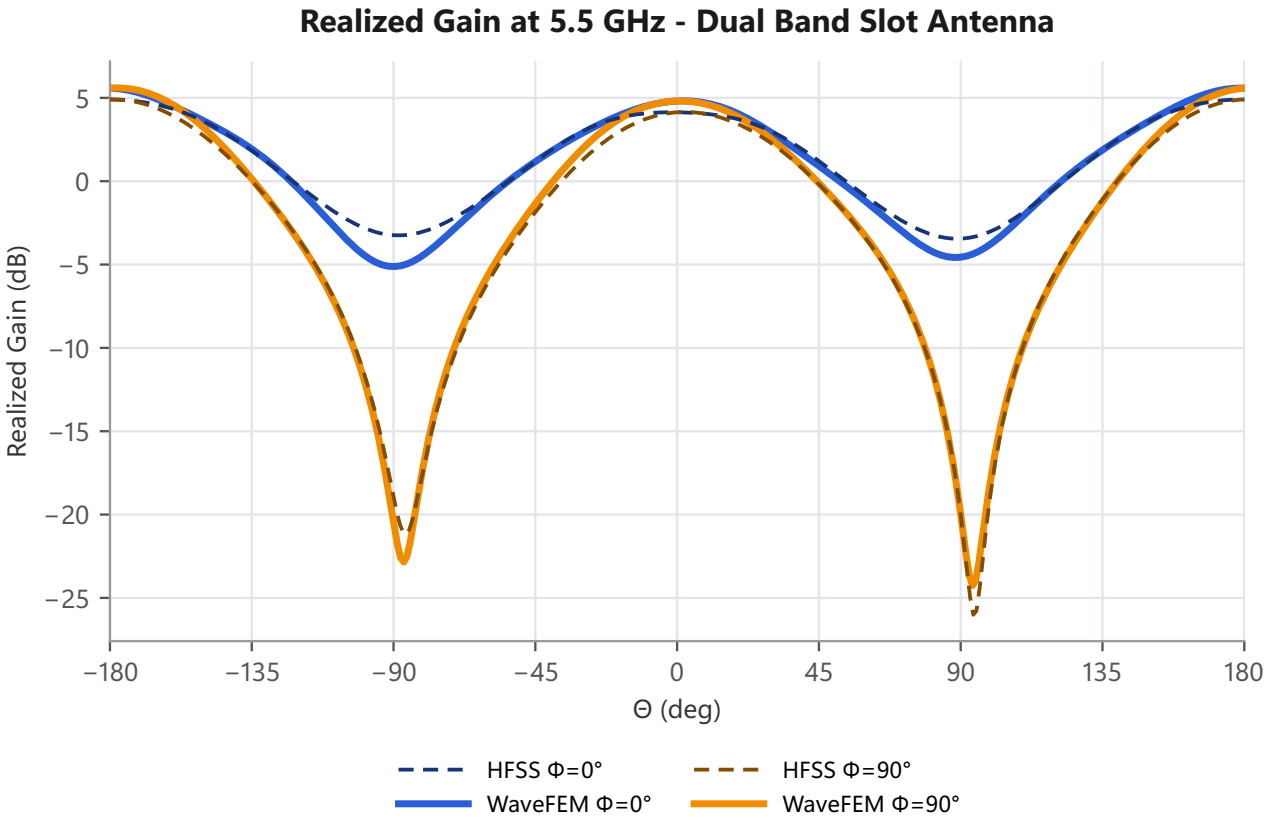


Figure 40 — Total realized gain cuts at 5.5 GHz.

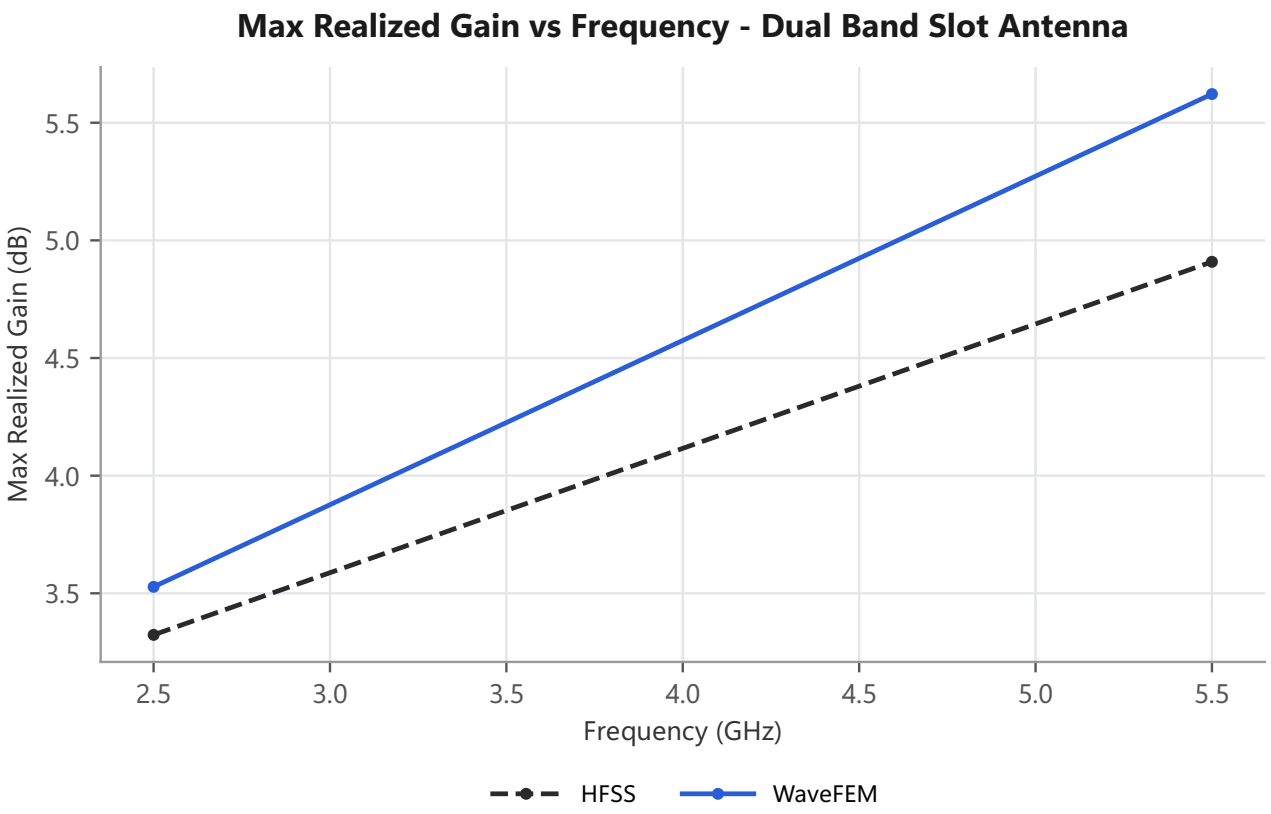


Figure 41 — Maximum realized gain against frequency.

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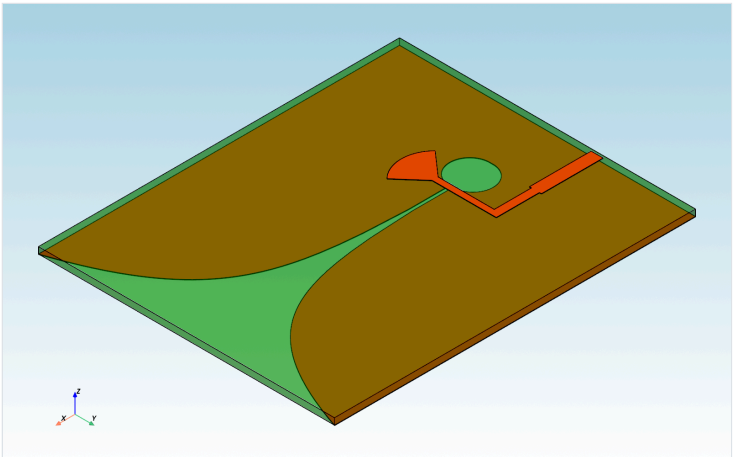
Planar Vivaldi

A wideband planar Vivaldi antenna, using radiation boundaries to terminate into free space.

Simulation Setup

Refinement frequency	11 GHz
Delta S	0.02
Frequency range	2 - 11 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:02:47	00:01:17
Adaptive meshing	00:00:46	00:00:15
Frequency sweep	00:01:55	00:00:58
Adaptive passes	7	4
Final elements	54,475	15,177
Matrix size	323,782	95,399

Convergence

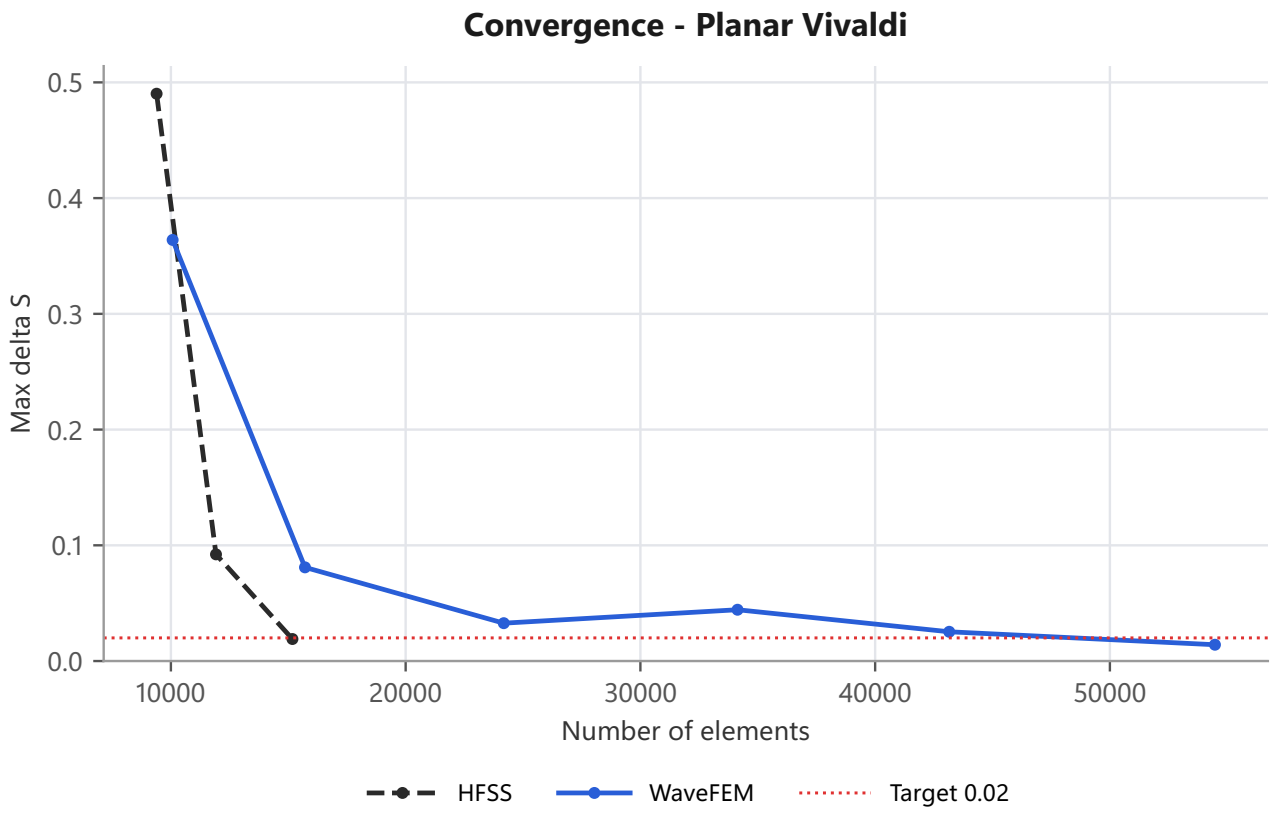


Figure 42 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

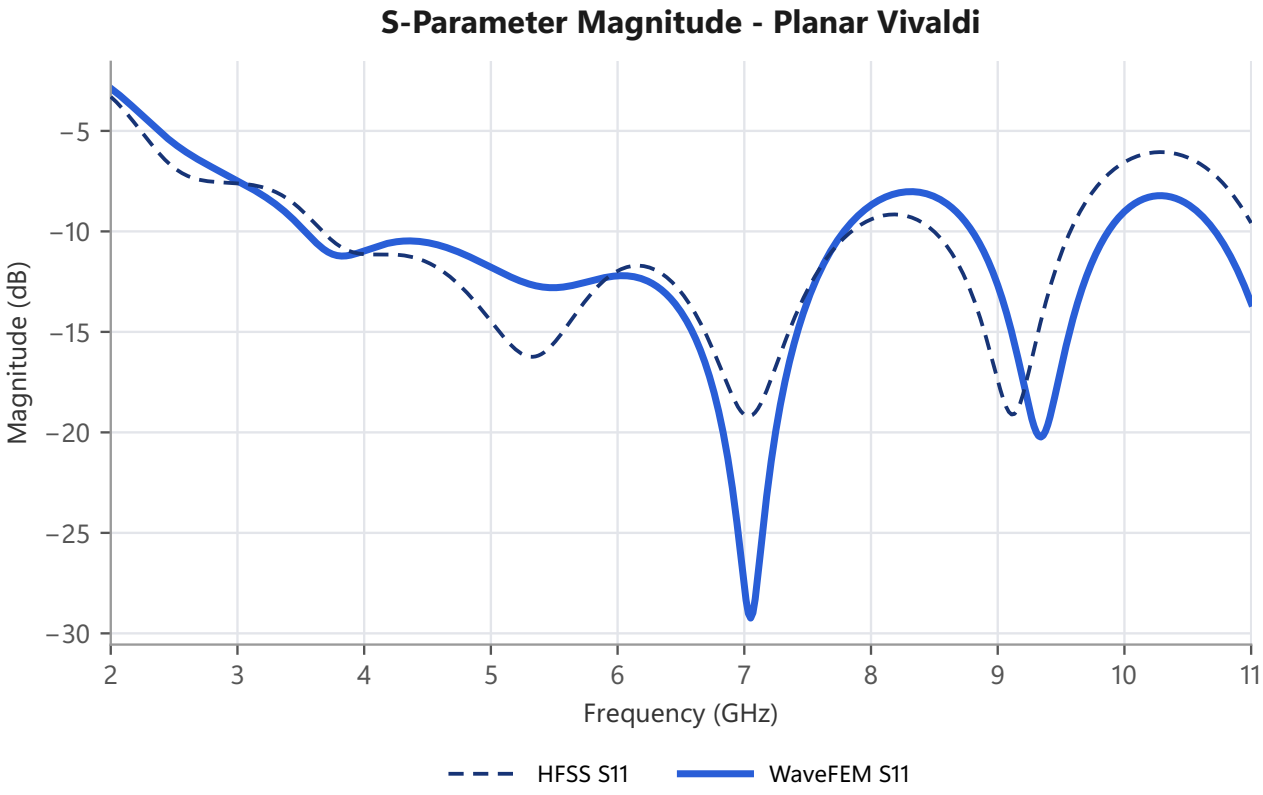


Figure 43 — S-parameter magnitude.

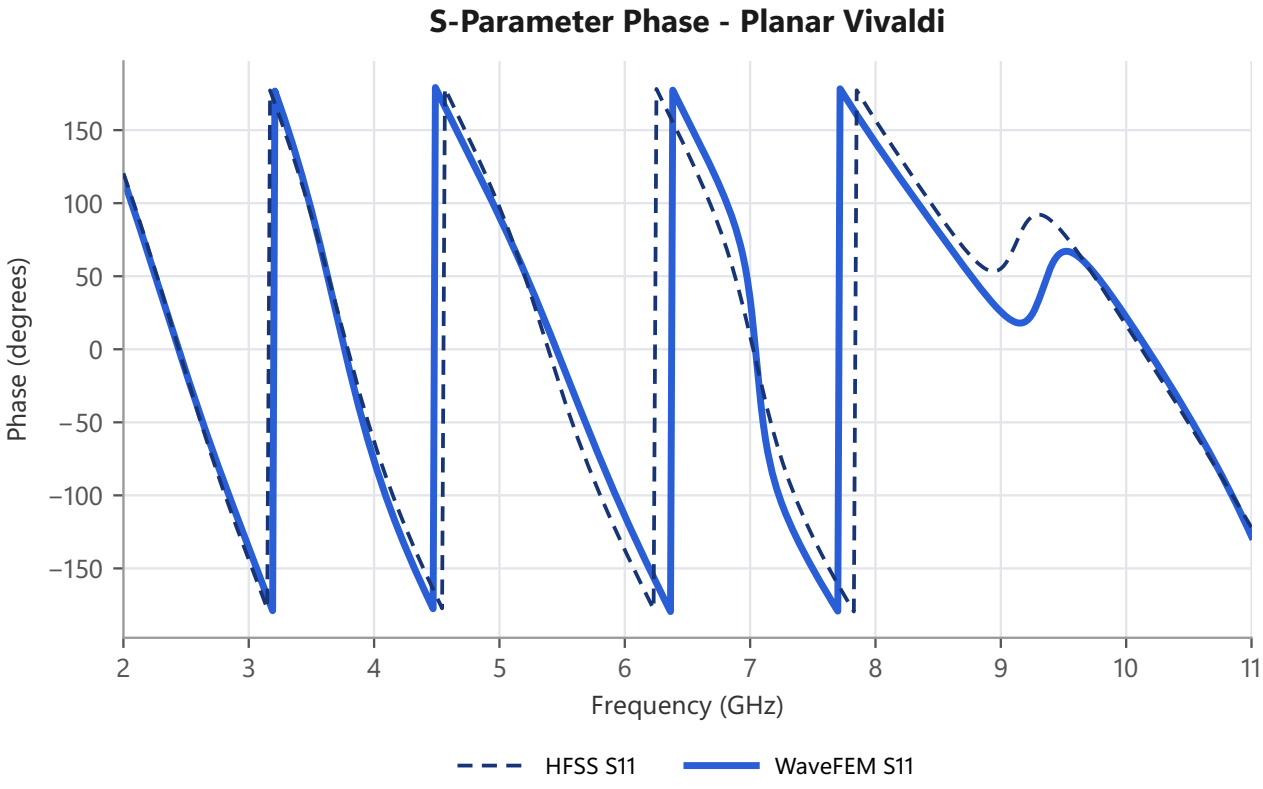


Figure 44 — S-parameter phase.

Far Field

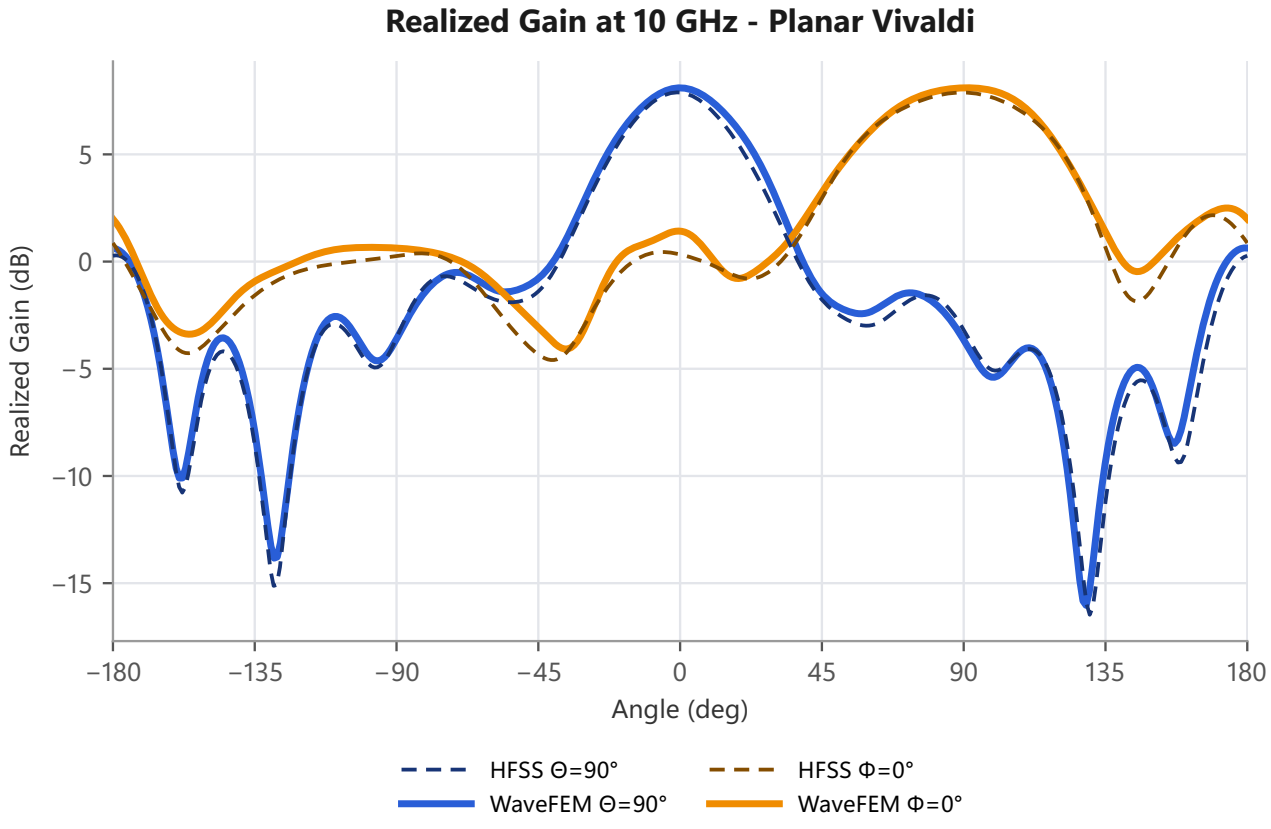


Figure 45 — Total realized gain cuts at 10 GHz.

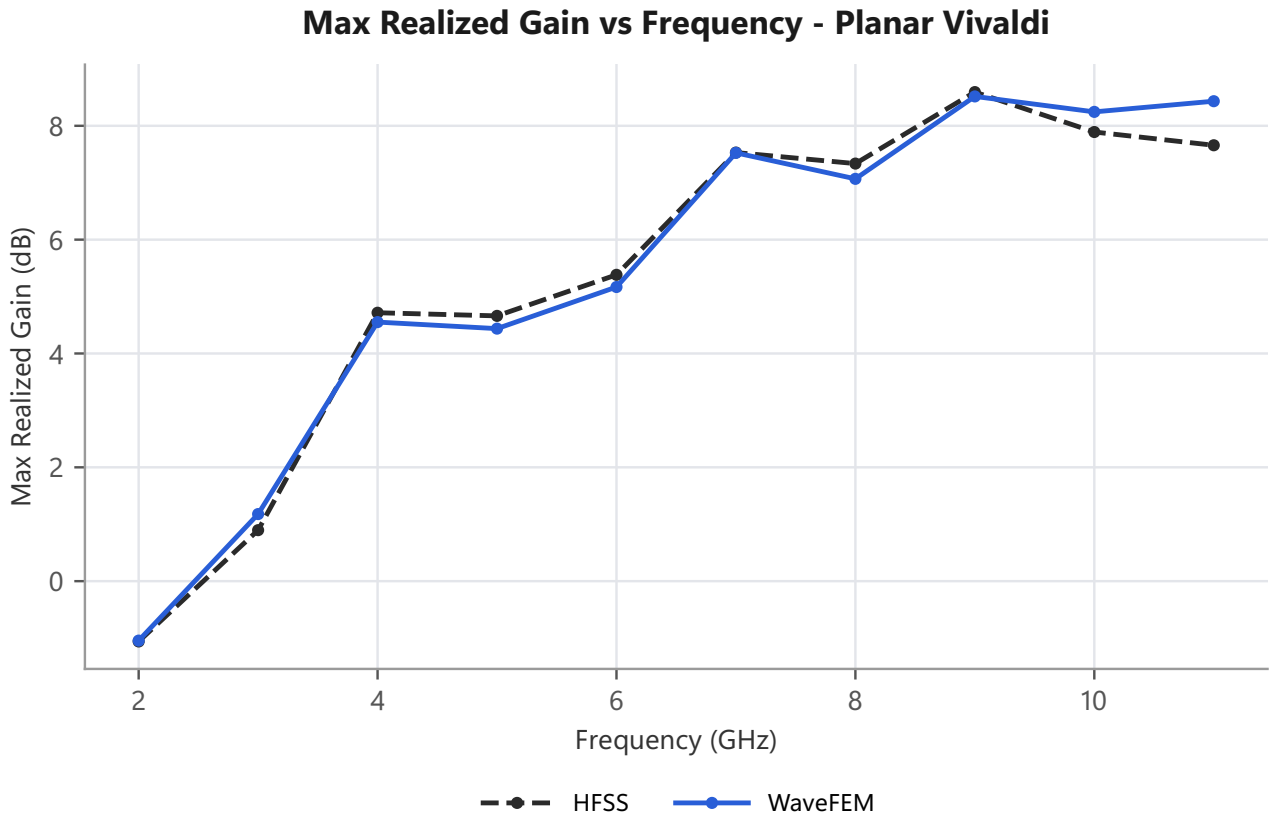


Figure 46 — Maximum realized gain against frequency.

Last updated 2026-11-02

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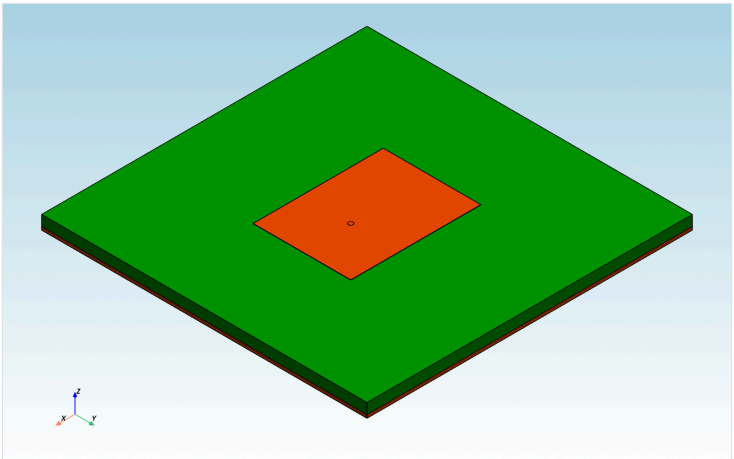
Probe Fed Patch

A coaxial probe fed rectangular patch antenna resonant at 2.4 GHz.

Simulation Setup

Refinement frequency	2.4 GHz
Delta S	0.02
Frequency range	2 - 3 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:00:29	00:01:14
Adaptive meshing	00:00:19	00:00:46
Frequency sweep	00:00:07	00:00:22
Adaptive passes	10	13
Final elements	26,213	19,159
Matrix size	151,810	115,581

Convergence

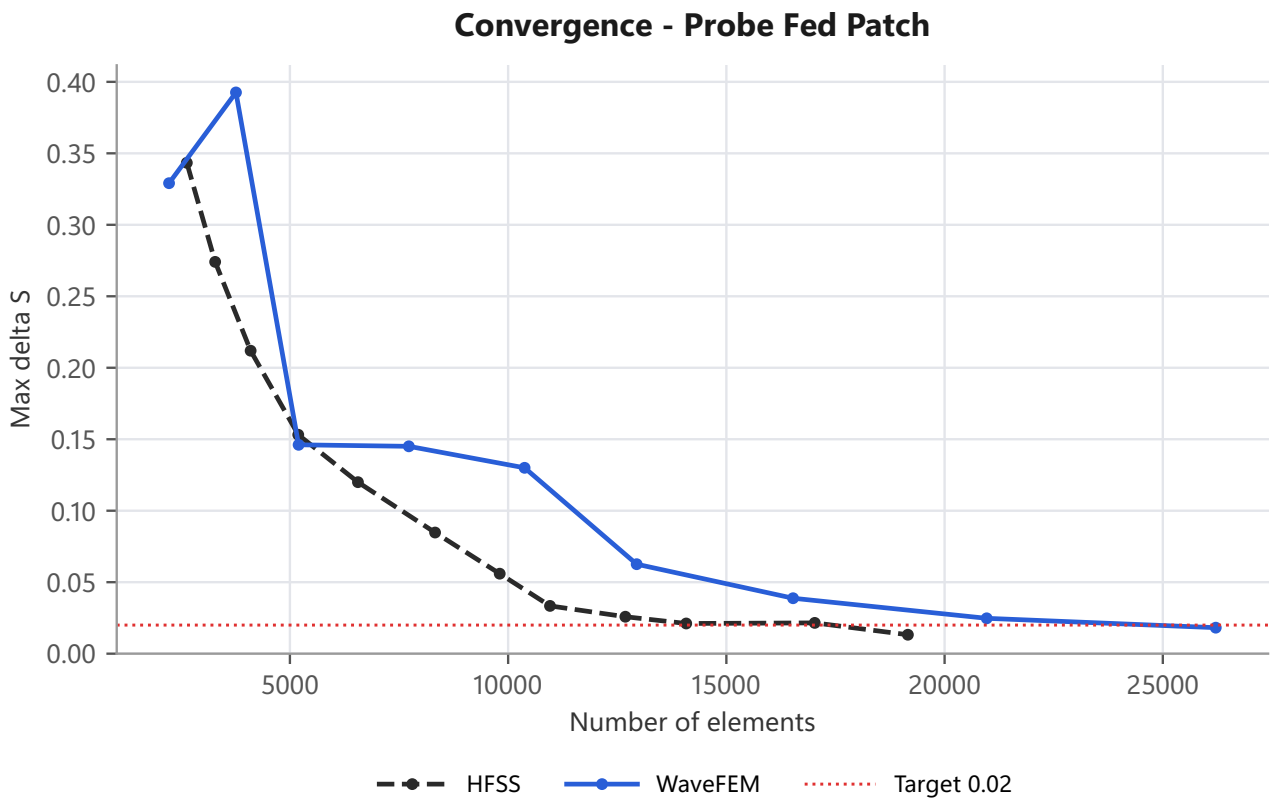


Figure 47 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

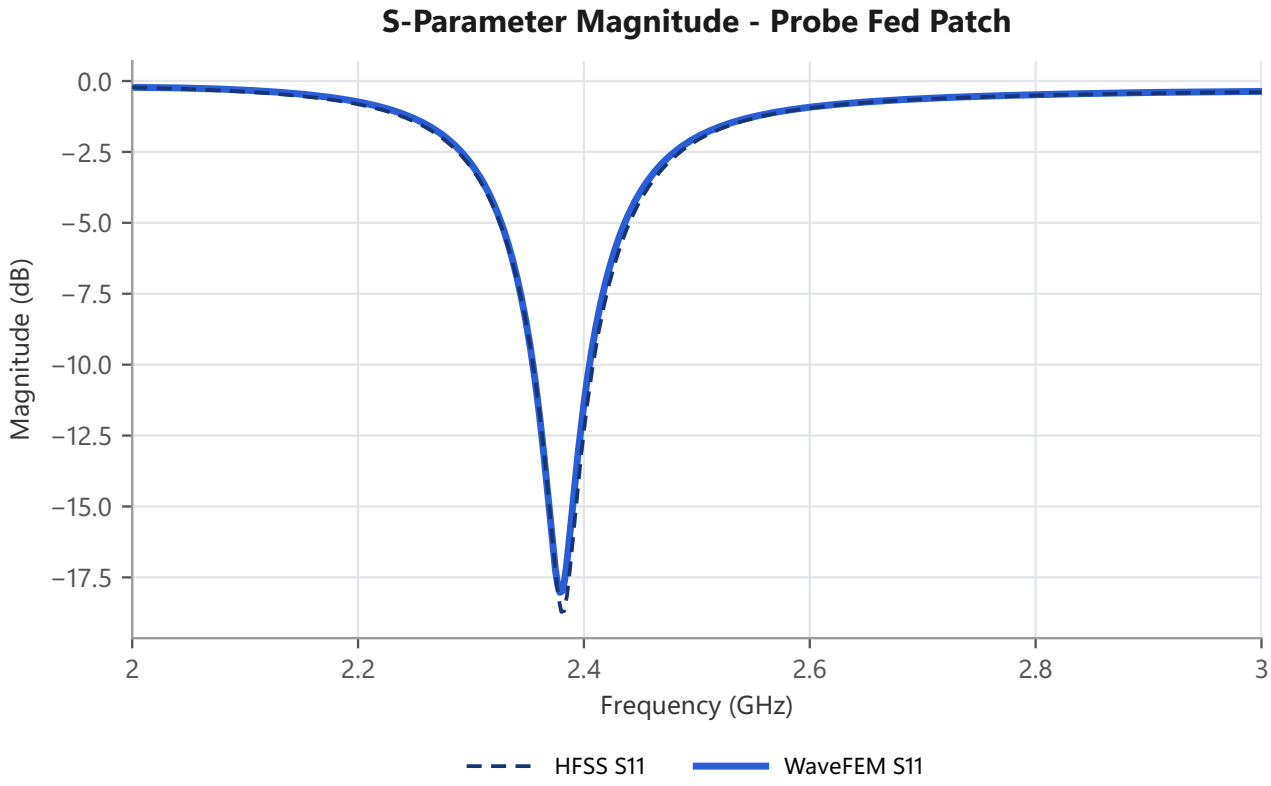


Figure 48 — S-parameter magnitude.

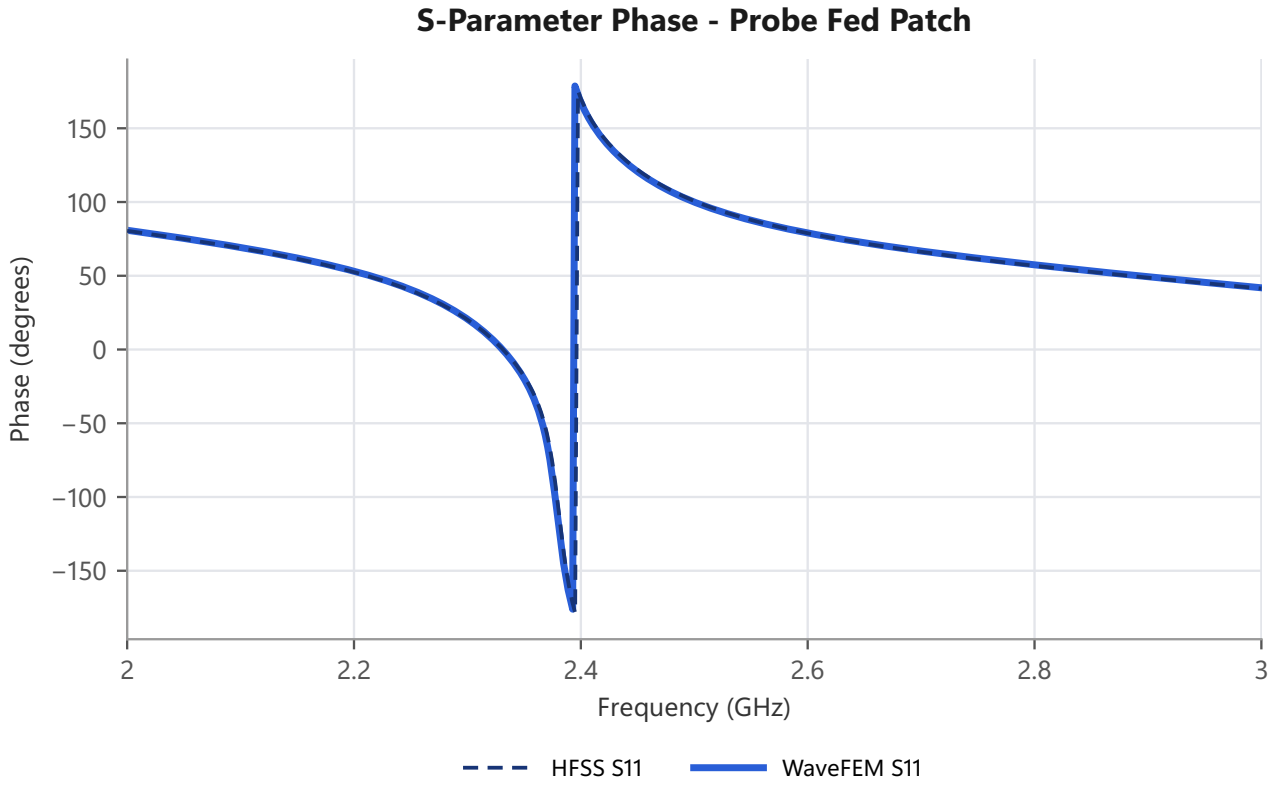


Figure 49 — S-parameter phase.

Far Field

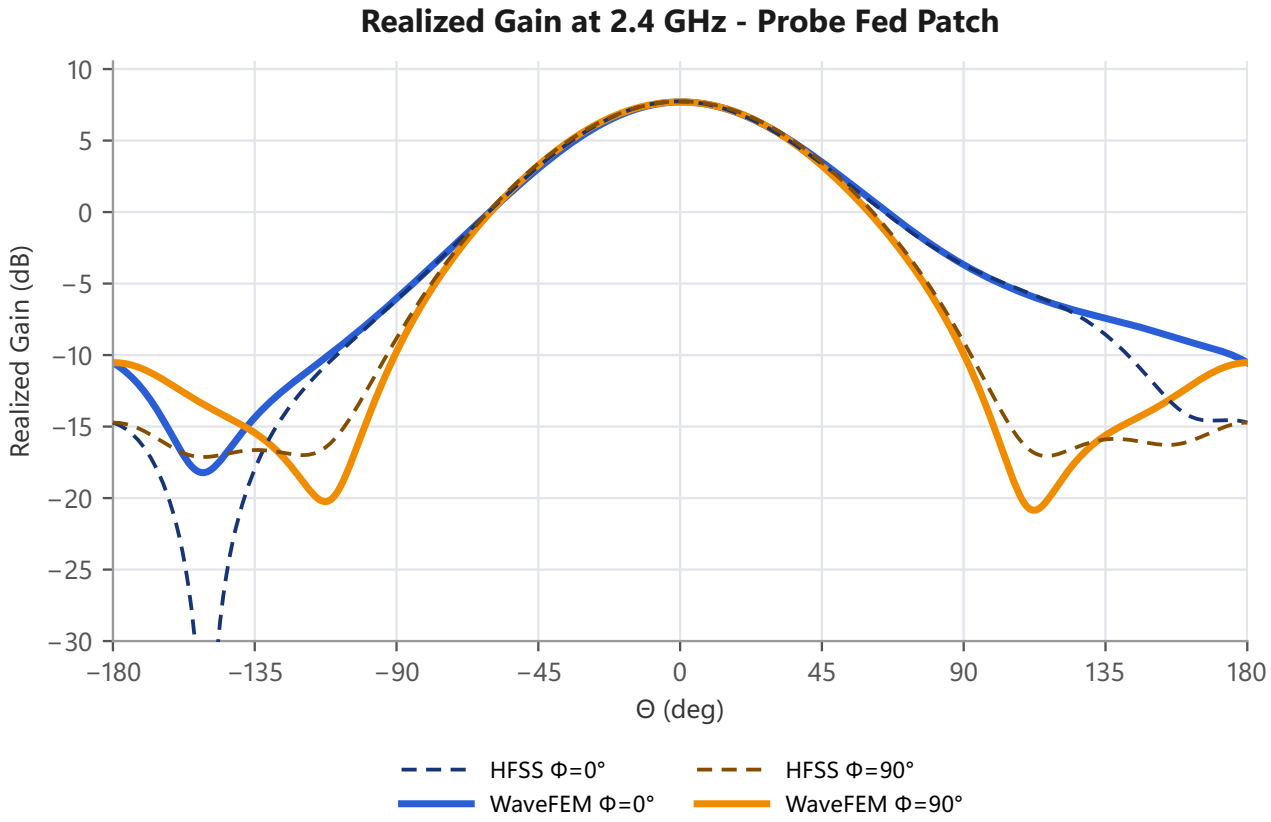


Figure 50 — Total realized gain cuts at 2.4 GHz.

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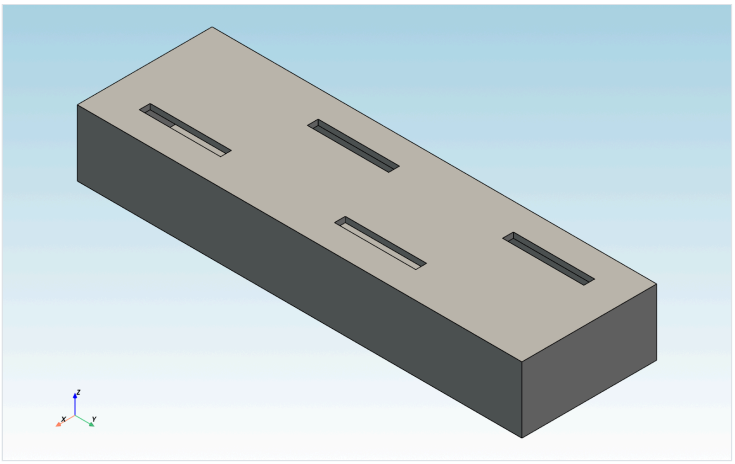
Slotted Waveguide Antenna

A WR-90 waveguide with four longitudinal slots offset alternately from the centreline of the broad wall, shorted at the far end

Simulation Setup

Refinement frequency	10 GHz
Delta S	0.02
Frequency range	9 - 11 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:00:05	00:00:31
Adaptive meshing	00:00:03	00:00:10
Frequency sweep	00:00:02	00:00:19
Adaptive passes	4	4
Final elements	5,409	4,496
Matrix size	31,512	27,725

Convergence

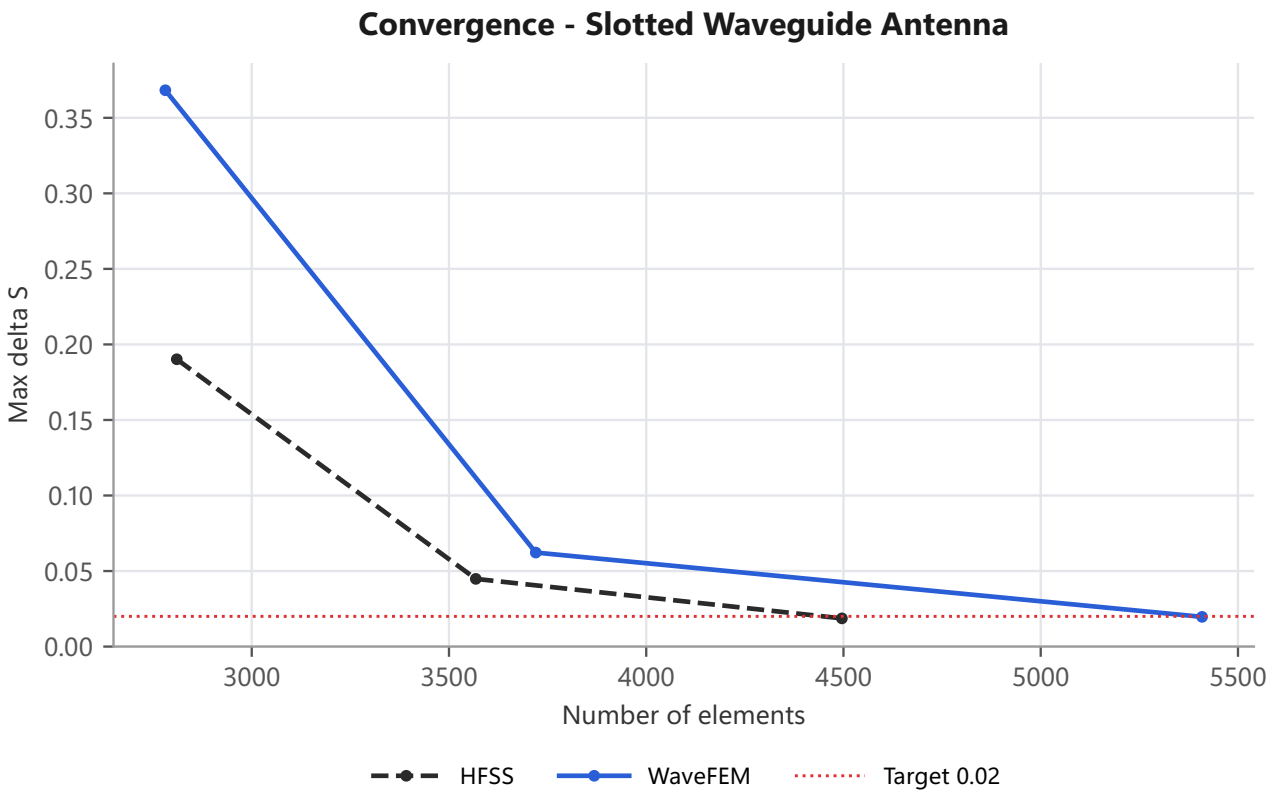


Figure 51 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

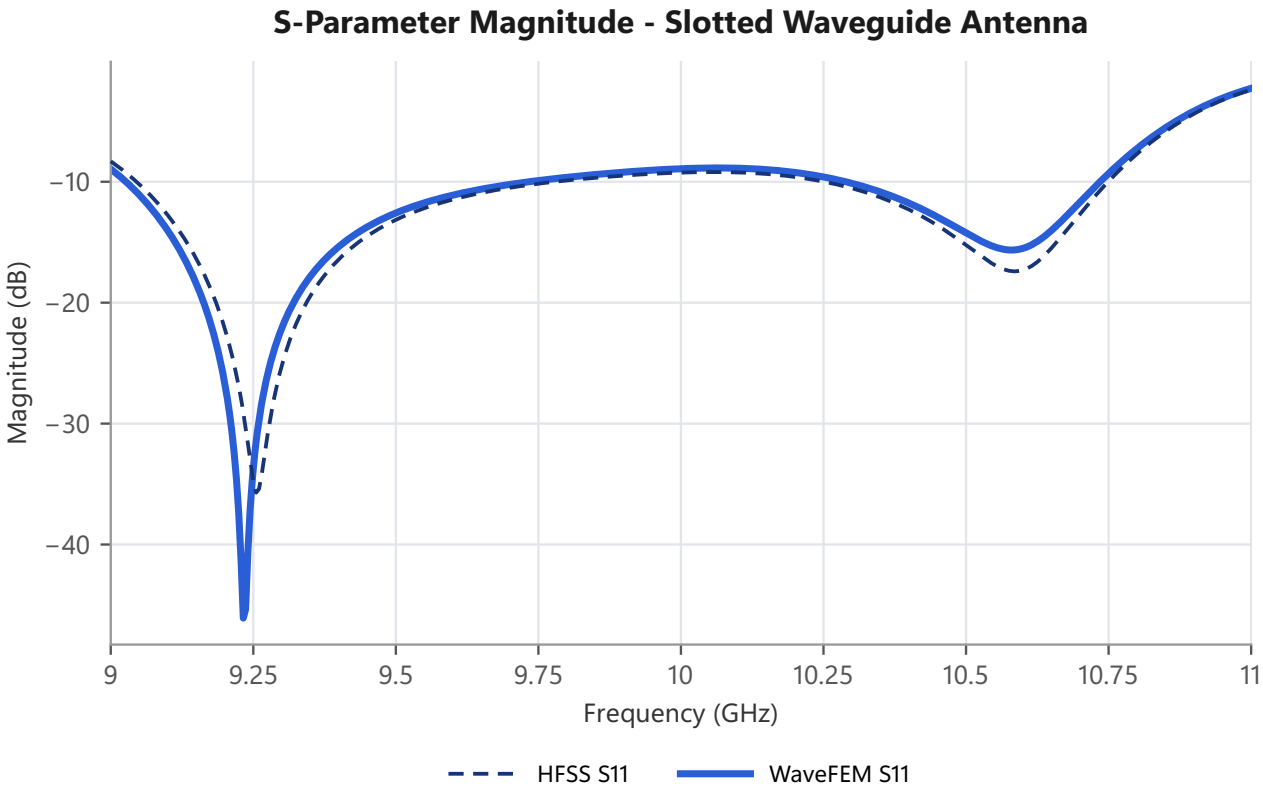


Figure 52 — S-parameter magnitude.

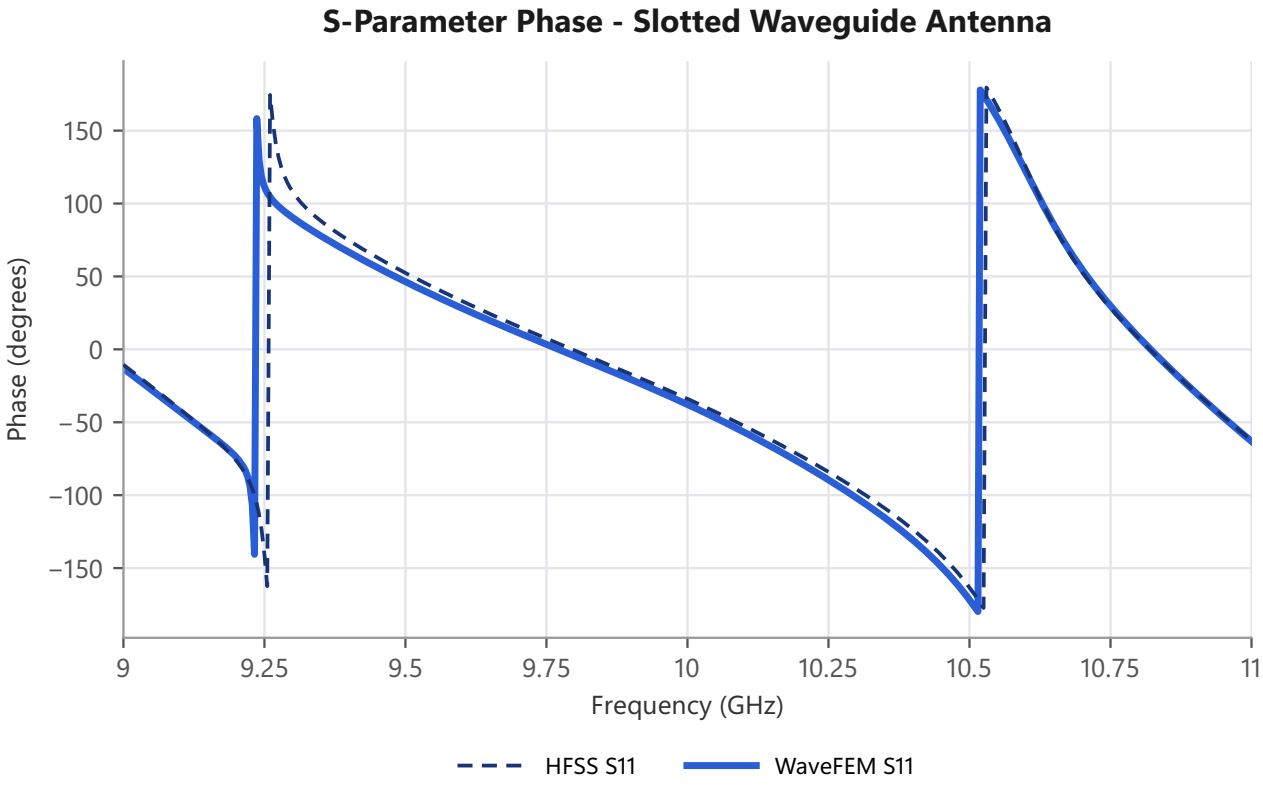


Figure 53 — S-parameter phase.

Far Field

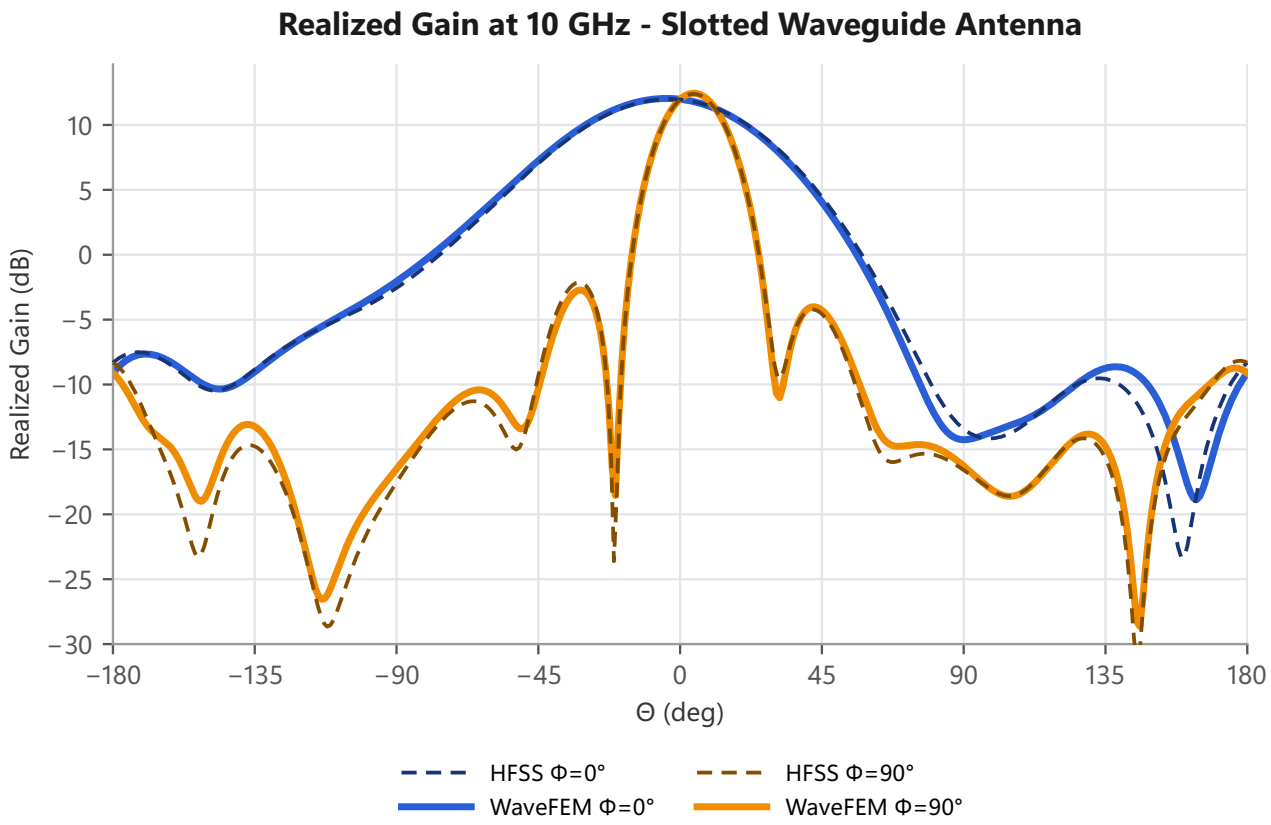


Figure 54 — Total realized gain cuts at 10 GHz.

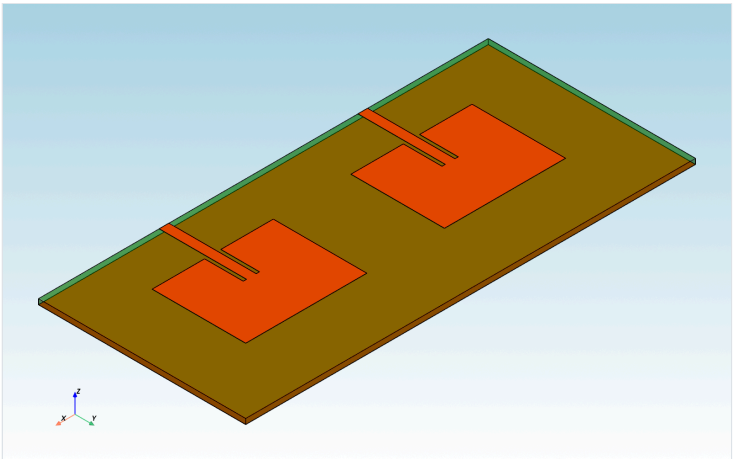
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Two Element Patch Array

Simulation Setup

Refinement frequency	2.45 GHz
Delta S	0.02
Frequency range	2 - 3 GHz
Excitation	port 1: 1 W, 0°; port 2: 1 W, 0°

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:01:54	00:02:16
Adaptive meshing	00:01:05	00:01:27
Frequency sweep	00:00:29	00:00:47
Adaptive passes	15	18
Final elements	73,331	42,337
Matrix size	415,834	253,708

Convergence

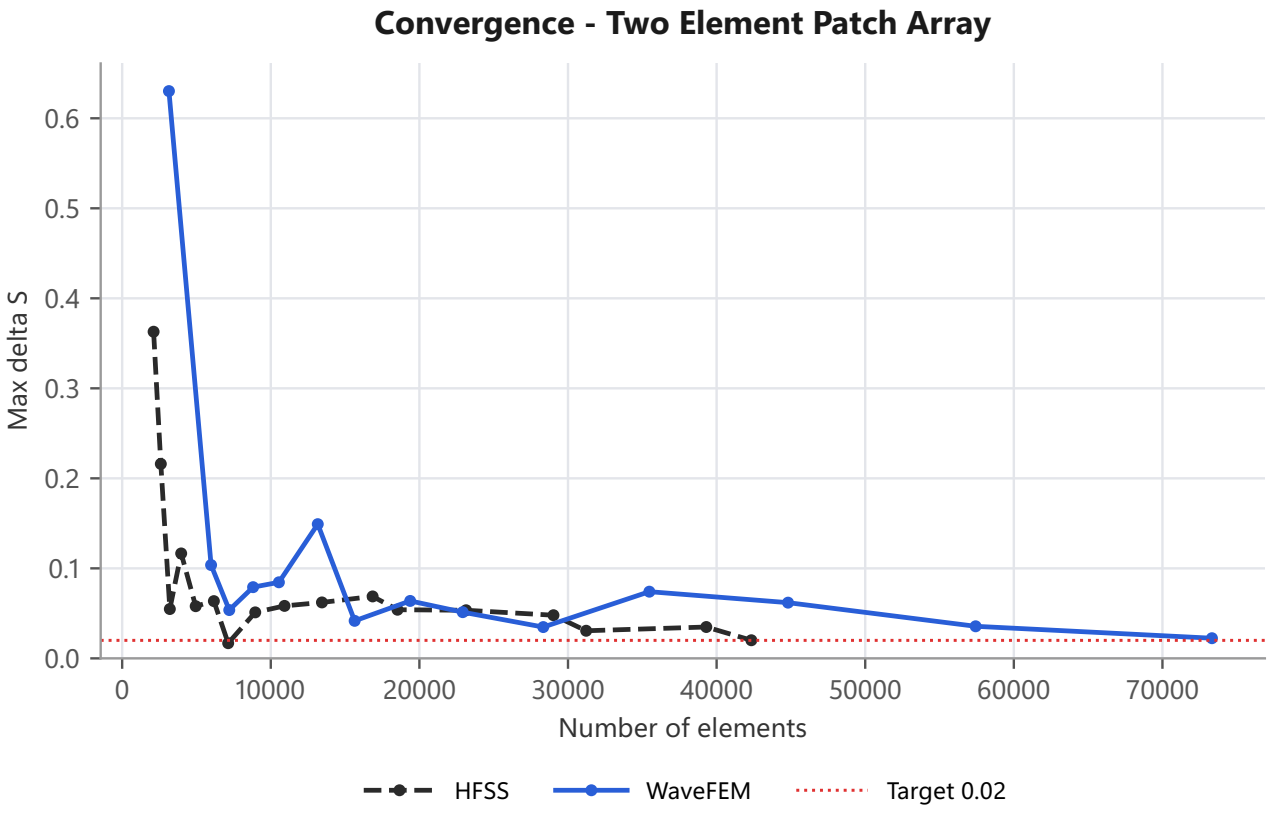


Figure 55 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

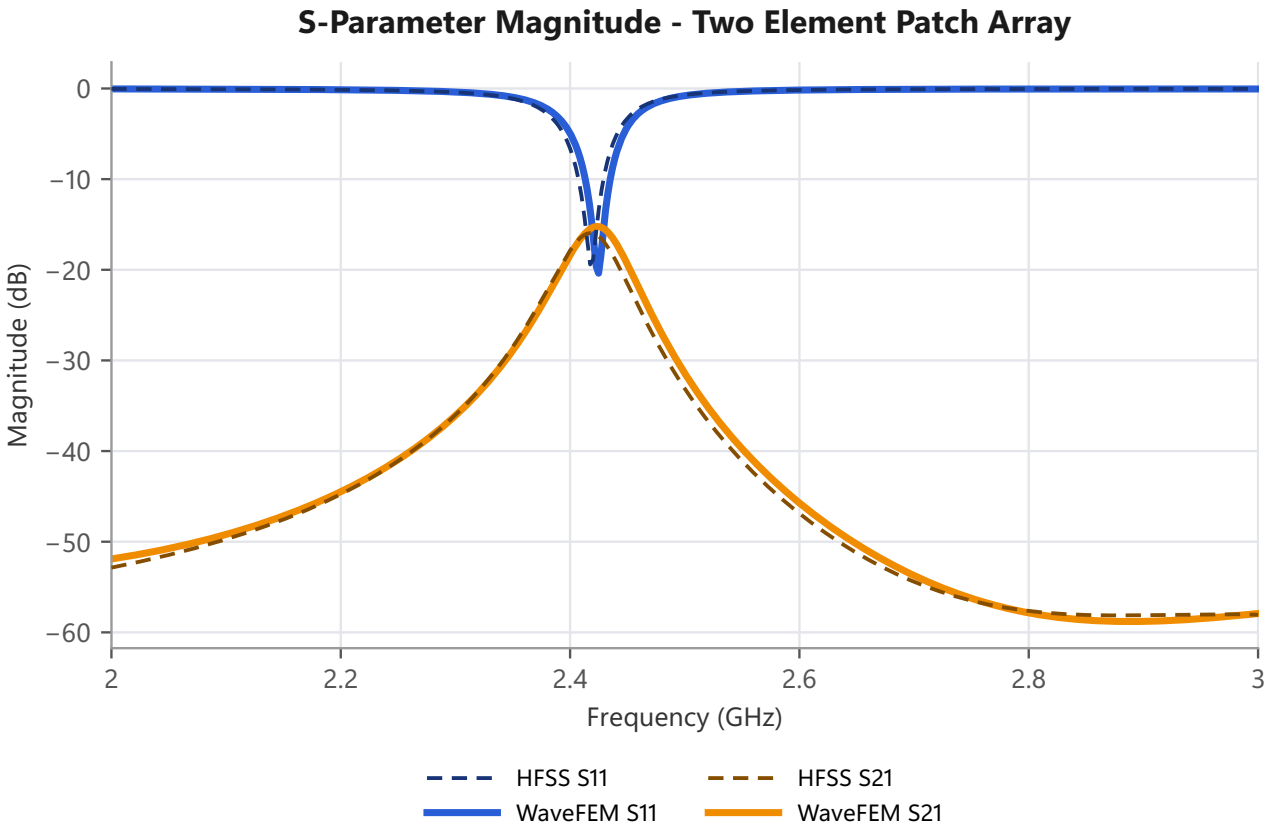


Figure 56 — S-parameter magnitude.

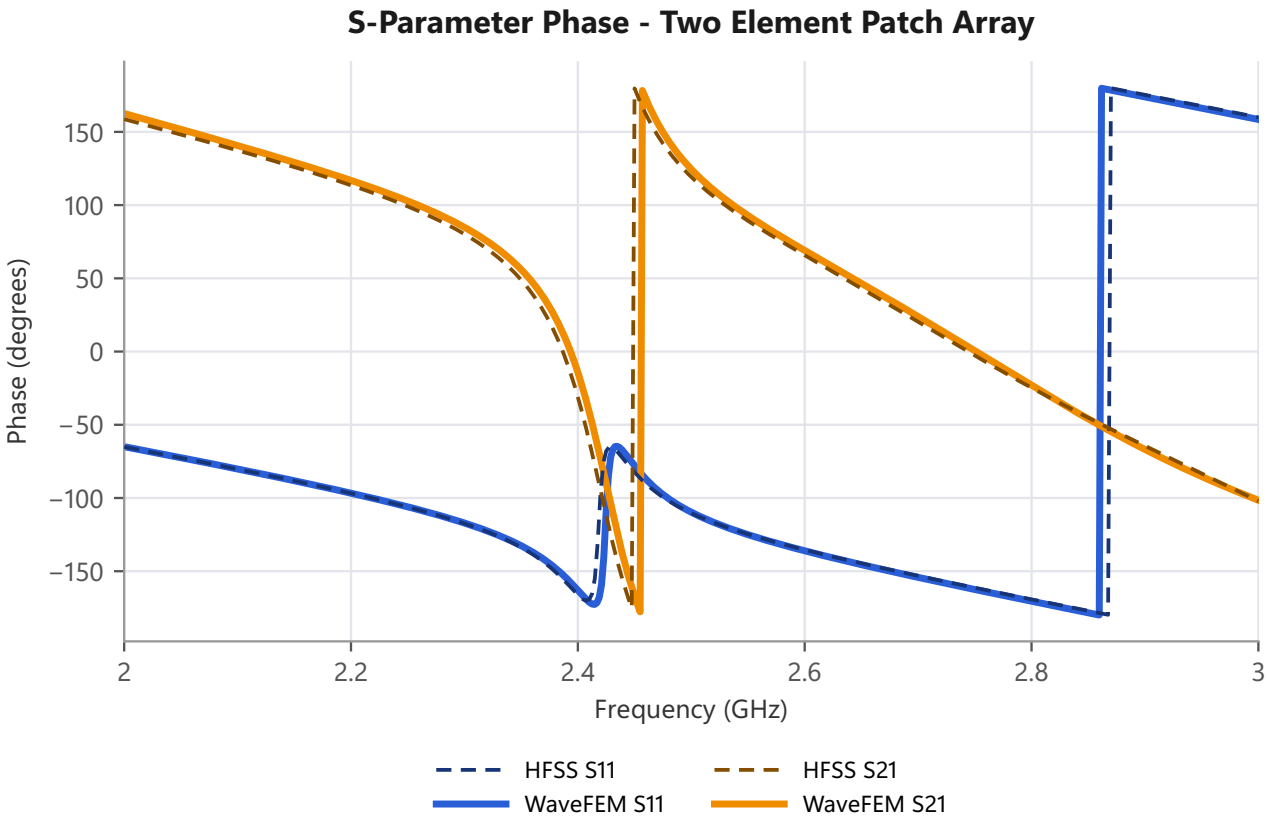


Figure 57 — S-parameter phase.

Far Field

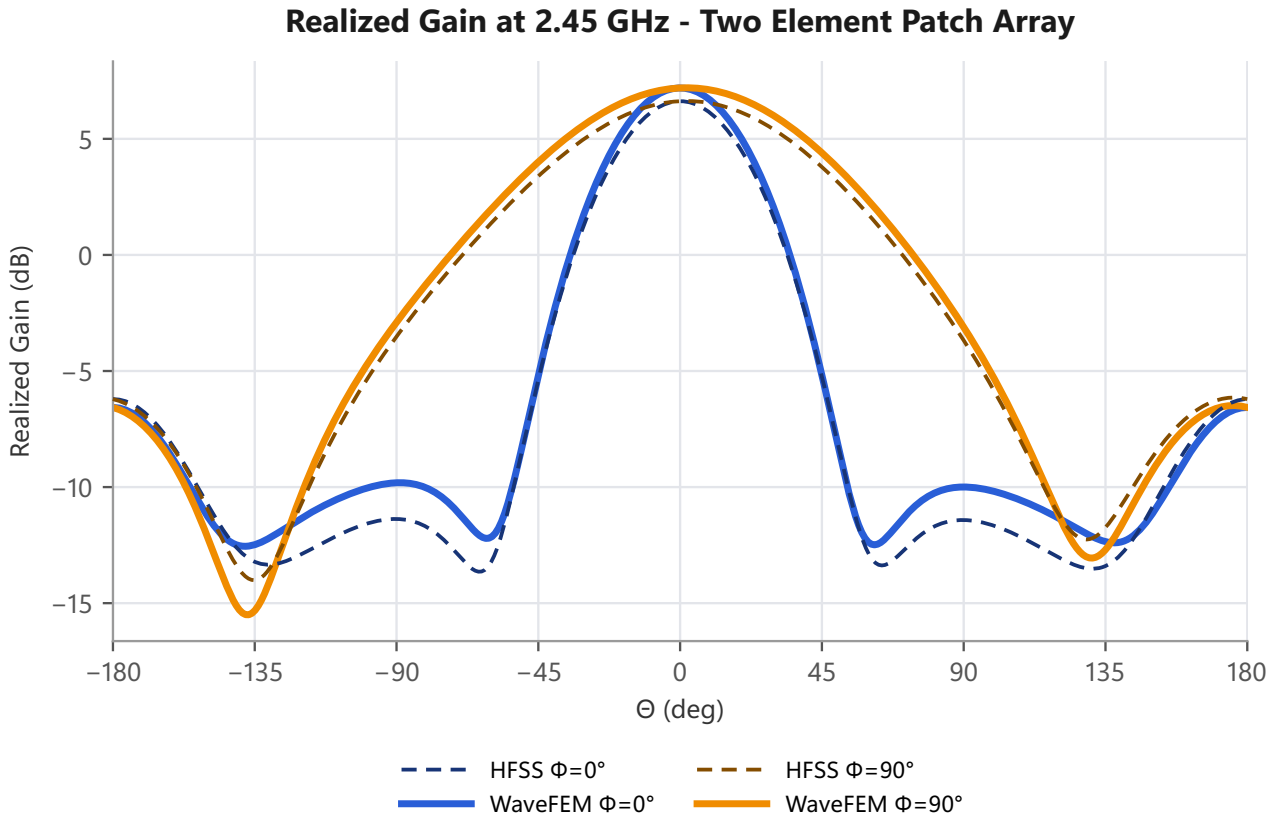


Figure 58 — Total realized gain cuts at 2.45 GHz.

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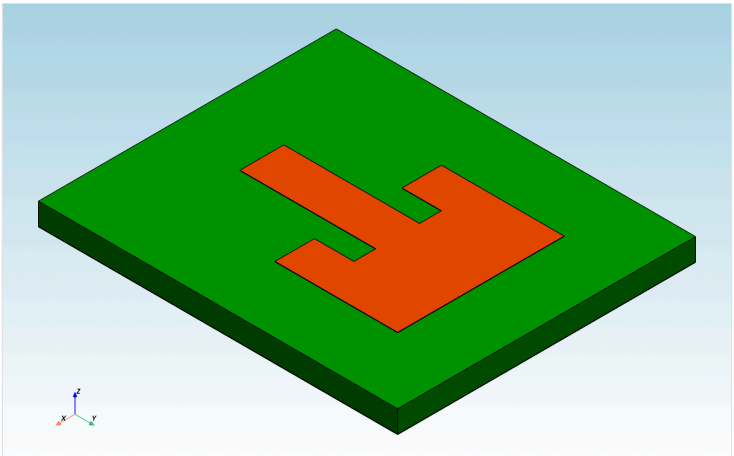
Inset Fed Patch

An inset fed microstrip patch on a 5 mm FR4 substrate (e_r=4.4), resonant at 2.41 GHz and fed by a lumped port at the end of the feed line.

Simulation Setup

Refinement frequency	2.4 GHz
Delta S	0.02
Frequency range	2 - 3 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:00:31	00:01:26
Adaptive meshing	00:00:20	00:00:45
Frequency sweep	00:00:08	00:00:39
Adaptive passes	13	14
Final elements	23,642	24,003
Matrix size	136,664	147,219

Convergence

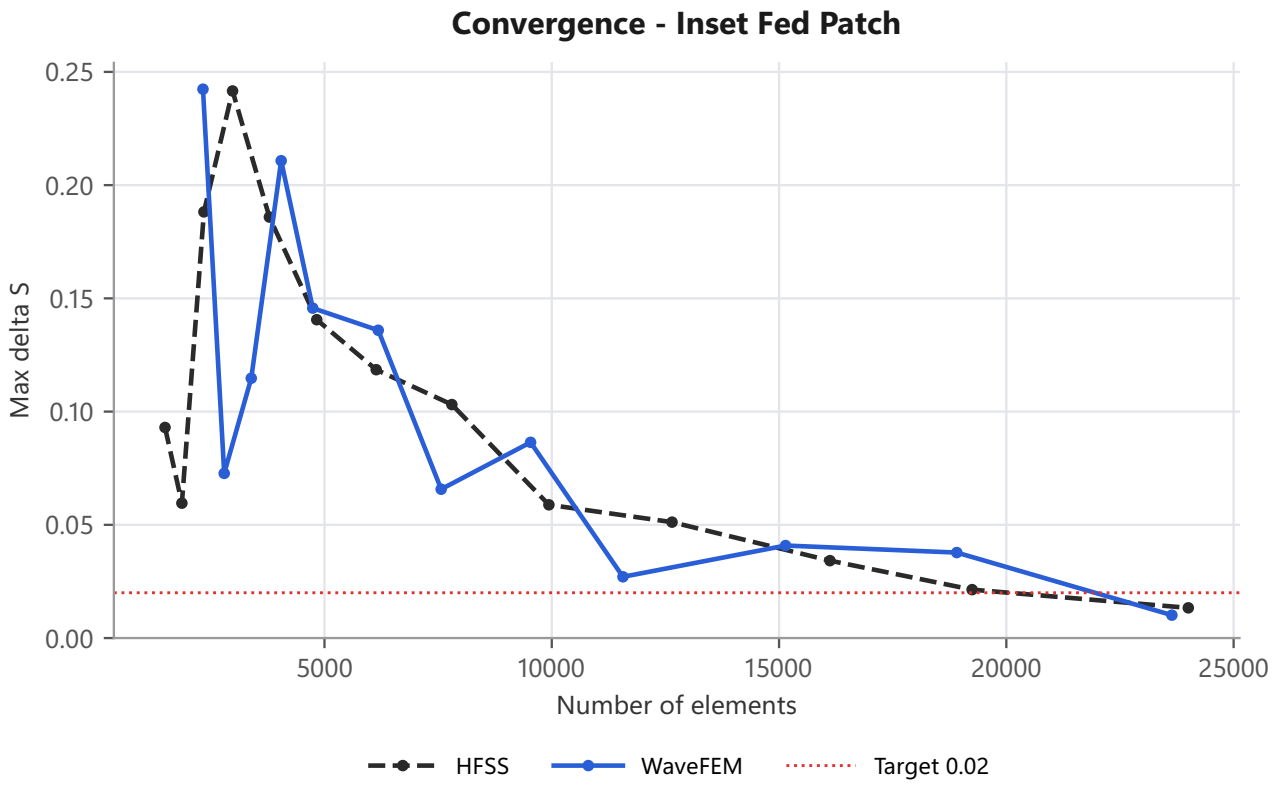


Figure 59 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

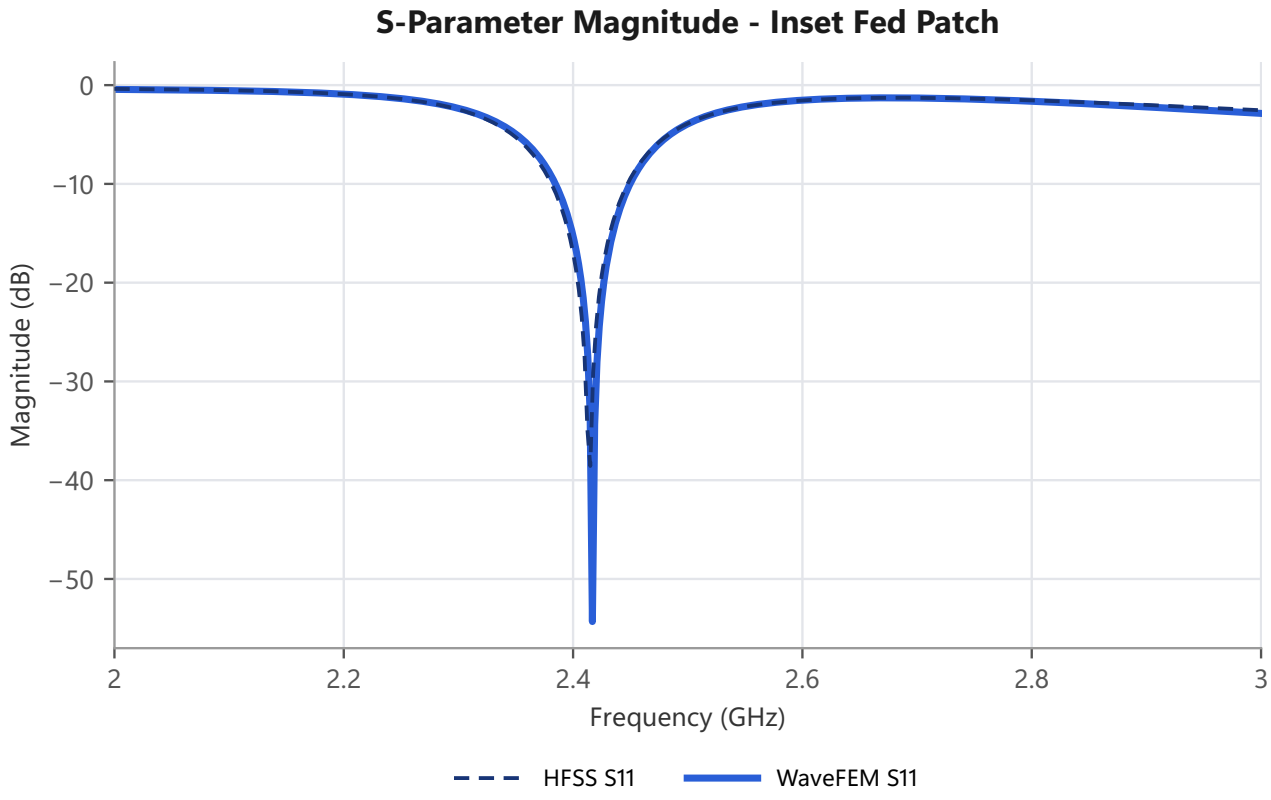


Figure 60 — S-parameter magnitude.

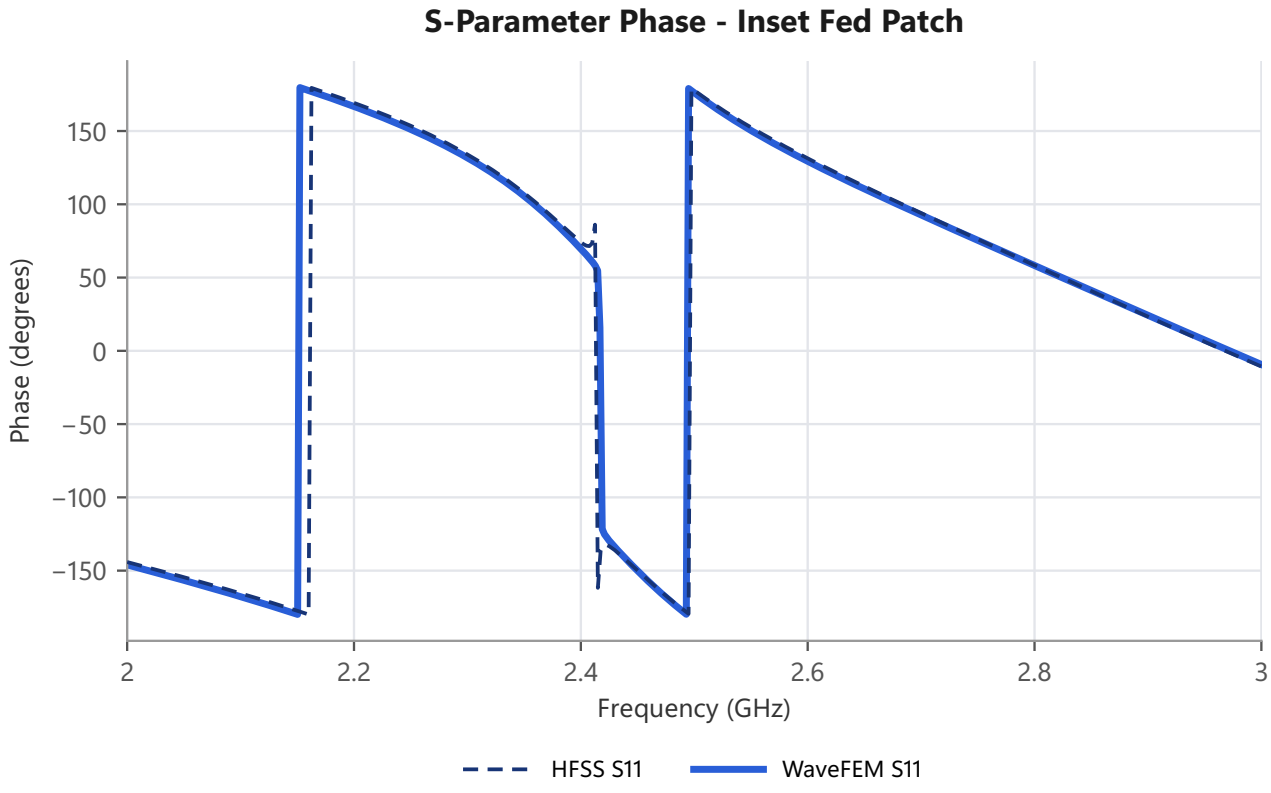


Figure 61 — S-parameter phase.

Far Field

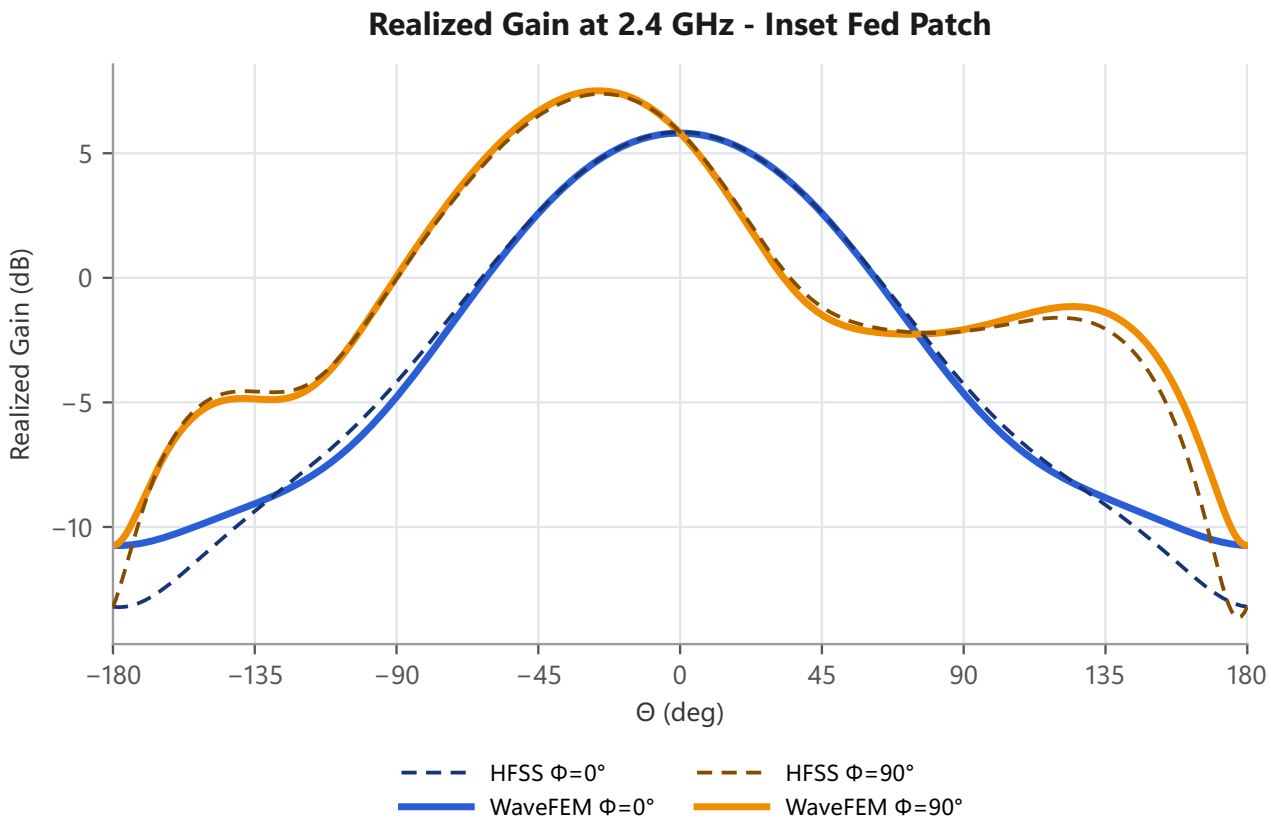


Figure 62 — Total realized gain cuts at 2.4 GHz.

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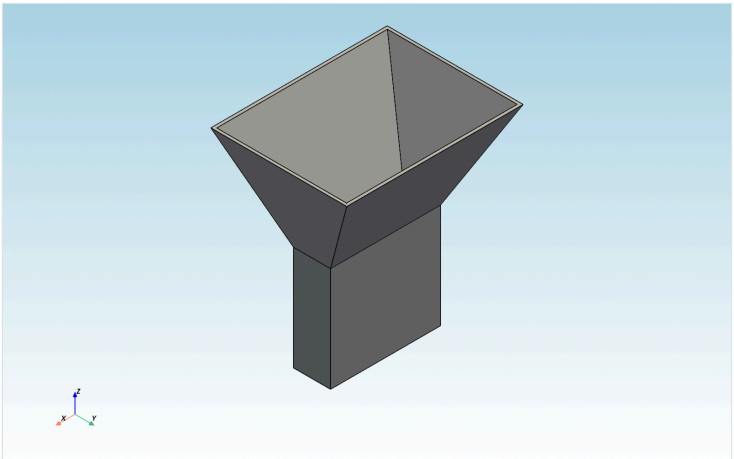
Pyramidal Horn

A pyramidal horn fed by a 62.3 x 18.9 mm rectangular waveguide.

Simulation Setup

Refinement frequency	6 GHz
Delta S	0.02
Frequency range	3 - 6 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:00:10	00:00:36
Adaptive meshing	00:00:03	00:00:05
Frequency sweep	00:00:07	00:00:28
Adaptive passes	3	2
Final elements	7,991	5,248
Matrix size	49,660	33,387

Convergence

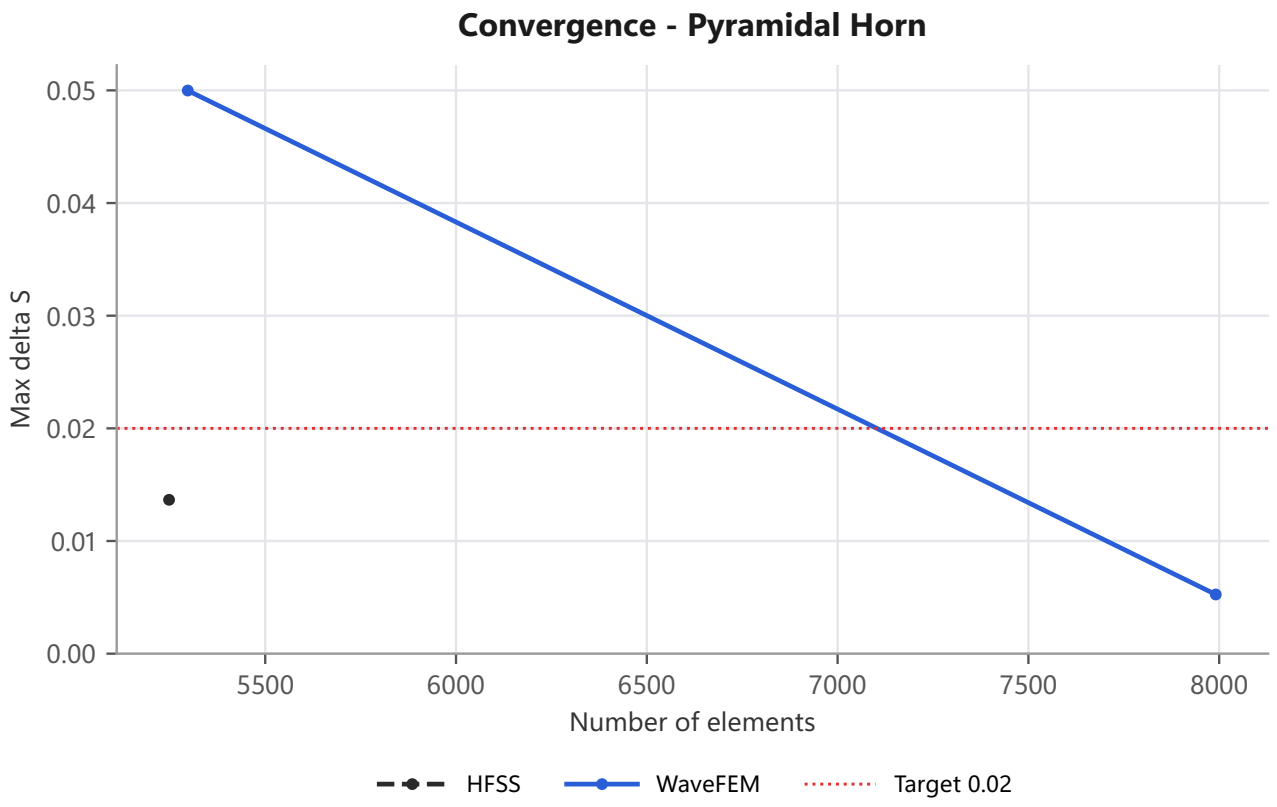


Figure 63 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

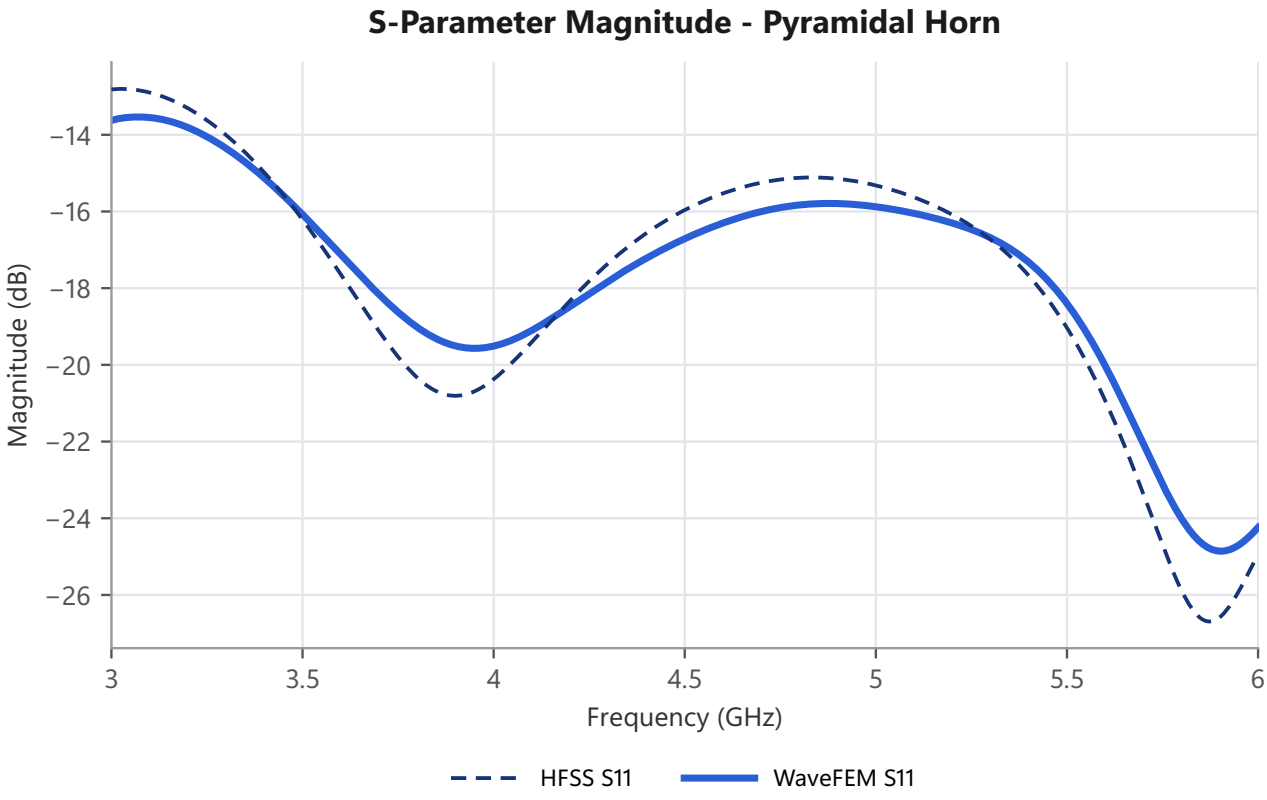


Figure 64 — S-parameter magnitude.

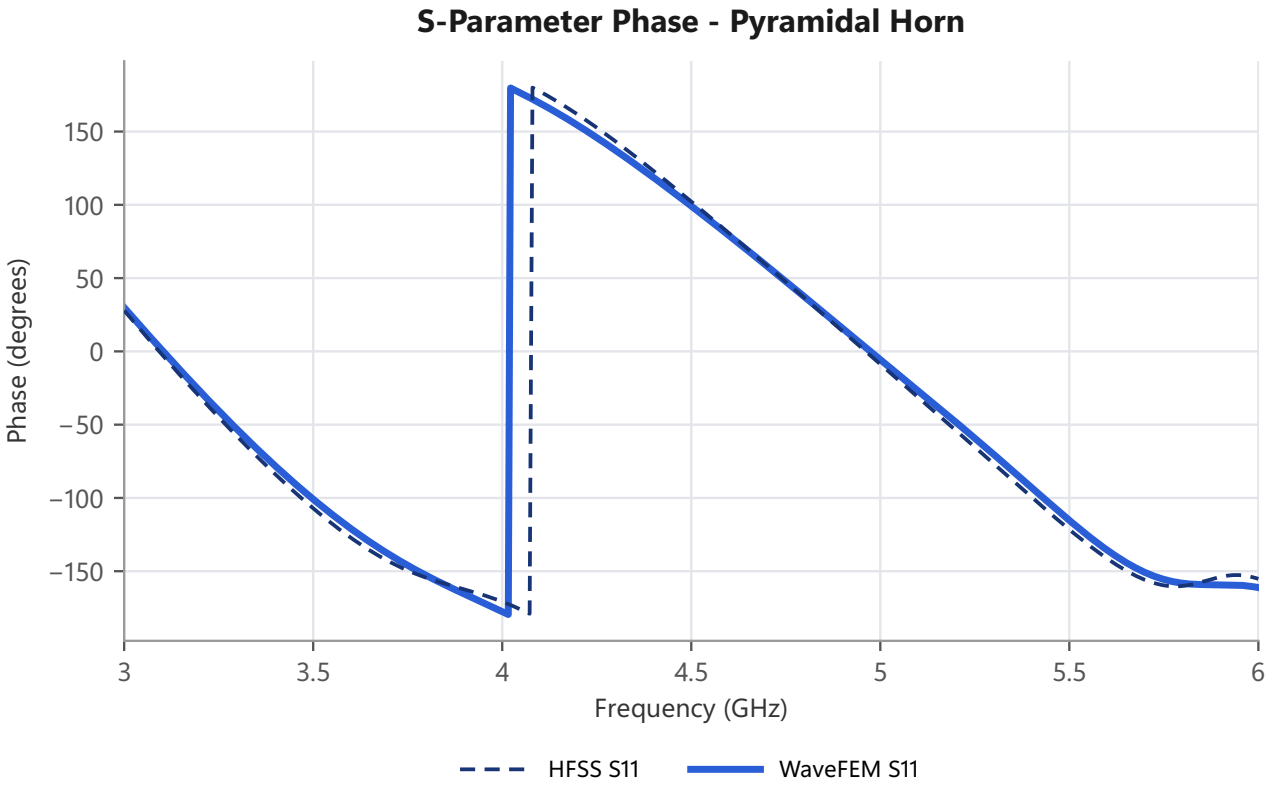


Figure 65 — S-parameter phase.

Far Field

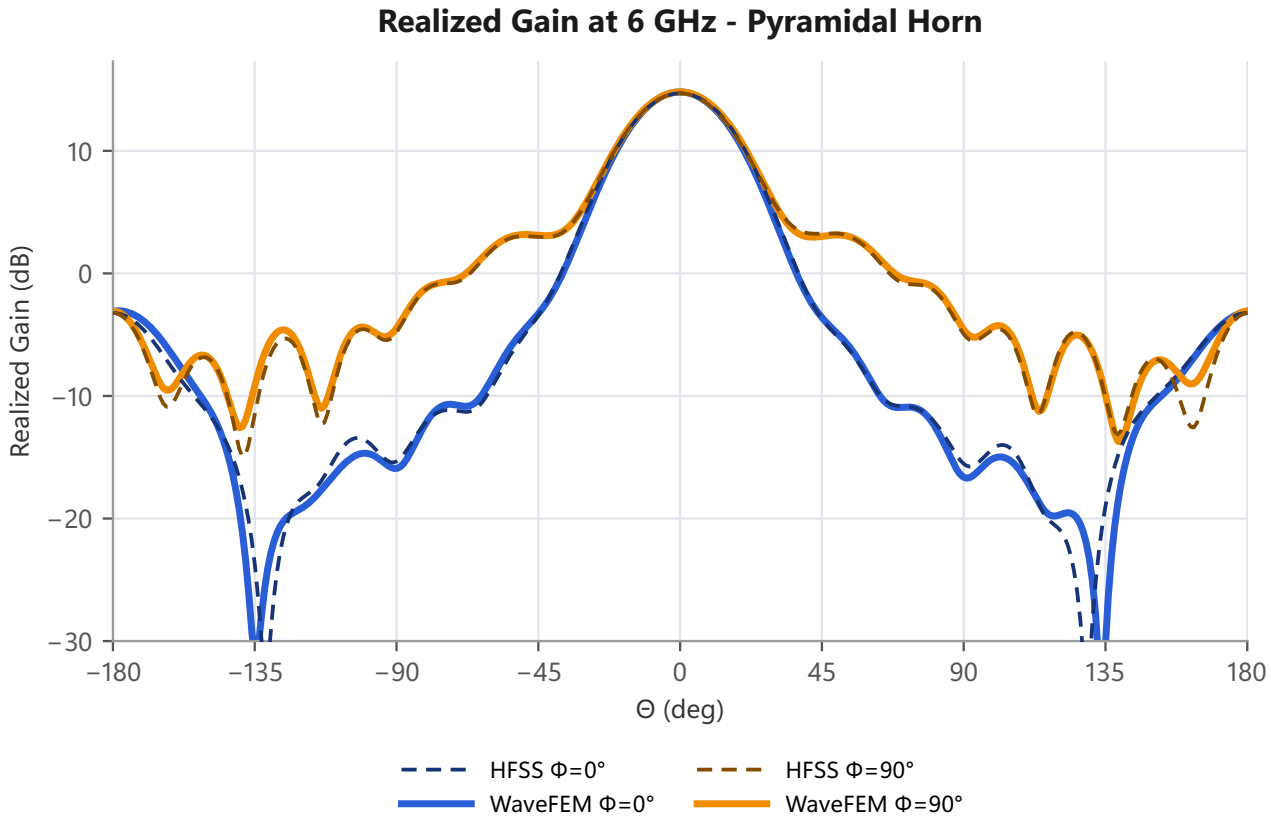


Figure 66 — Total realized gain cuts at 6 GHz.

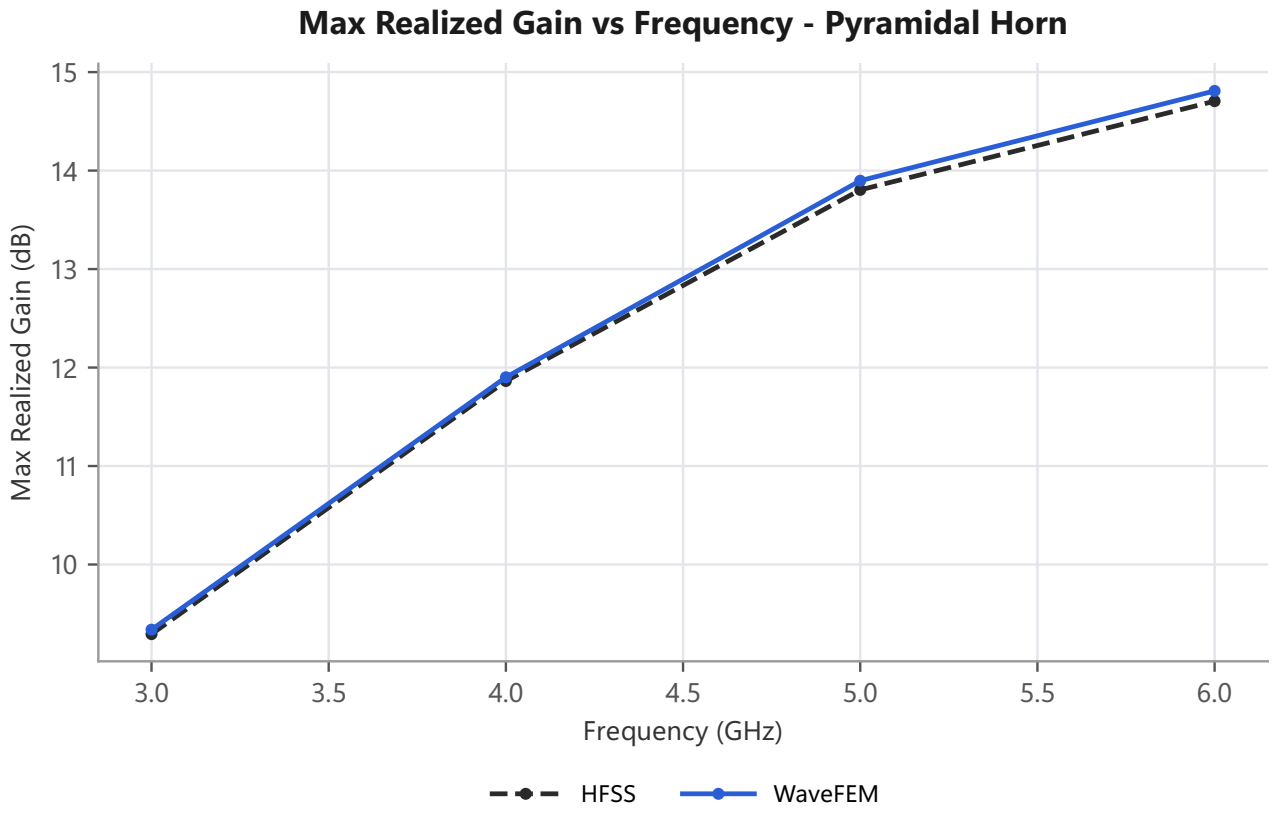


Figure 67 — Maximum realized gain against frequency.

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ANTENNA

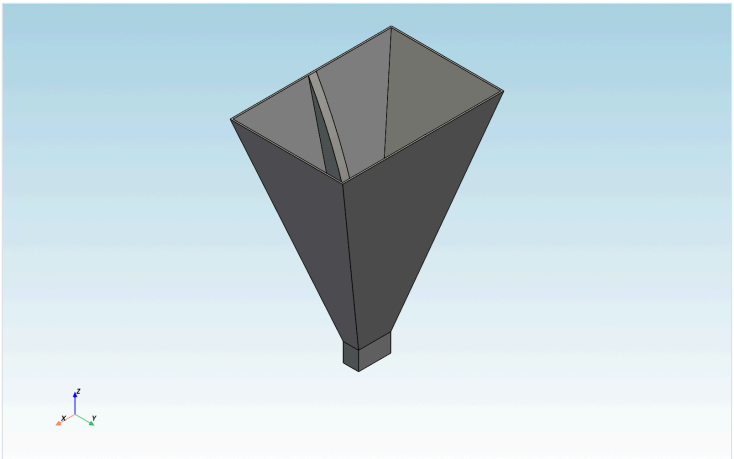
Exponential Horn

A horn with an exponential flare, fed by a 42.3 x 18.2 mm rectangular waveguide.

Simulation Setup

Refinement frequency	5 GHz
Delta S	0.02
Frequency range	4 - 5 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:00:11	00:01:16
Adaptive meshing	00:00:05	00:00:19
Frequency sweep	00:00:05	00:00:49
Adaptive passes	2	3
Final elements	14,197	23,138
Matrix size	89,458	145,281

Convergence

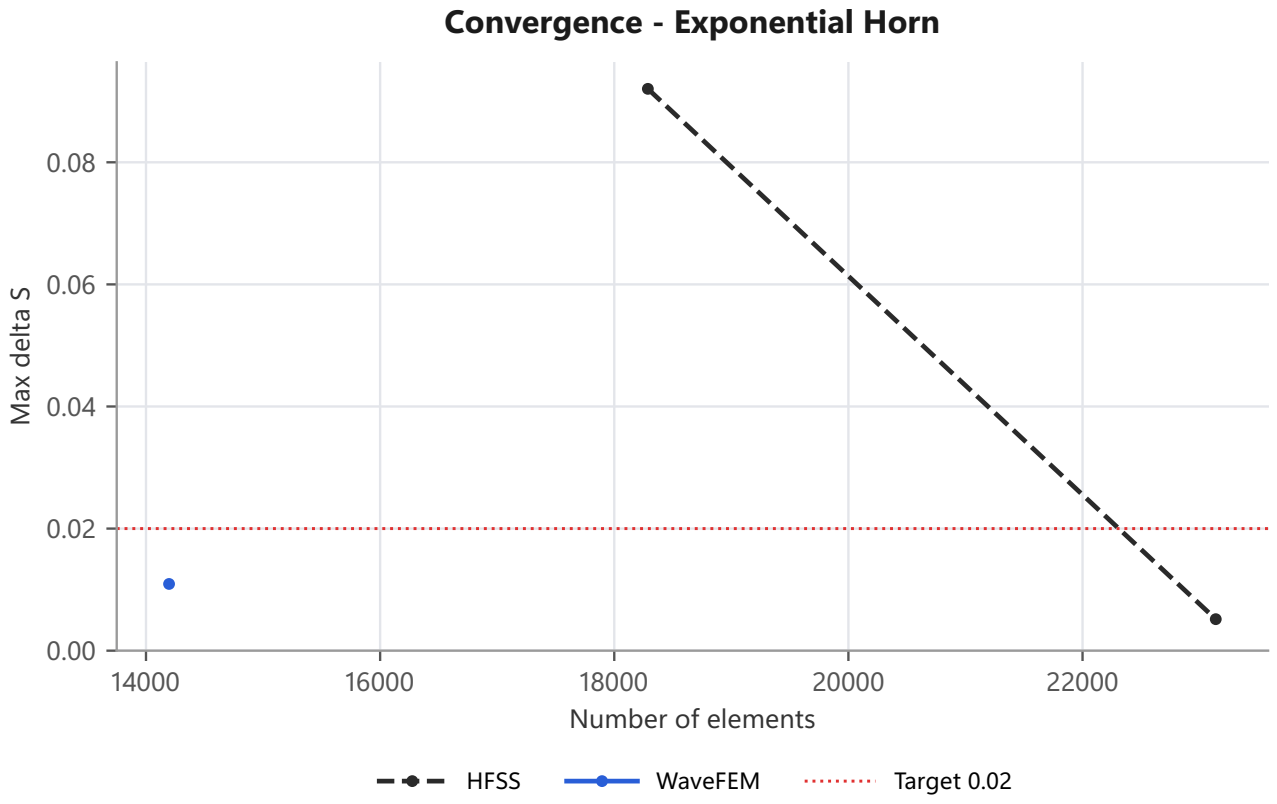


Figure 68 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

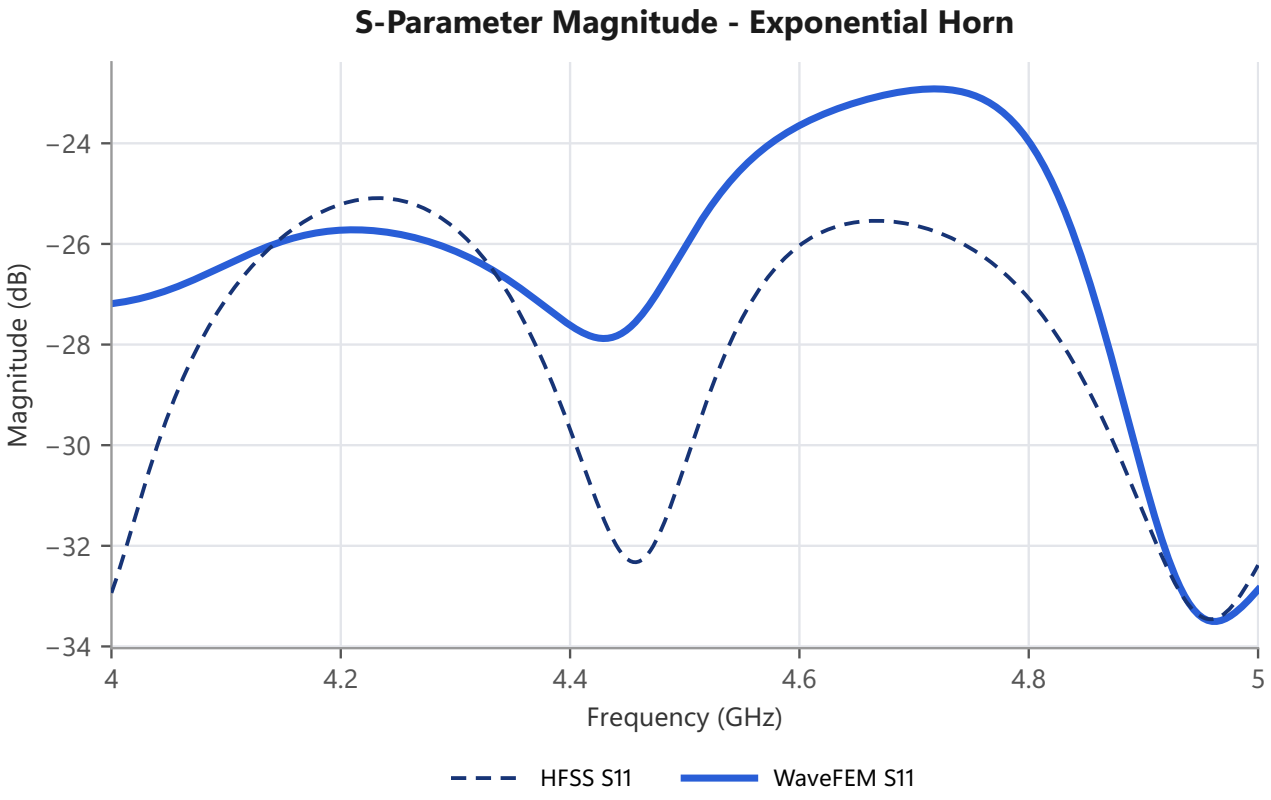


Figure 69 — S-parameter magnitude.

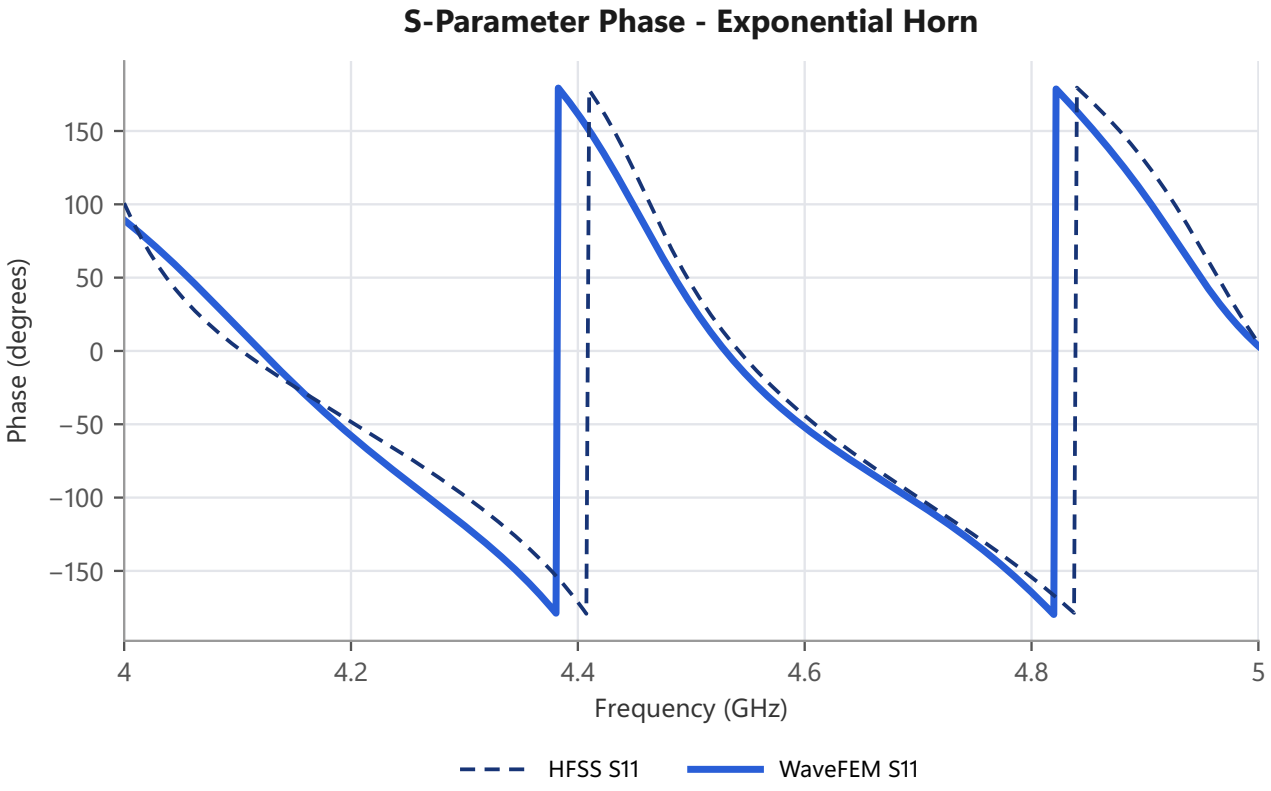


Figure 70 — S-parameter phase.

Far Field

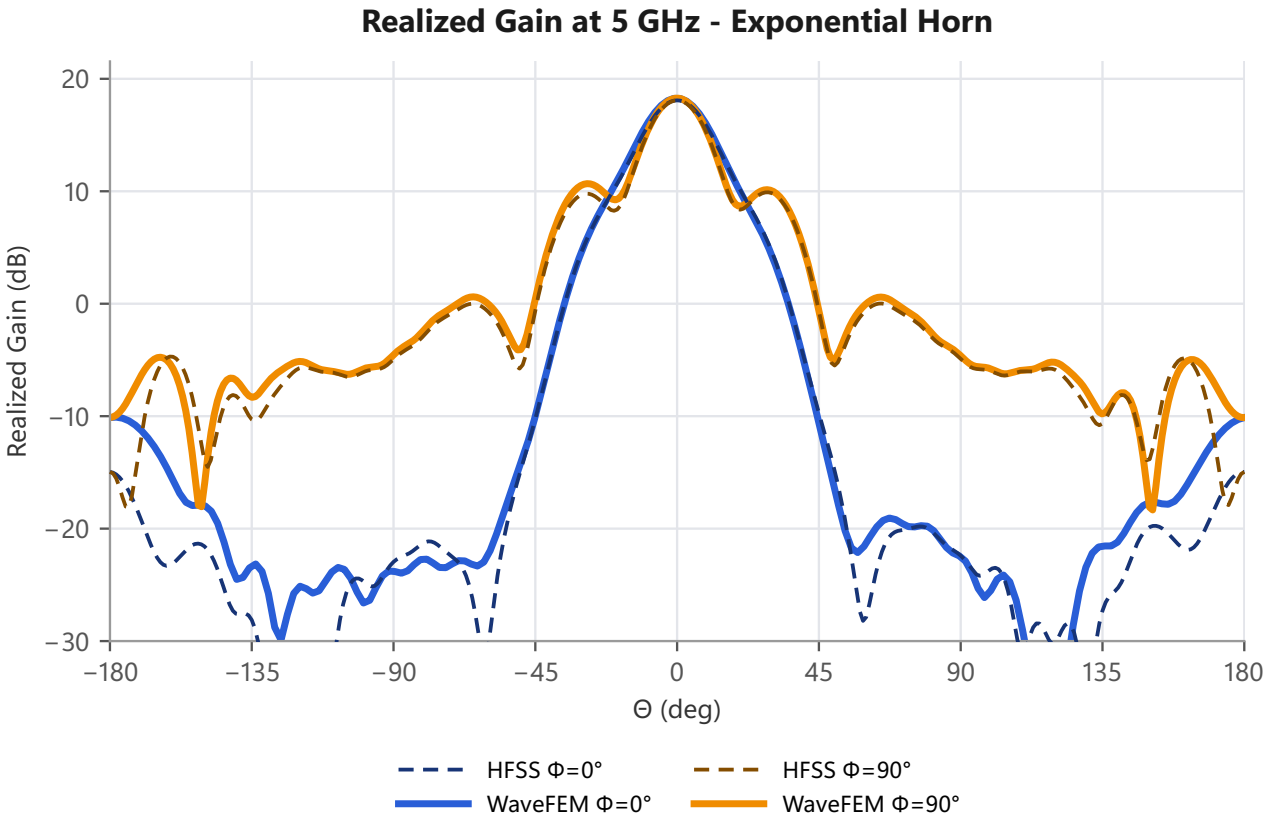


Figure 71 — Total realized gain cuts at 5 GHz.

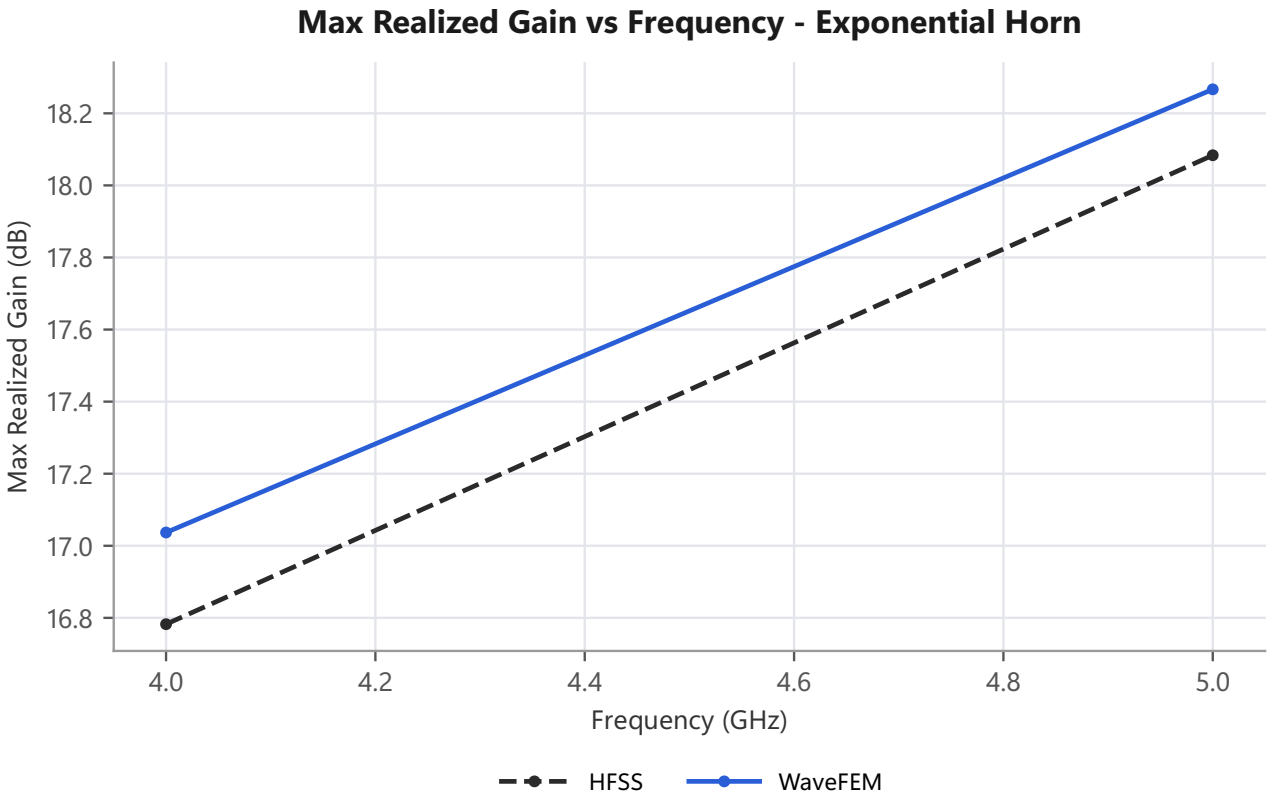


Figure 72 — Maximum realized gain against frequency.

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ANTENNA

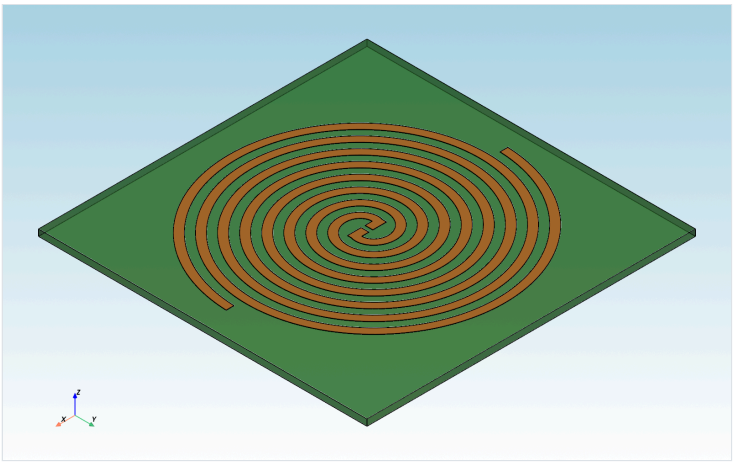
Spiral Antenna

A two-arm Archimedean spiral (4 turns, 1 mm traces) in copper on 0.813 mm RO4003C, fed by a 120 ohm lumped port at the centre gap. Wideband: S11 below -10 dB over most of 2-8 GHz.

Simulation Setup

Refinement frequency	8 GHz
Delta S	0.02
Frequency range	2 - 8 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:04:21	00:05:24
Adaptive meshing	00:00:53	00:00:38
Frequency sweep	00:03:24	00:04:31
Adaptive passes	5	4
Final elements	40,711	36,454
Matrix size	275,390	242,613

Convergence

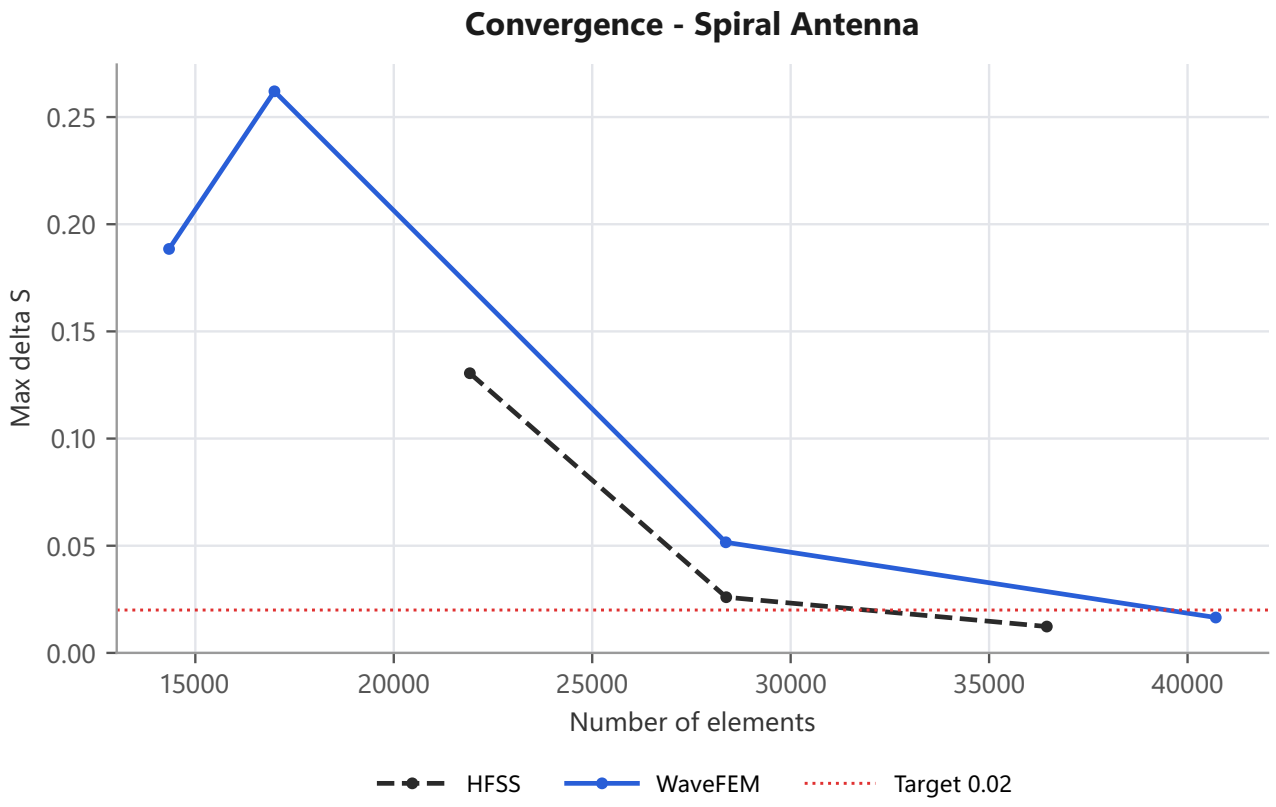


Figure 73 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

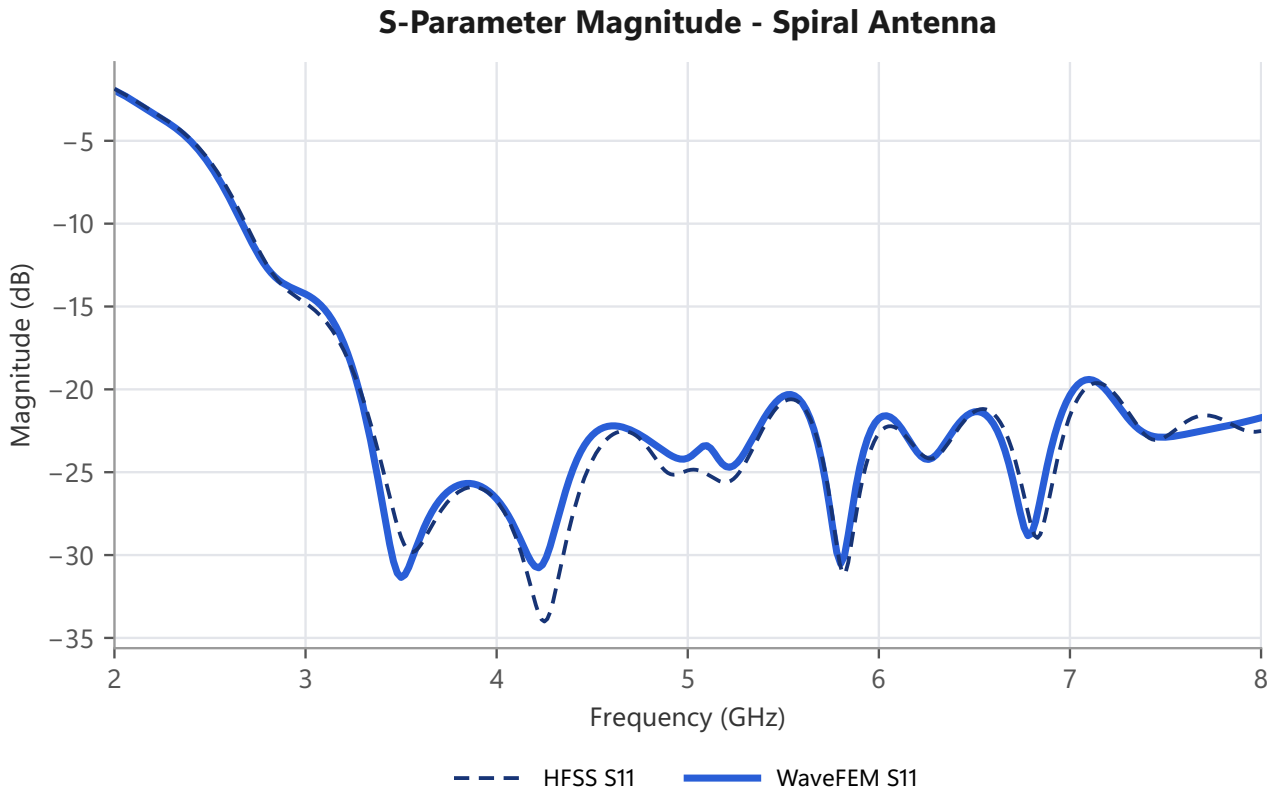


Figure 74 — S-parameter magnitude.

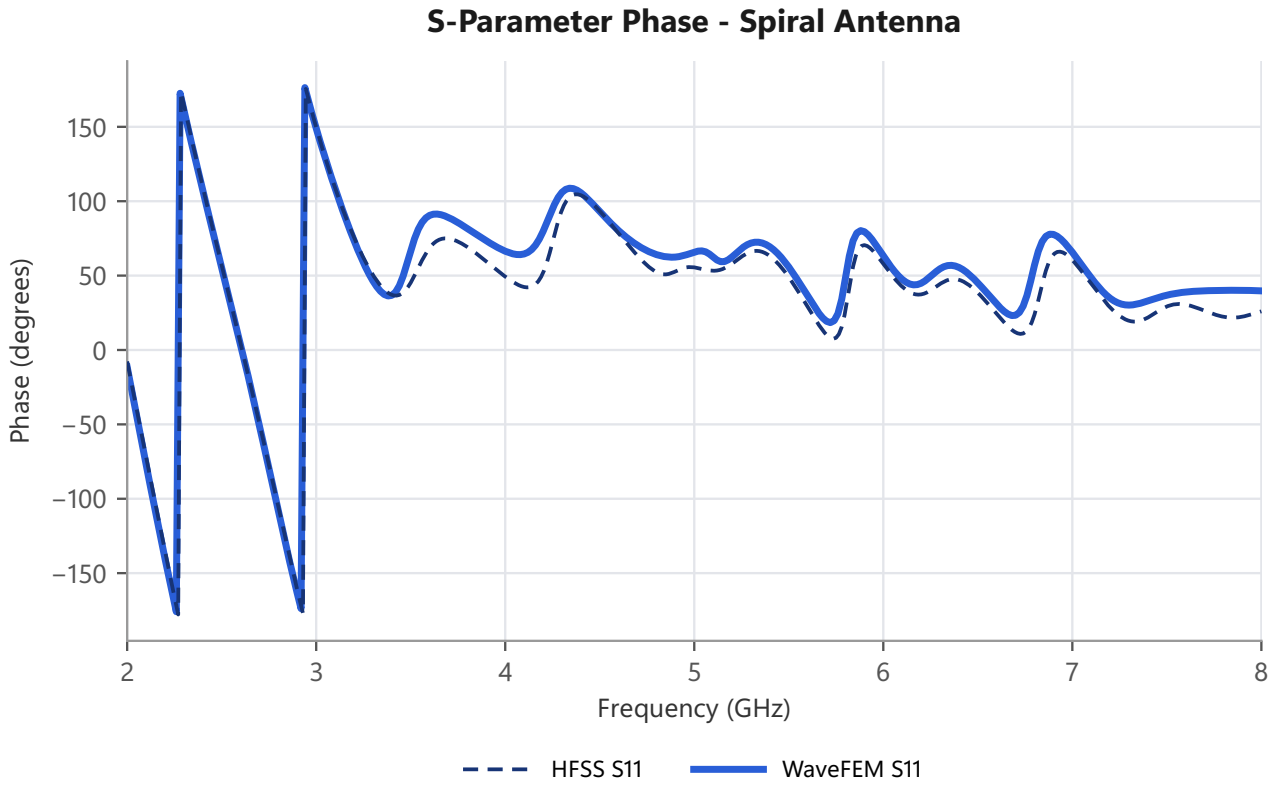


Figure 75 — S-parameter phase.

Far Field

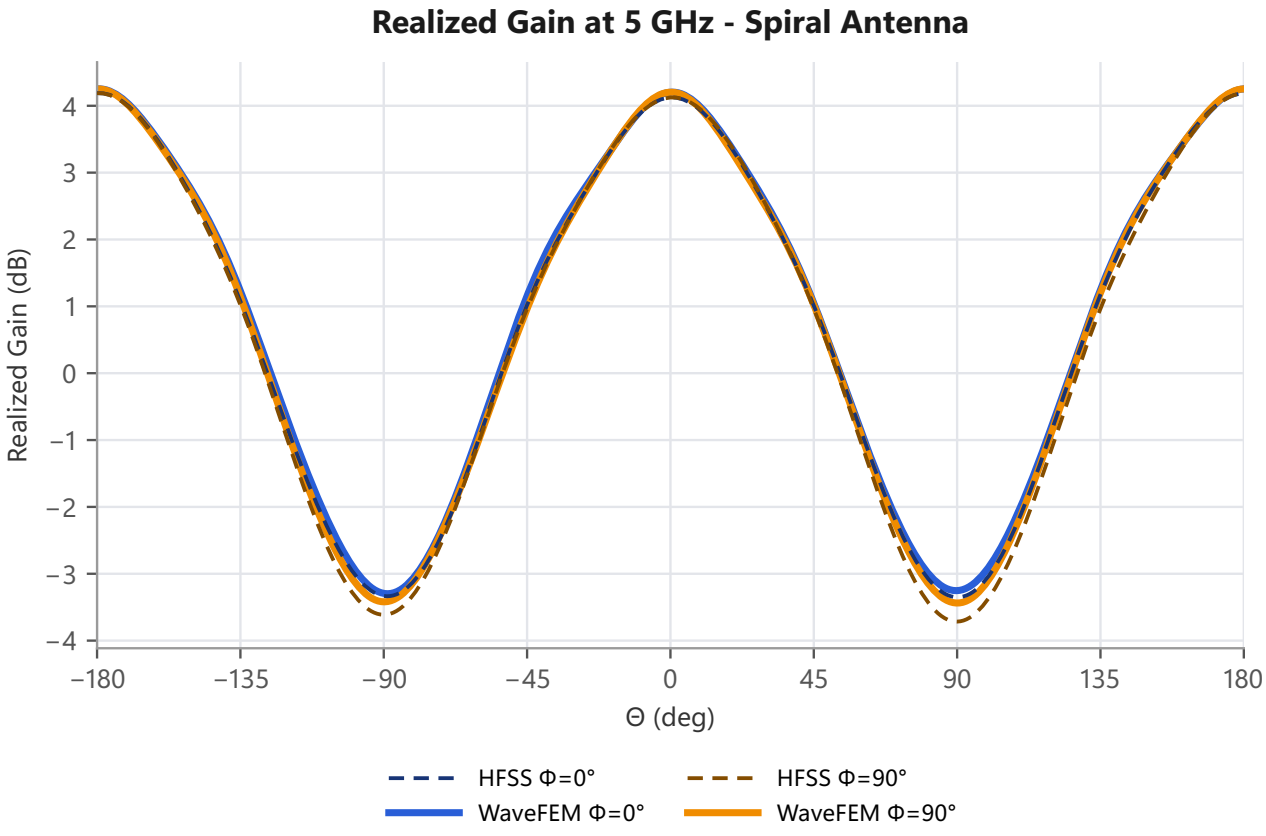


Figure 76 — Total realized gain cuts at 5 GHz.

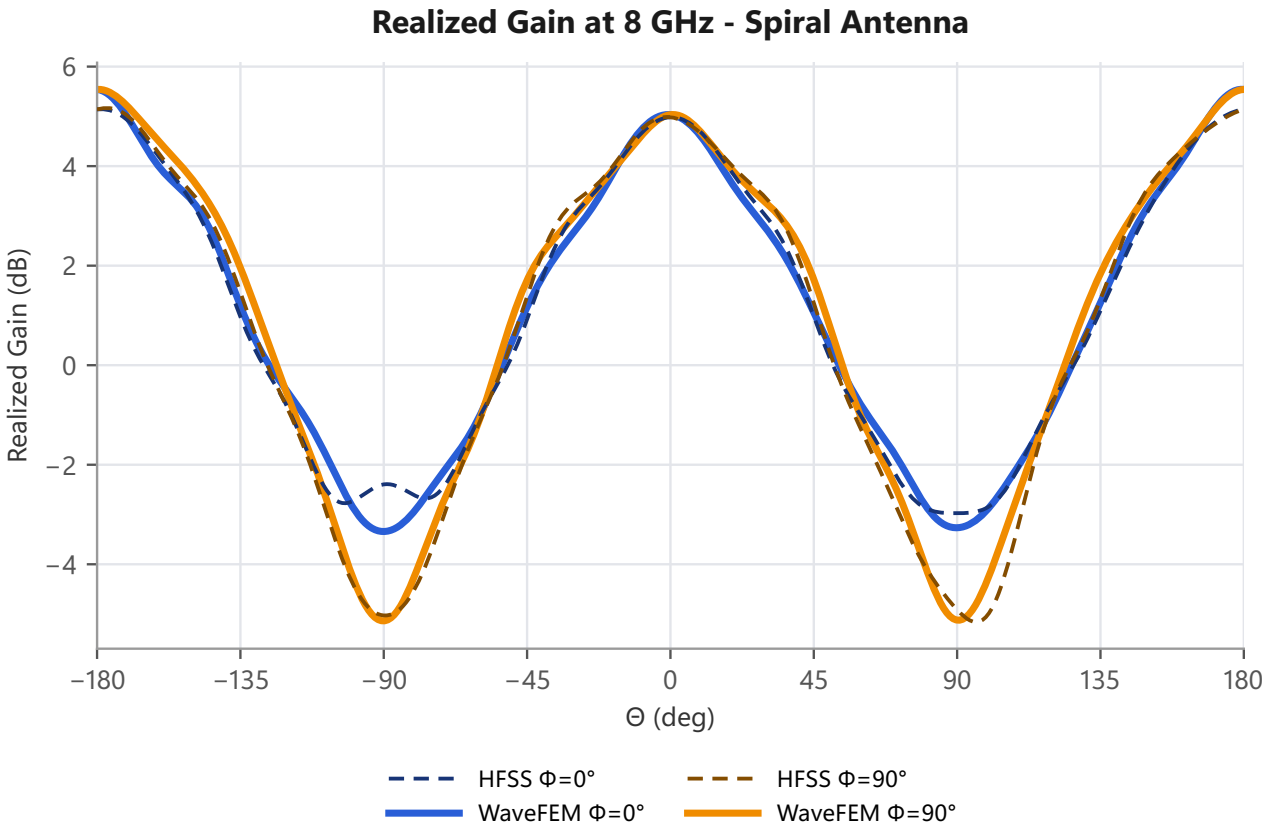


Figure 77 — Total realized gain cuts at 8 GHz.

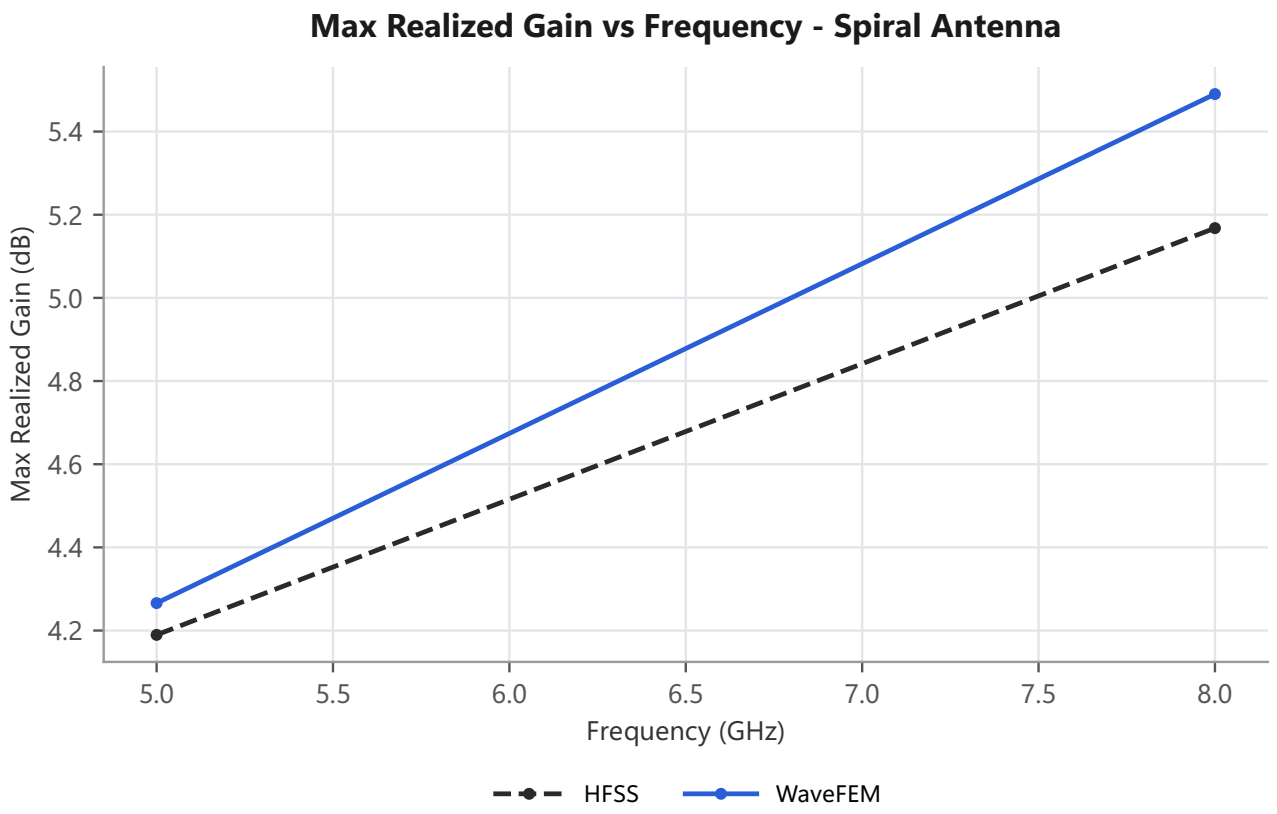


Figure 78 — Maximum realized gain against frequency.

PLANAR

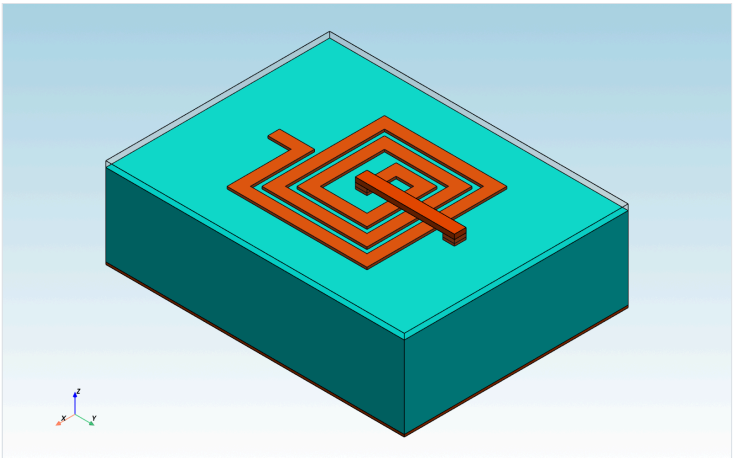
Spiral Inductor

A three-turn square spiral inductor (10 µm lines, 5 µm gaps) with an air bridge on a second metal layer, on a 70 µm substrate over ground.

Simulation Setup

Refinement frequency	70 GHz
Delta S	0.02
Frequency range	1 - 70 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:00:16	00:00:49
Adaptive meshing	00:00:09	00:00:26
Frequency sweep	00:00:05	00:00:21
Adaptive passes	7	10
Final elements	8,423	8,894
Matrix size	58,470	58,410

Convergence

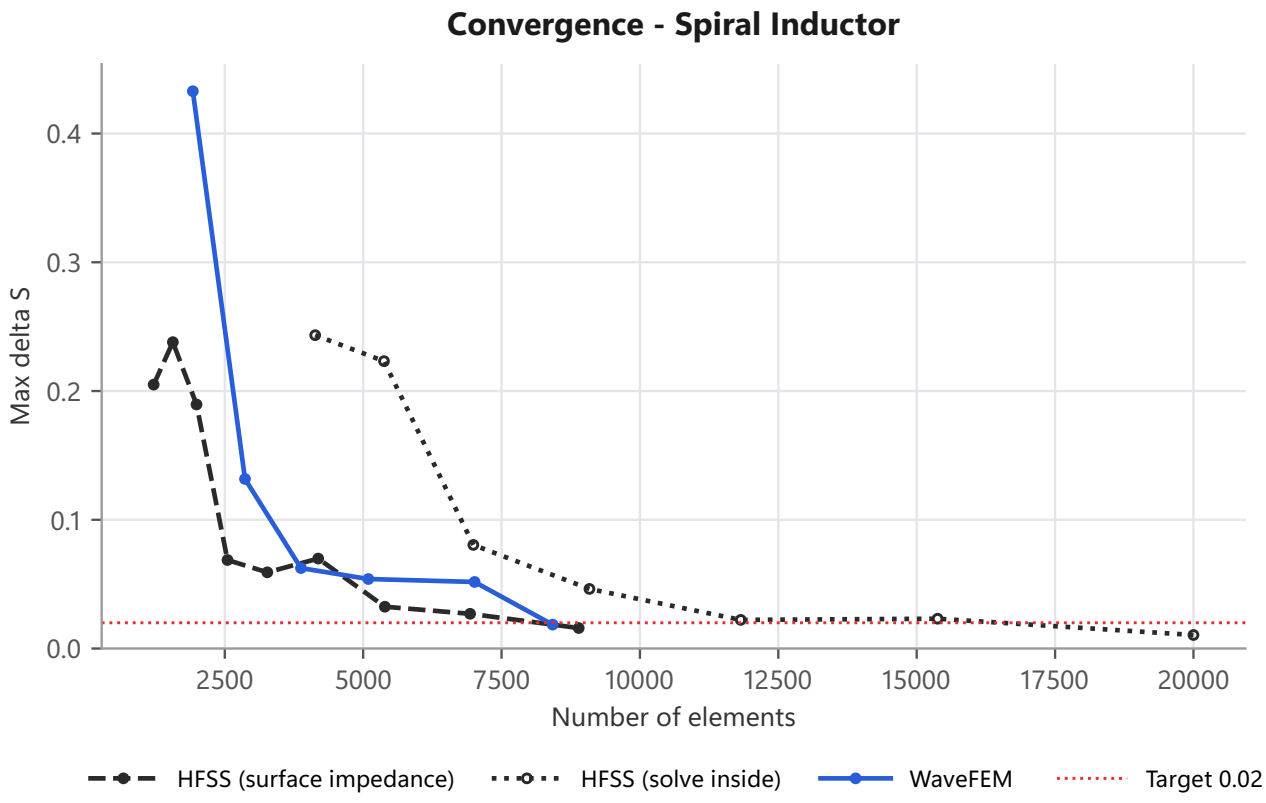


Figure 79 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

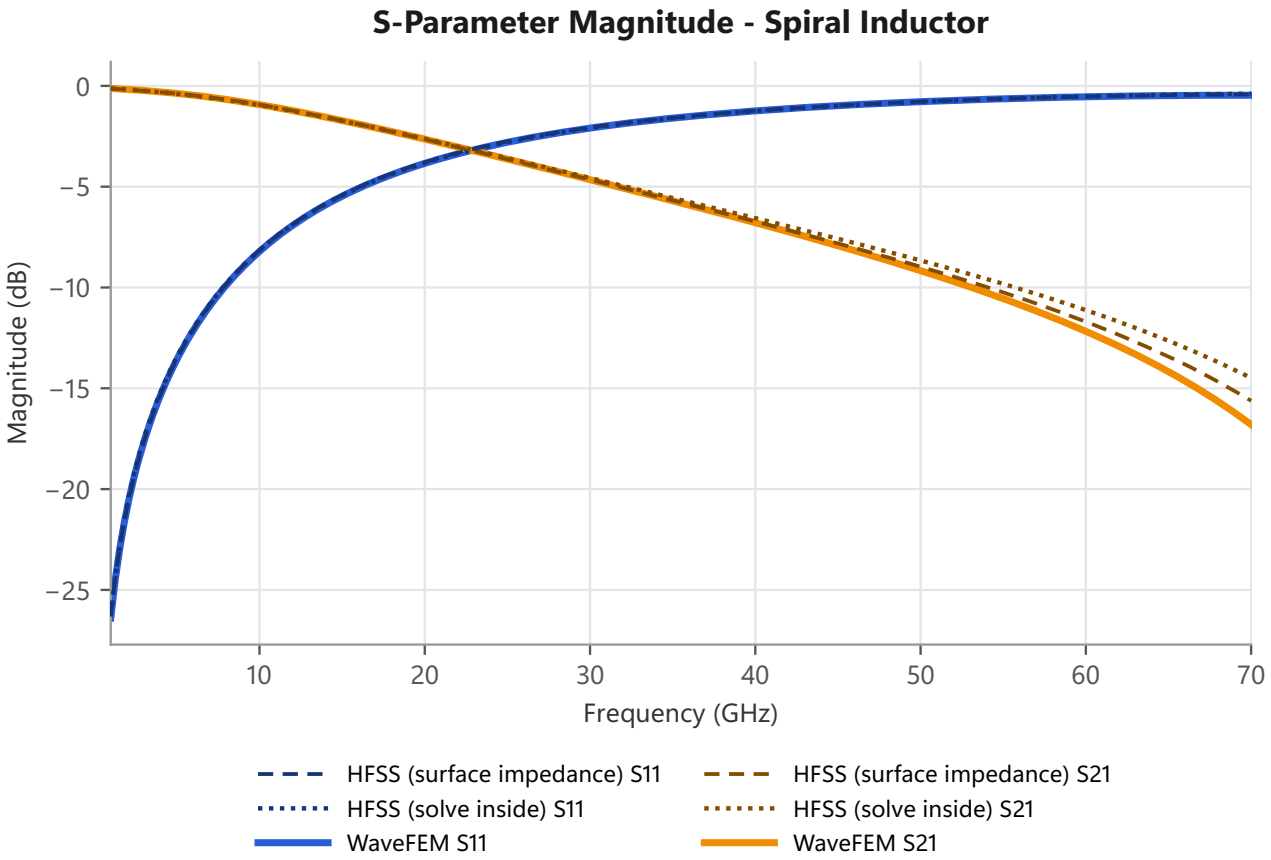


Figure 80 — S-parameter magnitude.

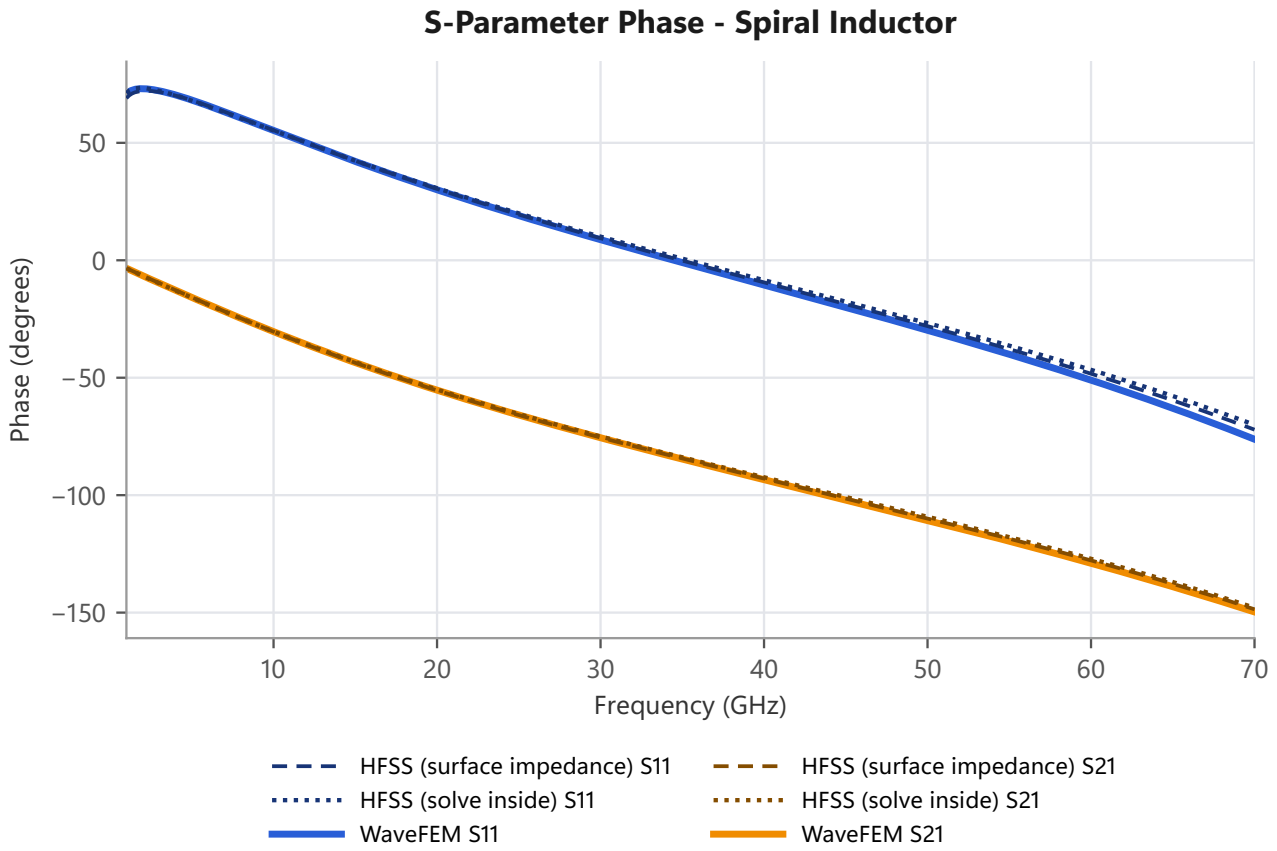


Figure 81 — S-parameter phase.

Inductance and Q

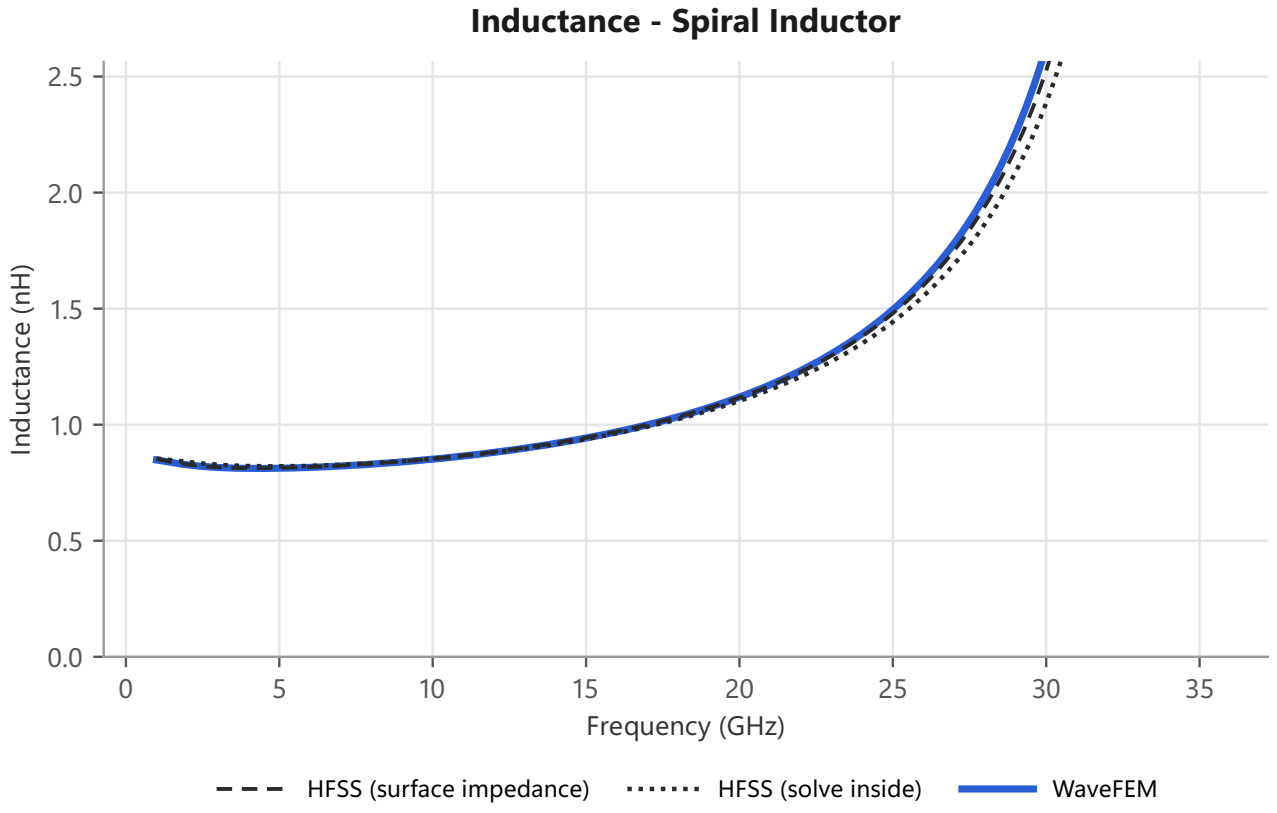


Figure 82 — Inductance Im(Z) / ω, Z = 1 / Y11 (port 2 grounded).

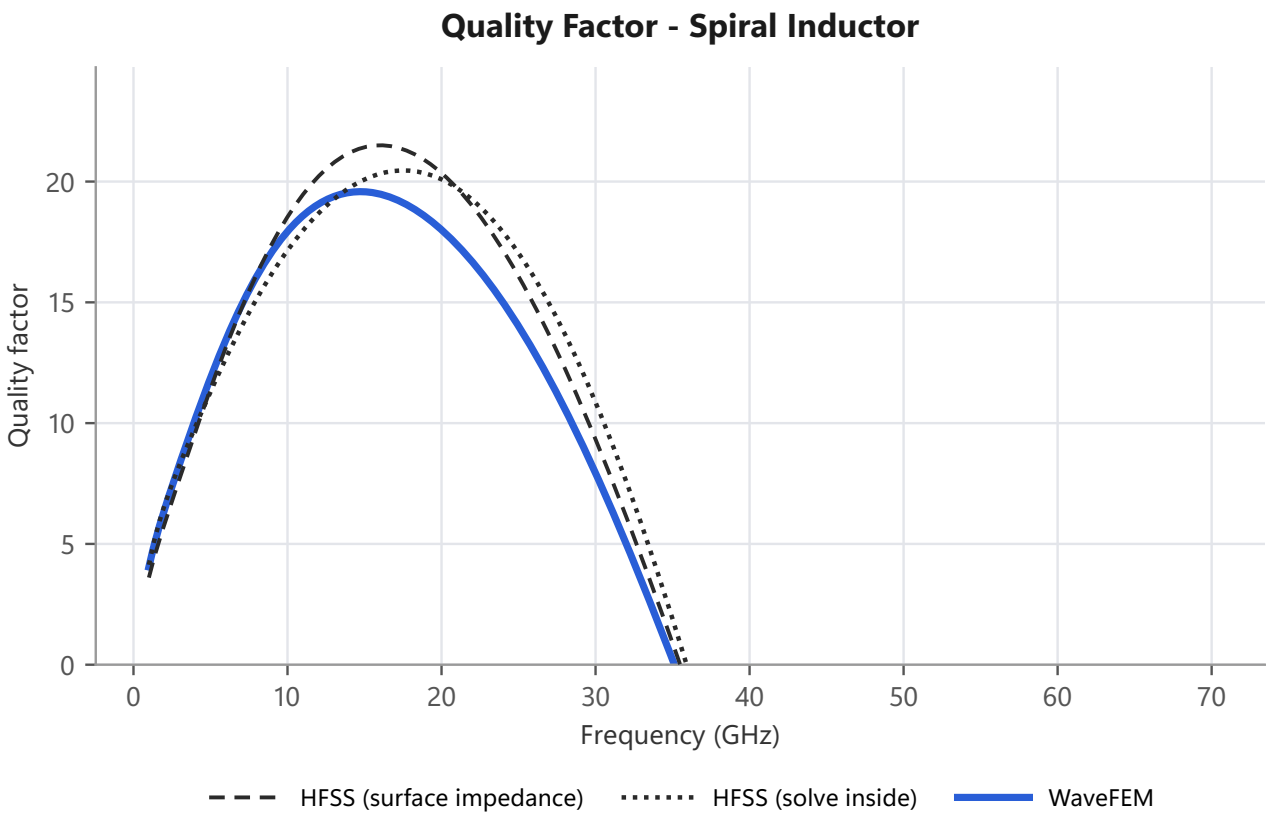


Figure 83 — Quality factor Im(Z) / Re(Z), Z = 1 / Y11 (port 2 grounded). It falls to zero at the self-resonance.

TRANSMISSION LINE

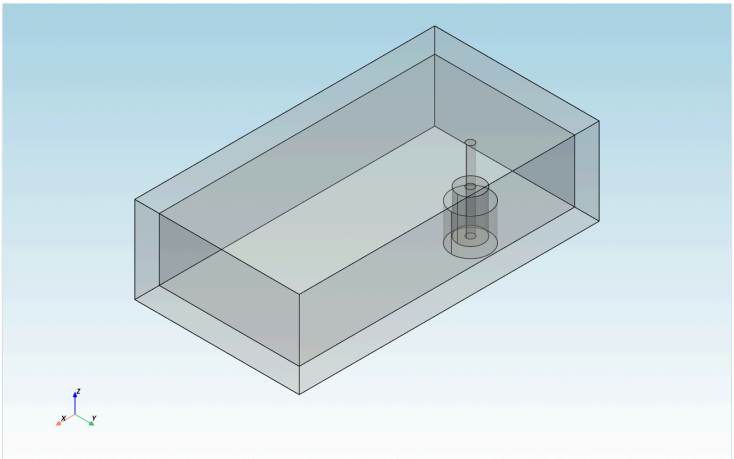
Coax to Waveguide Adapter

A transition between a coaxial cable and a rectangular waveguide, optimised for 9-10 GHz

Simulation Setup

Refinement frequency	10 GHz
Delta S	0.02
Frequency range	8 - 13 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:00:05	00:00:38
Adaptive meshing	00:00:02	00:00:15
Frequency sweep	00:00:02	00:00:19
Adaptive passes	4	6
Final elements	3,486	3,732
Matrix size	19,308	21,270

Convergence

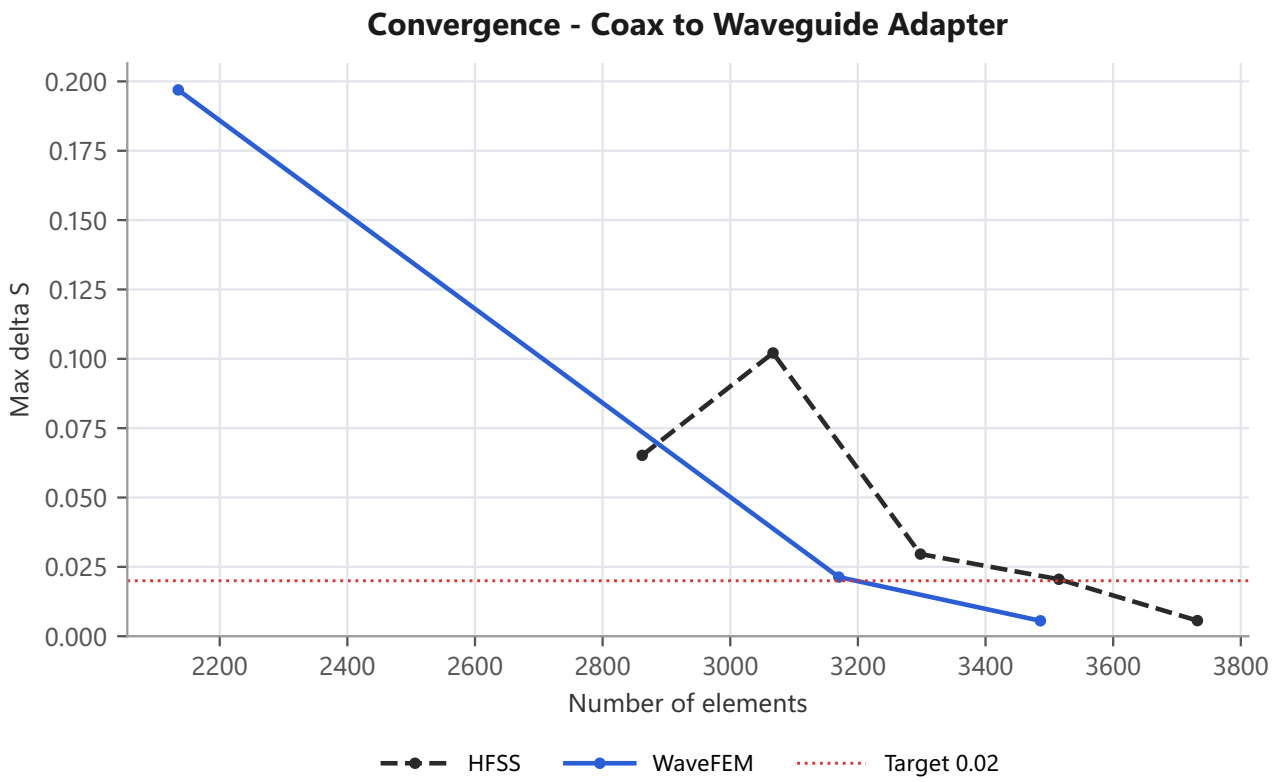


Figure 84 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

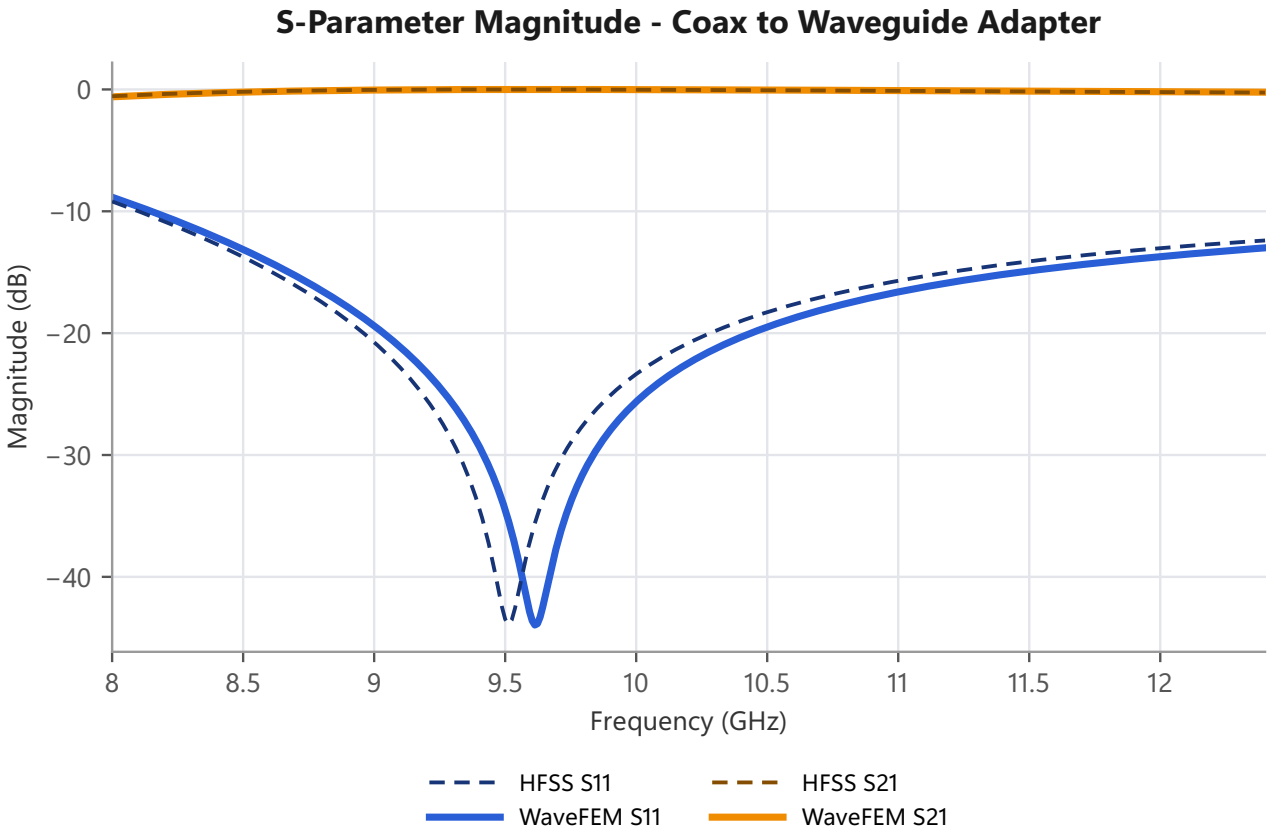


Figure 85 — S-parameter magnitude.

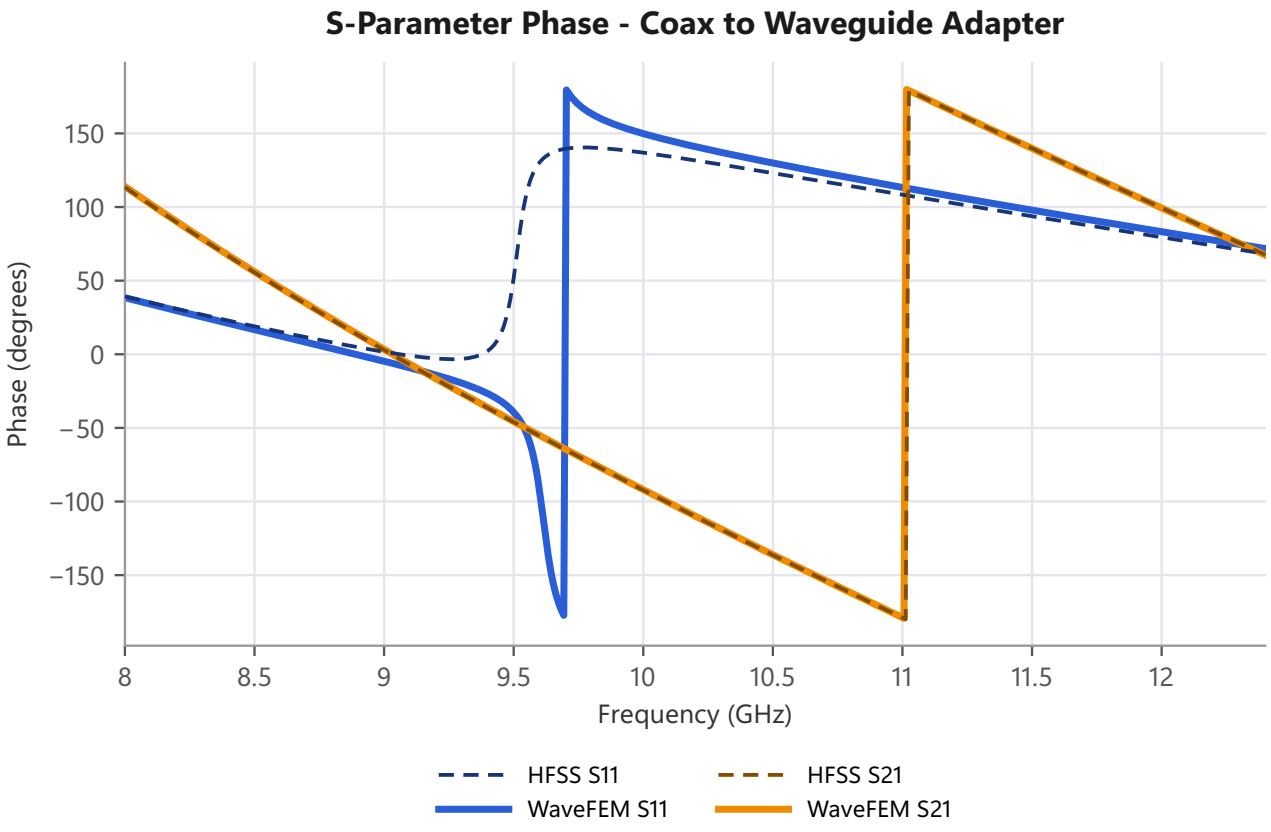


Figure 86 — S-parameter phase.

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TRANSMISSION LINE

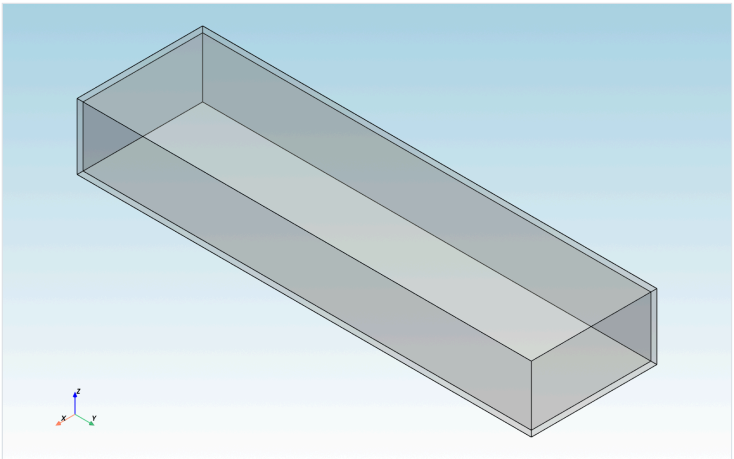
Waveguide

A straight WR159 waveguide, 150mm long

Simulation Setup

Refinement frequency	10 GHz
Delta S	0.02
Frequency range	2 - 10 GHz

Geometry



Statistics

	WaveFEM	HFSS
Total time	00:00:10	00:00:45
Adaptive meshing	00:00:02	00:00:07
Frequency sweep	00:00:06	00:00:34
Adaptive passes	3	3
Final elements	4,909	5,915
Matrix size	34,928	41,448

Convergence

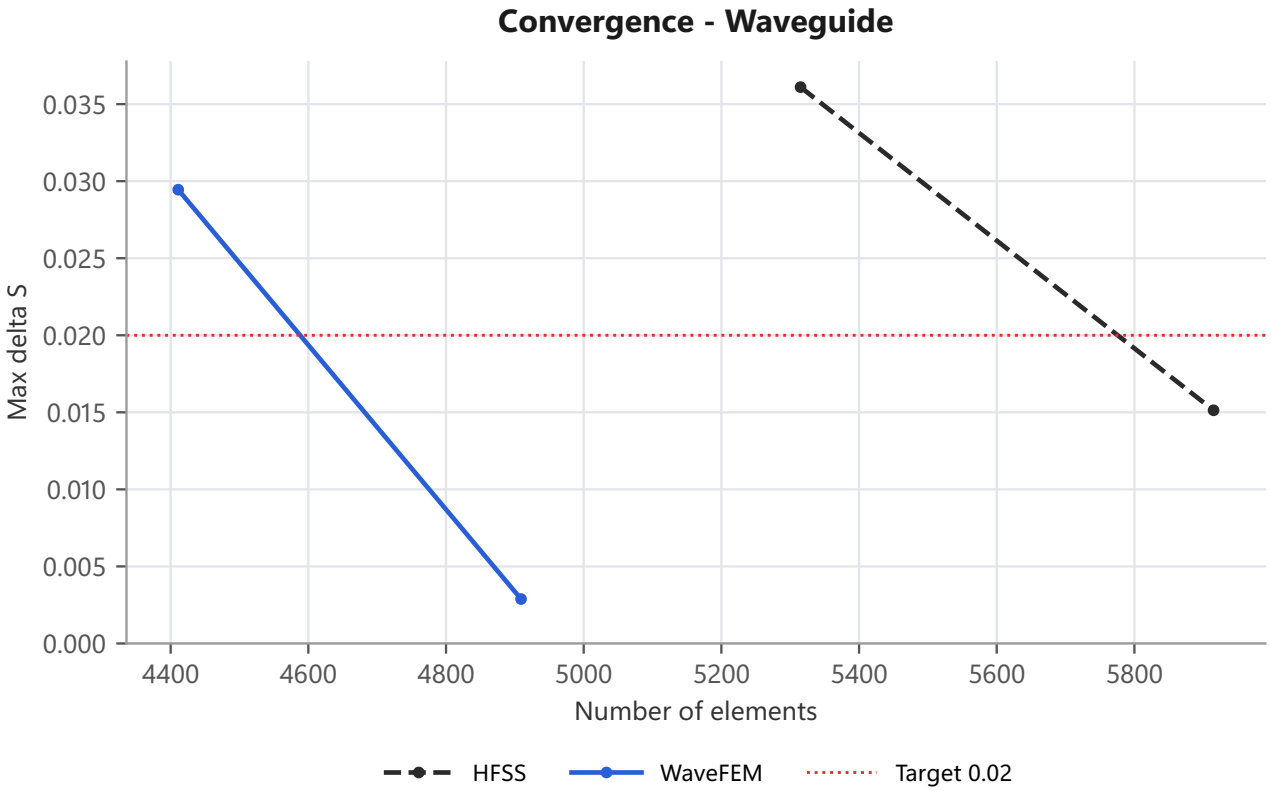


Figure 87 — Adaptive mesh convergence: maximum delta S and solved elements per pass.

S-Parameters

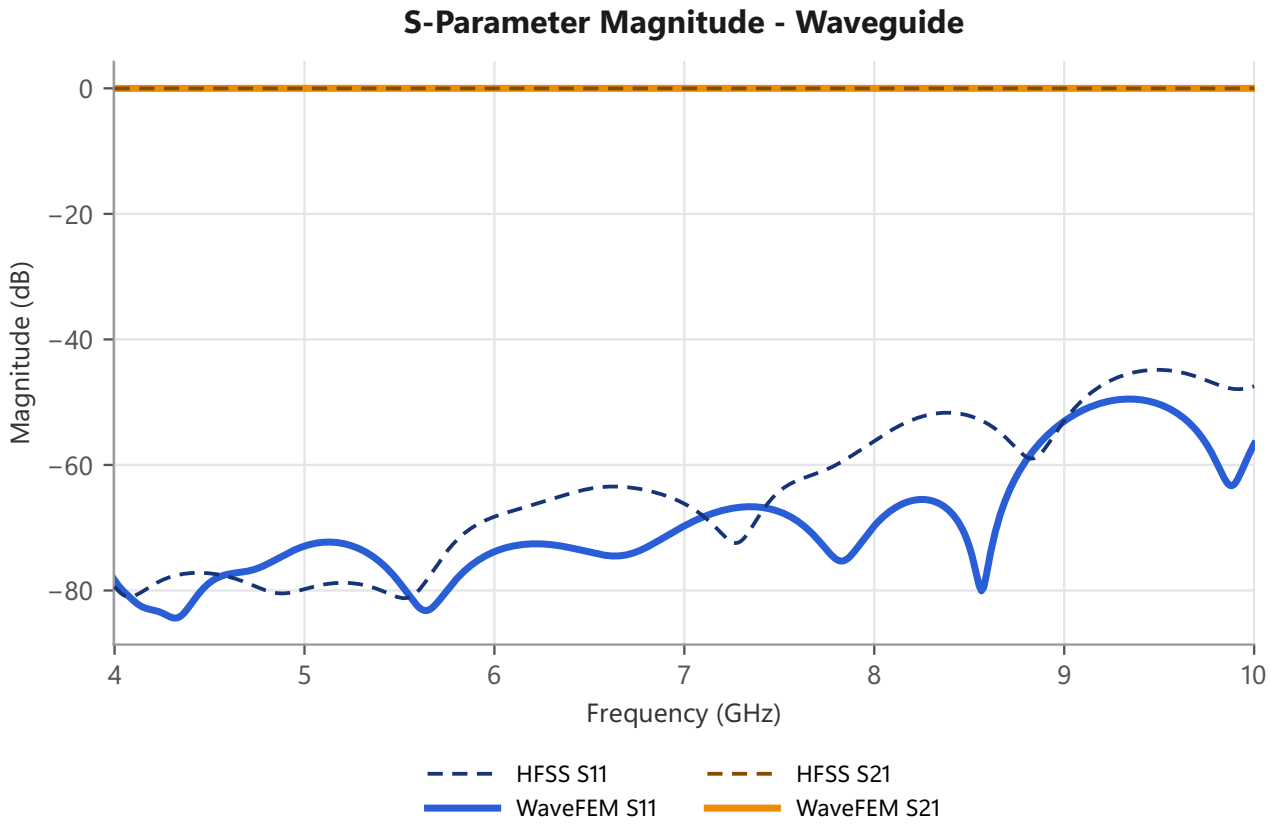


Figure 88 — S-parameter magnitude.

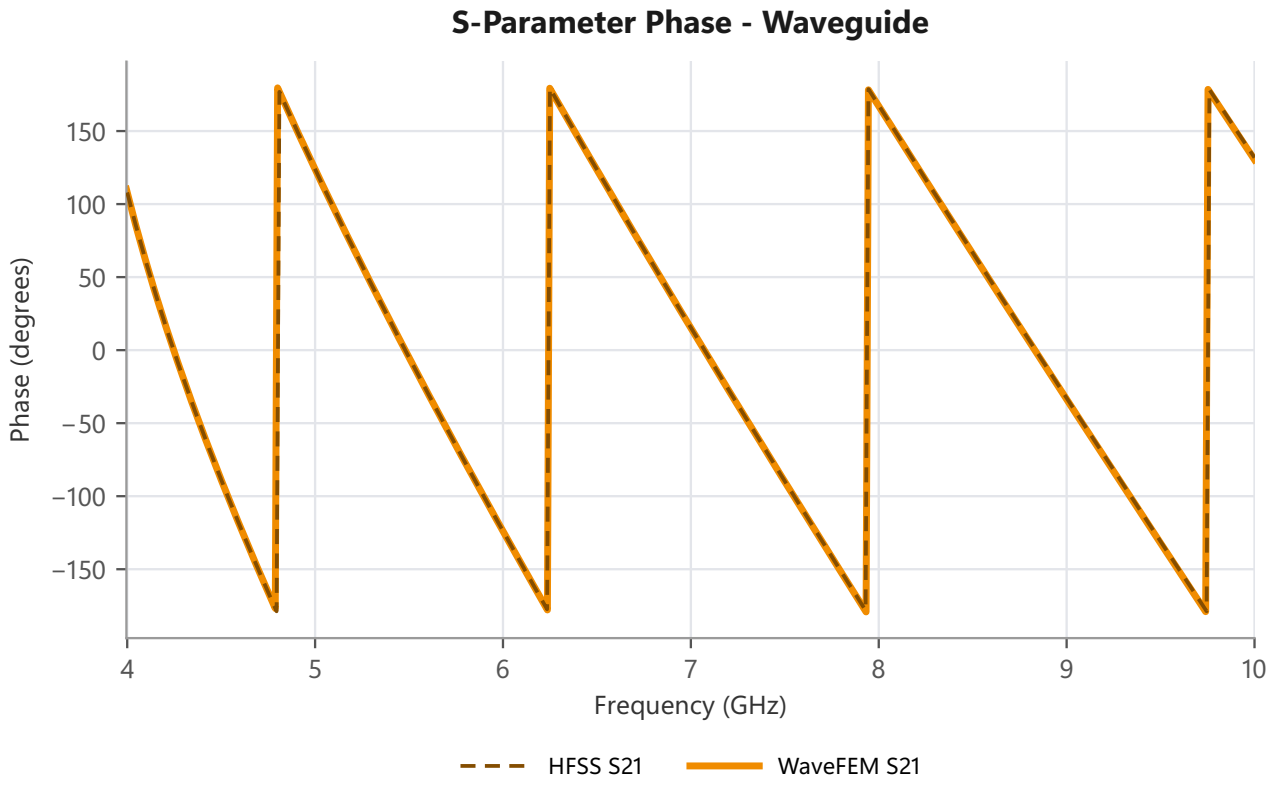


Figure 89 — S-parameter phase.

Last updated 2026-11-02